Question Bank

for AGRICET Preparation

Best Objective Book for AGRICET Aspirants

✓ All 17 Diploma in Agriculture Subjects ✓ Topic-wise MCQs
✓ 1750+ Questions with Answer Keys ✓ Previous Year AGRICET Papers
✓ General Agriculture Section ✓ Two-Column Print Layout

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HOW TO USE THIS BOOK

- This book covers all 17 subjects of the PJTAU Diploma in Agriculture programme relevant to AGRICET.
- Questions are arranged TOPIC-WISE within each subject for systematic revision.
- Each question has four options (a, b, c, d). Choose the single best answer.
- Answer keys are provided at the END of each subject section in the format: 1-A, 2-C, etc.
- For maximum benefit, attempt all questions in a subject before checking the answer key.
- Questions are compiled from official PJTAU mock test series and new practice sets aligned with AGRICET syllabus.
- Use the two-column print layout to save paper — print double-sided for best results.

SUBJECT 1: PRINCIPLES OF AGRONOMY

Code: DA 101 | Total Questions: 100

Topic 1: Definition, Scope & Research Institutes

1. The term Agriculture is derived from *Latin words* ager meaning soil and cultura meaning
   a) management
   b) science
   c) cultivation
   d) growth

2. The term Agronomy is derived from *Greek words Agros meaning* field and nomos meaning
   a) crop
   b) to manage
   c) soil
   d) science

3. Agronomy is a branch of agricultural science dealing with principles and practices of
   a) animal husbandry
   b) horticulture
   c) forestry
   d) soil water and crop management

4. Central Rice Research Institute CRRI is located at
   a) Hyderabad
   b) Cuttack Orissa
   c) New Delhi
   d) Coimbatore

5. Indian Agriculture Research Institute IARI is located at
   a) Nagpur
   b) Lucknow
   c) Bangalore
   d) Pusa New Delhi

6. Central Research *Institute for* Dryland Agriculture CRIDA is located at
   a) Jodhpur
   b) Hyderabad Telangana
   c) Karnal
   d) Bhopal

7. Central Tobacco Research Institute CTRI is located at
   a) Mysore
   b) Hyderabad
   c) Coimbatore
   d) Rajahmundry AP

8. Indian Institute of Horticultural Research IIHR is located at
   a) Bangalore Karnataka
   b) Hyderabad
   c) Lucknow
   d) New Delhi

9. Indian Institute of Rice Research IIRR is located at
   a) Hyderabad Telangana
   b) Cuttack
   c) Coimbatore
   d) Pusa

10. Indian Institute of Millets Research IIMR is located at
   a) Jodhpur
   b) Bhopal
   c) Lucknow
   d) Hyderabad Telangana

11. Central Potato Research Institute CPRI is located at
   a) Nagpur
   b) Shimla HP

14. Central *Institute for* Cotton Research CICR is located at
   a) Coimbatore
   b) Hyderabad
   c) Nagpur Maharashtra
   d) Bhopal

15. Central Arid Zone Research Institute CAZRI is located at
   a) Jaipur
   b) Bikaner
   c) Ajmer
   d) Jodhpur Rajasthan

16. Indian Institute of Oilseeds Research IIOR is located at
   a) Bhopal
   b) Kanpur
   c) Hyderabad Telangana
   d) Lucknow

17. Indian Institute of Pulse Research IIPR is located at
   a) Kanpur UP
   b) Nagpur
   c) Bhopal
   d) Hyderabad

18. Central Soil Salinity Research Institute CSSRI is located at
   a) Jhansi
   b) Dehradun
   c) Karnal Haryana
   d) Bhopal

19. National Plant Protection Training Institute NPPTI is located at
   a) Bangalore
   b) New Delhi
   c) Pune
   d) Hyderabad Telangana

20. Forest Research Institute FRI is located at
   a) Dehradun UP
   b) Shimla
   c) Nainital
   d) Mussoorie

21. Central Food Technological Research Institute CFTRI is located at
   a) Hyderabad
   b) Mysore Karnataka
   c) New Delhi
   d) Pune

22. ICAR stands for
   a) International *Centre for* Agronomy Research
   b) Indian Council of Agricultural Research
   c) Indian *Centre for* Agricultural Resources
   d) Indian *Commission for* Agronomic Research

23. The aim of an agronomist is to obtain
   a) minimum production at maximum cost
   b) maximum production at minimum cost
   c) average production at any cost
   d) maximum production at maximum cost

24. Agriculture is influenced by factors beyond human control such as
   a) irrigation
   b) climate
   c) soil type

c) Jhansi
d) Karnal

12. Sugarcane Breeding Institute SBI is located at
a) Lucknow
b) Coimbatore Tamil Nadu
c) Hyderabad
d) Mysore

13. National Research *Centre for* Groundnut NRCG is located at
a) Rajkot
b) Hyderabad
c) Jodhpur
d) Junagadh Gujarat

d) fertilizers

25. Central Plantation Crops Research Institute CPCRI is located at
a) Coimbatore
b) Kasargod Kerala
c) Hyderabad
d) New Delhi

Topic 2: Agricultural Meteorology & Agro-Climatic Zones

26. Agricultural *Meteorology studies* relationship between weather climate and
a) agricultural crops and operations
b) soil only
c) insects only
d) irrigation only

27. Rainfall is measured by
a) Barometer
b) Anemometer
c) Hygrometer
d) *Rain gauge*

28. *Robinson cup* anemometer is used to measure
a) rainfall
b) humidity
c) temperature
d) wind velocity

29. USWB open pan evaporimeter is used to measure
a) evaporation
b) rainfall
c) wind speed
d) solar radiation

30. *Wet and* dry bulb thermometers are used to measure
a) wind velocity
b) rainfall
c) soil temperature
d) atmospheric humidity

31. *Maximum thermometer* records the
a) minimum temperature of day
b) average temperature
c) maximum temperature of day
d) soil temperature

32. The lowest layer of atmosphere is called
a) Stratosphere
b) Mesosphere
c) Exosphere
d) Troposphere

33. *Solar radiation* reaching earth surface is called
a) Albedo
b) Radiation
c) Insolation
d) Conduction

34. Photoperiod is the period of
a) daylight
b) darkness
c) 24 hour temperature cycle
d) rainfall duration

39. Vernalization is promotion of flowering by
a) low temperature treatment
b) light
c) high temperature
d) fertilizers

40. *Relative humidity* is expressed in
a) percentage
b) mm
c) grams per litre
d) mg per cubic metre

41. The three types of rainfall based on origin are convectional orographic and
a) cyclonic or frontal
b) monsoon
c) continental
d) local

42. Fog is condensation of water vapour occurring at
a) high altitude
b) stratosphere
c) sea surface only
d) ground level

43. *Hygrograph records*
a) temperature
b) relative humidity continuously
c) wind speed
d) solar radiation

44. *Dust storms* occur mainly in
a) coastal areas
b) arid and semi-arid zones
c) humid tropics
d) hill stations

45. *Wind direction* is measured by
a) Anemometer
b) *Wind vane*
c) *Rain gauge*
d) Barometer

46. Agro-climatic zones are classified based on
a) soil type only
b) crop varieties
c) fertilizer use
d) temperature and rainfall patterns

47. Long day plants require day length
a) more than critical period to flower
b) less than critical period to flower
c) exactly 12 hours
d) total darkness of 24 hours

35. Short day plants flower when day length is
a) less than critical period
b) more than critical period
c) exactly 12 hours
d) exactly 24 hours

36. *Thermograph records*
a) humidity only
b) temperature continuously over time
c) wind speed
d) solar radiation

37. *Barometer measures*
a) humidity
b) wind speed
c) atmospheric pressure
d) rainfall

38. Climatology is the science dealing with
a) long-term weather patterns
b) weather prediction only
c) soil formation
d) crop diseases

48. *Dew point* is the temperature at which air becomes saturated with
a) dust
b) water vapour
c) salt
d) nitrogen

49. *Heat wave* is a condition of
a) very low temperature
b) abnormally high temperature for extended period
c) very high humidity
d) very low rainfall

50. *Psychrometer uses* wet and dry bulb thermometers to measure
a) temperature only
b) wind velocity
c) solar radiation
d) relative humidity and dew point

Topic 3: Tillage Sowing & Weed Management

51. Mechanical manipulation of soil for crop production is called
a) irrigation
b) tillage
c) harvesting
d) threshing

52. Primary tillage involves
a) deep ploughing of soil
b) shallow cultivation
c) weed control only
d) fertilizer application

53. Secondary tillage refers to
a) deep ploughing
b) sub-soil ploughing
c) ridge making only
d) shallow lighter tillage for seedbed preparation

54. *Ideal soil* physical condition for seed germination is called
a) Tilth
b) Texture
c) Structure
d) Porosity

55. Puddling is tillage done under
a) dry conditions
b) flooded conditions for rice cultivation
c) sandy soils only
d) upland crops

56. *Weeds are* plants growing where they are
a) desired
b) planted intentionally
c) needed by farmer
d) not desired

57. *Herbicides are* chemicals used for control of
a) insects
b) weeds
c) diseases
d) rats

58. 2,4-D herbicide is used mainly against
a) grassy weeds in paddy
b) broadleaf weeds in cereals

64. *Cyperus rotundus* purple nutsedge is difficult to control because of its
a) underground tubers and rhizomes
b) tall growth
c) resistance to all herbicides
d) shallow roots

65. *Zero tillage* means
a) deep ploughing
b) minimum tillage
c) no tillage with direct seeding
d) ridge and furrow method

66. *Minimum tillage* is practiced to
a) conserve soil moisture and reduce costs
b) increase soil erosion
c) increase weed population
d) improve drainage only

67. *Striga witch* weed is a parasitic weed that infests
a) soybean only
b) vegetables only
c) fruit crops only
d) cereals like sorghum and millets

68. *Drilling method* of sowing places seeds in
a) furrows at regular depth and spacing
b) broadcast randomly
c) hills on mounds
d) pits only

69. Seed rate is quantity of seed required per
a) plant
b) unit area of land
c) season
d) variety

70. *Critical period* of crop-weed competition is when
a) crops are harvested
b) irrigation is applied
c) fertilizer is added
d) weeds must be controlled to prevent yield loss

71. The implement used for primary tillage is
a) Cultivator
b) *Disc harrow*

c) all weeds equally
d) perennial grasses only

59. Integrated Weed *Management combines*
a) chemical mechanical and biological methods
b) only chemical methods
c) only mechanical methods
d) only biological methods

60. Allelopathy is suppression of one plant by another through
a) chemical substances released into environment
b) competition for water only
c) pest attack
d) soil erosion

61. Pre-emergence herbicides are applied
a) before sowing
b) before weed emergence but after sowing
c) after weed emergence
d) during crop harvesting

62. Post-emergence herbicides are applied
a) before sowing
b) before weeds emerge
c) during land preparation
d) after emergence of weeds or crop

63. *Parthenium hysterophorus* is commonly called
a) *Cyperus rotundus*
b) *Cynodon dactylon*
c) *Congress grass* or *Carrot weed*
d) *Echinochloa crusgalli*

c) *Blade harrow*
d) *Mould board* plough

72. Butachlor is a pre-emergence herbicide used in
a) wheat
b) maize
c) transplanted paddy
d) sorghum

73. *Echinochloa crusgalli* barnyard grass is a major weed in
a) wheat fields
b) cotton fields
c) paddy fields
d) groundnut fields

74. Thinning in crops is done to
a) increase plant population
b) reduce plant competition where plants are too dense
c) improve soil structure
d) control weeds only

75. Harrowing is used to
a) deep plough soil
b) apply fertilizers
c) harvest crops
d) break clods and prepare seedbed after primary tillage

Topic 4: Irrigation Cropping Systems & Organic Farming

76. *Drip irrigation* is also called
a) sprinkler irrigation
b) flood irrigation
c) furrow irrigation
d) trickle irrigation

77. The irrigation method with highest water use efficiency is
a) drip irrigation
b) flood irrigation
c) furrow irrigation
d) basin irrigation

78. *Sprinkler irrigation* is most suitable for
a) rice cultivation
b) heavy clay soils
c) waterlogged areas
d) uneven topography and light sandy soils

79. *Field capacity* is soil moisture retained after
a) rainfall only
b) plant wilting
c) excess water drains by gravity 1 to 3 days after saturation
d) deep irrigation only

80. Drainage is removal of excess water from
a) atmosphere
b) soil and land surface
c) plants only
d) air only

81. Mixed cropping is growing two or more crops simultaneously in same field
a) with definite row arrangement
b) without definite row arrangement
c) in alternate years
d) in separate fields

89. *Critical stage* of irrigation in wheat is
a) crown root initiation CRI stage
b) germination
c) harvesting
d) grain filling only

90. *Contour farming* means cultivating land
a) in straight rows down slope
b) along contour lines across slope to reduce erosion
c) along the slope
d) in any direction

91. Green manuring involves incorporation of
a) crop residues after harvest
b) FYM only
c) green plant material into soil before decomposition
d) compost only

92. Crop rotation grows different crops on same land in
a) same season only
b) a planned sequence over years
c) random order
d) same year always

93. *Contingency crop* planning is prepared for
a) normal rainfall
b) excess rainfall only
c) frost only
d) delayed or deficient rainfall

94. Mulching is done to
a) increase soil evaporation
b) increase surface runoff
c) conserve soil moisture and control weeds
d) compact the soil

95. Permanent wilting point is soil moisture at which plants
a) wilt permanently and cannot recover

82. Intercropping is growing two or more crops simultaneously with
a) no row arrangement
b) random mixing
c) definite row arrangement
d) different seasons

83. *Relay cropping* means sowing second crop
a) after harvest of first crop
b) in different field
c) before harvest of first crop
d) in pots

84. Dryland agriculture is practiced where annual rainfall is less than
a) 750 mm
b) 1200 mm
c) 500 mm
d) 1500 mm

85. Watershed is an area of land draining to a common
a) outlet point or stream
b) field
c) reservoir only
d) well

86. *Organic farming* avoids
a) synthetic chemicals and fertilizers
b) seeds
c) irrigation
d) all machinery

87. *Sustainable agriculture* meets present needs without compromising
a) crop yield only
b) market prices
c) future generations ability to meet their needs
d) land area

88. *Water harvesting* collects runoff water for
a) drinking only
b) industrial use only
c) productive use in agriculture
d) evaporation

b) show luxuriant growth
c) just wilt temporarily
d) need maximum irrigation

96. NPOP stands for
a) National *Plan for* Organic Produce
b) National Policy on Organic Products
c) National *Programme for* Organic Production
d) National *Protocol for* Organic Plants

97. *Furrow irrigation* applies water in
a) basins
b) overhead sprinklers
c) underground pipes only
d) small channels between crop rows

98. The main problem in dryland agriculture is
a) excess water
b) moisture stress or drought
c) soil salinity
d) waterlogging

99. Integrated Farming *System combines* crop production with
a) only chemical inputs
b) only irrigation systems
c) only marketing
d) livestock fishery and other farm enterprises

100. *Basin irrigation* is most suitable for
a) wheat and maize only
b) cotton and groundnut
c) vegetables only
d) orchards and rice paddies

ANSWER KEY — PRINCIPLES OF AGRONOMY

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	B	D	B	D	B	D	A	A	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	B	D	C	D	C	A	C	D	A
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	B	B	B	B	A	D	D	A	D
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
C	D	C	A	A	B	C	A	A	A
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
A	D	B	B	B	D	A	B	B	D
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
B	A	D	A	B	D	B	B	A	A
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
B	D	C	A	C	A	D	A	B	D
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
D	C	C	B	D	D	A	D	C	B
81	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**
B	C	C	A	A	A	C	C	A	B
91	**92**	**93**	**94**	**95**	**96**	**97**	**98**	**99**	**100**
C	B	D	C	A	C	D	B	D	D

SUBJECT 2: CROP PRODUCTION I – KHARIF CROPS

Code: DA 102 | Total Questions: 100

Topic 1: Paddy – Nursery Transplanting and Varieties

1. Botanical name of rice is
   a) *Triticum aestivum*
   b) *Zea mays*
   c) *Sorghum bicolor*
   d) *Oryza sativa*

2. Rice belongs to family
   a) Leguminosae
   b) Poaceae Gramineae
   c) Solanaceae
   d) Malvaceae

3. SRI stands for
   a) Systematic Rice Improvement
   b) System of Rice Intensification
   c) Standard Rice Irrigation
   d) Sequential Rice Intercropping

4. In SRI method seedlings are transplanted at age of
   a) 25 to 30 days
   b) 8 to 12 days
   c) 45 days
   d) 60 days

5. *Wet nursery* for paddy requires
   a) dry raised seedbed
   b) upland dry bed
   c) flooded or puddled seedbed
   d) container nursery

6. Duration of paddy nursery before transplanting is generally
   a) 5 to 7 days
   b) 45 to 60 days
   c) 10 to 12 days
   d) 20 to 25 days

7. *Breaking seed* dormancy in paddy is done by
   a) soaking in plain water
   b) soaking in 1 percent KNO3 or dry heat treatment
   c) soaking in acid
   d) exposure to light only

8. Seed rate for paddy transplanting method is approximately
   a) 100 kg per ha
   b) 20 to 25 kg per ha
   c) 75 kg per ha
   d) 50 kg per ha

9. BPT 5204 Samba Mahsuri is a popular variety of
   a) maize
   b) paddy
   c) groundnut
   d) cotton

10. MTU 1010 is a variety of
   a) wheat
   b) maize
   c) paddy rice
   d) sorghum

11. *Spacing recommended* for transplanting paddy is
   a) 5cm x 5cm
   b) 50cm x 50cm
   c) 20cm x 10cm
   d) 100cm x 50cm

12. *Clipping tips* of paddy seedlings before transplanting eliminates

14. Paddy is a
   a) long day plant
   b) short day plant
   c) day neutral plant
   d) biennial plant

15. *Zinc deficiency* in paddy causes
   a) blast
   b) neck rot
   c) khaira disease yellowing and stunting
   d) brown spot

16. *Submergence tolerant* variety of paddy is
   a) BPT 5204
   b) MTU 1010
   c) NLR 34449
   d) Swarna Sub 1

17. *Seedling rate* per hill in SRI method is
   a) 3 to 4 seedlings
   b) 5 to 6 seedlings
   c) 1 seedling
   d) 2 to 3 seedlings

18. *Spacing used* in SRI method is
   a) 10cm x 10cm
   b) 5cm x 5cm
   c) 25cm x 25cm
   d) 50cm x 50cm

19. Types of paddy nursery commonly practised are wet dry and
   a) container nursery
   b) raised bed nursery
   c) dapog nursery
   d) pot nursery

20. The age of healthy paddy nursery for transplanting should not exceed
   a) 10 days
   b) 60 days
   c) 25 to 30 days
   d) 45 days

21. *Dapog nursery* requires a seed rate of approximately
   a) 20 kg per ha
   b) 100 to 120 kg per ha
   c) 50 kg per ha
   d) 5 kg per ha

22. Transplanting by machine ensures
   a) uniform spacing and depth
   b) random planting
   c) uneven spacing
   d) deeper planting only

23. *Iron deficiency* in paddy causes
   a) blast disease
   b) white tip
   c) neck rot
   d) bronzing or yellowing of leaves

24. Rice is primarily grown in regions that are
   a) arid regions
   b) humid tropical and subtropical
   c) temperate only
   d) alpine regions

25. *Dry nursery* for paddy is suitable when

a) leaf diseases
b) weeds
c) tillers
d) egg masses of stem borer

13. Main field preparation for paddy involves
a) puddling under flooded conditions
b) dry ploughing only
c) only surface levelling
d) only harrowing

a) water is abundant
b) summer season only
c) water supply is limited or in rabi season
d) any season freely

Topic 2: Paddy – Nutrient Water and Weed Management

26. INM stands for
a) Indian Nitrogen Management
b) International Nutrient Management
c) Intensive Nutrient Mix
d) Integrated Nutrient Management

27. *Recommended nitrogen* dose for high yielding paddy is generally
a) 10 to 20 kg per ha
b) 200 kg per ha
c) 100 to 120 kg per ha
d) 50 kg per ha

28. *Split application* of nitrogen in paddy means applying in
a) single dose at sowing
b) 2 to 3 splits at different growth stages
c) only at panicle initiation
d) only at transplanting

29. *Fertilizer applied* as basal dose is incorporated
a) after transplanting
b) at harvesting
c) at or before transplanting
d) at flowering

30. *Critical stage* for irrigation in paddy is
a) only at germination
b) only at maturity
c) only at transplanting
d) tillering and flowering

31. *Butachlor herbicide* is used to control weeds in
a) wheat
b) transplanted paddy
c) maize
d) sorghum

32. Blue Green *Algae and Azolla are* used as biofertilizer in
a) upland crops
b) dryland crops
c) flooded paddy fields
d) forest nurseries

33. Leaf Color Chart LCC is used to estimate
a) water requirement
b) disease severity
c) weed population
d) nitrogen status of paddy leaves

34. AWD Alternate *Wetting and* Drying in paddy saves
a) water without significant yield loss
b) labour
c) fertilizer
d) seeds

35. *Urea contains* nitrogen percentage of
a) 20.6 percent
b) 34 percent
c) 46 percent

39. *Bacterial leaf* blight of paddy is caused by
a) *Pyricularia oryzae*
b) *Rhizoctonia solani*
c) *Xanthomonas oryzae* pv oryzae
d) Helminthosporium

40. *Sheath blight* of paddy is caused by
a) *Magnaporthe oryzae*
b) *Xanthomonas oryzae*
c) Helminthosporium
d) *Rhizoctonia solani*

41. Paddy is harvested when grain moisture content is approximately
a) 20 to 25 percent
b) 5 percent
c) 50 percent
d) 35 percent

42. Most common micronutrient deficiency in paddy cultivation in Telangana is
a) Iron
b) Zinc
c) Manganese
d) Copper

43. Green manure crop commonly used in paddy is
a) Wheat
b) Mustard
c) Sunflower
d) *Sesbania dhaincha*

44. Paddy reaches physiological maturity when
a) panicle turns golden and 80 percent grains are hard
b) all grains are milky
c) only tillering is complete
d) leaves are still green

45. Single Super Phosphate SSP contains P2O5 of
a) 18 percent
b) 46 percent
c) 32 percent
d) 48 percent

46. High C to N ratio of paddy straw causes
a) nitrogen immobilization in soil
b) nitrogen enrichment in soil
c) increased crop yield
d) no change in soil

47. *Water depth* maintained in paddy during vegetative stage is
a) 2 to 5 cm
b) 10 to 15 cm
c) 30 cm
d) deeply flooded always

48. *Echinochloa crusgalli* barnyard grass is a major weed in
a) wheat fields
b) paddy fields

d) 32 percent

36. DAP Di-Ammonium *Phosphate contains* N and P2O5 in ratio
   a) 18:46
   b) 46:0
   c) 14:14
   d) 0:46

37. Stem borer damage in paddy during vegetative stage causes
   a) white ear
   b) blast
   c) sheath blight
   d) dead heart

38. Blast disease in paddy is caused by
   a) *Xanthomonas oryzae*
   b) *Helminthosporium oryzae*
   c) *Rhizoctonia solani*
   d) *Magnaporthe oryzae*

c) cotton fields
d) groundnut fields

49. Topdressing of fertilizers is done
   a) after transplanting when crop is growing
   b) before transplanting
   c) at land preparation
   d) at harvest

50. *Neem coated* urea is used to
   a) increase nitrogen loss
   b) improve colour
   c) slow down nitrification and reduce nitrogen losses
   d) increase disease resistance

Topic 3: Maize Sorghum and Bajra

51. Botanical name of maize is
   a) *Sorghum bicolor*
   b) *Pennisetum glaucum*
   c) *Zea mays*
   d) *Oryza sativa*

52. Maize belongs to family
   a) Fabaceae
   b) Solanaceae
   c) Asteraceae
   d) Poaceae

53. Maize is a
   a) C3 plant
   b) CAM plant
   c) C2 plant
   d) C4 plant

54. Botanical name of Sorghum is
   a) *Zea mays*
   b) *Pennisetum glaucum*
   c) *Sorghum bicolor*
   d) *Oryza sativa*

55. Common name *Jowar refers* to
   a) Maize
   b) Sorghum
   c) Bajra
   d) Ragi

56. Botanical name of Bajra *Pearl millet* is
   a) *Pennisetum glaucum*
   b) *Sorghum bicolor*
   c) *Zea mays*
   d) *Eleusine coracana*

57. Seed rate for maize is approximately
   a) 20 to 25 kg per ha
   b) 5 kg per ha
   c) 100 kg per ha
   d) 75 kg per ha

58. Sorghum is called camel of crops because it is
   a) grown in deserts only
   b) a C3 plant
   c) requires high water
   d) highly drought tolerant

59. *Shoot fly* of sorghum causes the symptom
   a) dead heart

64. *Ergot disease* of sorghum is caused by
   a) *Rhizoctonia solani*
   b) *Pyricularia grisea*
   c) *Claviceps sorghi*
   d) *Alternaria spp*

65. *Sweet sorghum* is used for producing
   a) protein
   b) oil
   c) starch only
   d) sugar from stem juice for ethanol

66. Bajra is mainly grown in
   a) Heavy black cotton soil
   b) *Waterlogged soils*
   c) *Highly acidic* soils
   d) *Sandy loam* to loam well-drained soils

67. *Zinc deficiency* in maize causes
   a) iron chlorosis
   b) tip burn
   c) white bud symptom
   d) stem rot

68. Midge of sorghum belongs to order
   a) Diptera
   b) Coleoptera
   c) Lepidoptera
   d) Hemiptera

69. Downy mildew of bajra is caused by
   a) *Sclerospora graminicola*
   b) *Pyricularia grisea*
   c) Helminthosporium
   d) Alternaria

70. *Charcoal rot* of sorghum is caused by
   a) *Magnaporthe oryzae*
   b) *Macrophomina phaseolina*
   c) *Rhizoctonia solani*
   d) *Fusarium oxysporum*

71. *Fodder crops* are harvested at which stage for maximum nutrition
   a) full maturity
   b) seedling stage
   c) 50 percent flowering stage
   d) after seed formation

72. Stem borer of maize bores into

b) white ear
c) blast
d) leaf blight

60. Maize is pollinated by
a) self-pollination
b) cross-pollination by wind
c) insects only
d) water

61. Male flower of maize is called
a) tassel
b) ear
c) spikelet
d) panicle

62. Female flower of maize bearing grain is called
a) tassel
b) spike
c) ear or cob
d) umbel

63. *Recommended spacing* for maize is
a) 5cm x 5cm
b) 100cm x 100cm
c) 60cm x 20cm
d) 10cm x 10cm

a) roots only
b) stem and cobs
c) leaves only
d) soil

73. Popcorn is a special type of
a) sorghum
b) maize variety
c) wheat
d) rice

74. *Ragi botanical* name is
a) *Sorghum bicolor*
b) *Pennisetum glaucum*
c) *Eleusine coracana*
d) *Panicum miliaceum*

75. Green cob stage of maize is harvested at approximately
a) 30 to 40 days
b) 90 to 100 days
c) 55 to 65 days
d) 120 days

Topic 4: Cotton Groundnut and Sunflower

76. Botanical name of *American cotton* is
a) *Arachis hypogaea*
b) *Helianthus annuus*
c) *Cajanus cajan*
d) *Gossypium hirsutum*

77. Cotton belongs to family
a) Poaceae
b) Malvaceae
c) Fabaceae
d) Asteraceae

78. Bt cotton contains gene from
a) *Bacillus thuringiensis*
b) Rhizobium
c) Agrobacterium
d) Azotobacter

79. Pink bollworm is a major pest of
a) cotton
b) paddy
c) groundnut
d) sunflower

80. Cotton lint is seed covering made of
a) protein fibres
b) lignin
c) cellulose fibres
d) wax

81. Botanical name of groundnut is
a) *Arachis hypogaea*
b) *Glycine max*
c) *Vigna radiata*
d) *Cajanus cajan*

82. *Groundnut belongs* to family
a) Poaceae
b) Asteraceae
c) Leguminosae Fabaceae
d) Malvaceae

83. Geocarpy in groundnut means

89. Earthing up in groundnut facilitates
a) fertilizer uptake
b) peg penetration underground for pod development
c) weed control
d) irrigation

90. Maturity of groundnut is indicated by
a) yellowing of all leaves
b) stem turning red
c) all flowers dropping
d) inner wall of pod turning dark and seed filling

91. *American bollworm Helicoverpa armigera* attacks
a) roots only
b) leaves of paddy
c) bolls squares and flowers of cotton
d) stems of wheat

92. Seed treatment with Thiram or Captan in groundnut protects against
a) seed and soil-borne diseases
b) stem borer
c) nematodes only
d) virus diseases only

93. TMV 2 is a popular variety of
a) cotton
b) sunflower
c) groundnut
d) soybean

94. *Sunflower crop* duration is approximately
a) 30 to 40 days
b) 180 days
c) 60 to 70 days
d) 90 to 100 days

95. Cotton requires approximately from sowing to harvest
a) 2 to 3 months
b) 1 year
c) 5 to 6 months
d) 8 to 9 months

a) growing in water
b) aerial roots
c) pod development underground
d) stem tubers

84. Tikka disease of groundnut is caused by
a) *Pyricularia oryzae*
b) *Cercospora arachidicola*
c) *Rhizoctonia solani*
d) Xanthomonas

85. Botanical name of sunflower is
a) *Helianthus annuus*
b) *Brassica juncea*
c) *Sesamum indicum*
d) *Carthamus tinctorius*

86. *Sunflower belongs* to family
a) Asteraceae Compositae
b) Poaceae
c) Fabaceae
d) Solanaceae

87. Sunflower is mainly pollinated by
a) wind
b) self-pollination
c) water
d) bees and insects

88. Seed rate of groundnut kernel basis is approximately
a) 5 kg per ha
b) 200 kg per ha
c) 10 kg per ha
d) 100 to 120 kg per ha

96. *Sucking pests* like aphids jassids and whitefly are major pests of
a) paddy
b) cotton
c) wheat
d) sorghum

97. *Sunflower oil* is rich in
a) linoleic acid polyunsaturated
b) saturated fats
c) lauric acid
d) palmitic acid

98. India ranks first in world production of
a) soybean
b) sunflower
c) castor safflower sesame and niger
d) wheat

99. Technology Mission on Oilseeds TMO was initiated in India in
a) 1975
b) 2001
c) 1991
d) 1986

100. K-6 is a variety of
a) groundnut
b) cotton
c) sunflower
d) sorghum

ANSWER KEY — CROP PRODUCTION I – KHARIF CROPS

1	2	3	4	5	6	7	8	9	10
D	B	B	B	C	D	B	B	B	C
11	12	13	14	15	16	17	18	19	20
C	D	A	B	C	D	C	C	C	C
21	22	23	24	25	26	27	28	29	30
B	A	D	B	C	D	C	B	C	D
31	32	33	34	35	36	37	38	39	40
B	C	D	A	C	A	D	D	C	D
41	42	43	44	45	46	47	48	49	50
A	B	D	A	A	A	A	B	A	C
51	52	53	54	55	56	57	58	59	60
C	D	D	C	B	A	A	D	A	B
61	62	63	64	65	66	67	68	69	70
A	C	C	C	D	D	C	A	A	B
71	72	73	74	75	76	77	78	79	80
C	B	B	C	C	D	B	A	A	C
81	82	83	84	85	86	87	88	89	90
A	C	C	B	A	A	D	D	B	D
91	92	93	94	95	96	97	98	99	100
C	A	C	D	C	B	A	C	D	A

SUBJECT 3: PLANT BREEDING & SEED TECHNOLOGY

Code: DA 111 | Total Questions: 100

Topic 1: Introduction, Modes of Reproduction & Pollination

1. The process of bringing a wild species under human management is referred to as
   a) Introduction
   b) Collection
   c) Domestication
   d) Breeding

2. The oldest method of crop improvement is
   a) Domestication
   b) Introduction
   c) Selection
   d) *Pedigree method*

3. Double helix structure of the DNA molecule was proposed by
   a) Frankel-conrat
   b) Mendel
   c) Hugo de Vries
   d) *Watson and* Crick

4. Mutations in plants were discovered by
   a) Hugo de Vries
   b) Theophrastus
   c) Darwin
   d) Boveri

5. Father of green revolution in India is
   a) Norman Borlaug
   b) M S Swaminathan
   c) Darwin
   d) Watson

6. Norin 10 dwarfing gene was involved in improvement of
   a) Rice
   b) Maize
   c) Wheat
   d) Cotton

7. First hybrid variety of cotton developed is
   a) H5
   b) H6
   c) H7
   d) H4

8. *Malting quality* is an important objective of ________ crop improvement
   a) Wheat
   b) Barley
   c) Rice
   d) Maize

9. *Perfect flower* consists of
   a) *All four* whorls
   b) Only androecium and gynoecium
   c) Only gynoecium
   d) Only androecium

10. Which of the following is NOT an endospermic seed?
   a) Wheat
   b) Rice
   c) Castor
   d) Pea

11. The stalk of ovule is called as
   a) Pedicel
   b) Petiole
   c) Funicle
   d) Peduncle

14. A group of plants produced from a single plant through asexual reproduction is
   a) Hybrid
   b) Variety
   c) Clone
   d) *Tissue culture*

15. Father of plant tissue culture is
   a) Dokuchaev
   b) Mendel
   c) Haberlandt
   d) Williamson

16. AICRP on Rice was started at Hyderabad in
   a) 1968
   b) 1963
   c) 1962
   d) 1965

17. High Yielding Variety *Programme was* started in the year
   a) 1962
   b) 1963
   c) 1965
   d) 1966

18. Imperial Council of Agricultural *Research established* at Delhi in the year
   a) 1928
   b) 1929
   c) 1927
   d) 1930

19. Headquarters of Southern Telangana Zone is located at
   a) Jagtial
   b) Hyderabad
   c) Palem
   d) Warangal

20. Which of the following is NOT a method of crop improvement?
   a) Selection
   b) Hybridization
   c) Mutation
   d) Salinization

21. *Vegetative propagation* is seen in
   a) Banana
   b) Jasmine
   c) Rose
   d) *All the* above

22. Suckers of ________ are used in vegetative propagation
   a) Banana
   b) Onion
   c) Sugarcane
   d) Potato

23. *Gladiolus and* lily are examples of
   a) Corm
   b) *Tunicate bulbs*
   c) Non-tunicate bulbs
   d) Rhizome

24. A specialized stem structure in which the main axis grows horizontally just below the surface of the ground is
   a) Corm
   b) Stolon

12. The residual part of nucellus in the matured seed is called as
a) Endosperm
b) Micropyle
c) Perisperm
d) Chalaza

13. Example of asexually reproducing species include
a) Sugarcane
b) Potato
c) *Sweet potato*
d) All of the above

c) Rhizome
d) Bulb

25. For a variety to be considered as superior, the yield should be higher by ________ than the existing variety
a) 30%
b) 50%
c) 20-25%
d) 40%

Topic 2: Pollination, Fertilization & Breeding Systems

26. Protogyny is seen in
a) Maize
b) Sugarbeet
c) Sunflower
d) Bajra

27. The inability of fertile pollens to fertilize the same flower is referred to as
a) Male sterility
b) *Self incompatibility*
c) Cross pollination
d) Xenogamy

28. Which of the following will NOT promote cross pollination?
a) Dicliny
b) Male sterility
c) Herkogamy
d) Chasmogamy

29. *Tristyly condition* is seen in
a) Primula
b) Lathyrus
c) Linum
d) Both *Lathyrus and* Linum

30. *Promoting cross* pollination by snails is known as
a) Chiropterophily
b) Ornithophily
c) Anemophily
d) Malacophily

31. The origin of embryo from either synergids or antipodal cells of embryo sac is
a) Parthenogenesis
b) *Adventive embryony*
c) Apogamy
d) Apospory

32. *Condition where* flowers do not open at all and ensures complete self-pollination is
a) Chasmogamy
b) Cleistogamy
c) Homogamy
d) Heterogamy

33. *Condition when* pollen from a flower of one plant falls on the stigmas of other flowers of the same plant is
a) Autogamy
b) Xenogamy
c) Geitonogamy
d) *Self pollination*

34. *Bats are* the pollinating agents in ________ plant
a) *Cestrum nocturnum*
b) *Kigelia pinnata*
c) Vallisnaria

39. Which gametocide causes male sterility in cotton?
a) 57% ethyl alcohol
b) Ethrel
c) FW 450
d) Mentek

40. Which of the following causes male sterility?
a) *Cold temperature*
b) *Hot temperature*
c) Use of gametocides
d) Both cold and gametocides

41. *Anemophily refers* to pollination by
a) Bees
b) Water
c) Wind
d) Insects

42. *Entomophily refers* to pollination by
a) Wind
b) Water
c) Birds
d) Insects

43. *Hydrophily refers* to pollination by
a) Wind
b) Water
c) Birds
d) Insects

44. *Ornithophily refers* to pollination by
a) Wind
b) Water
c) Birds
d) Insects

45. *Chiropterophily refers* to pollination by
a) Snails
b) Bats
c) Wind
d) Water

46. The transfer of one or more characters into a single variety from other varieties is
a) *Bulk method*
b) *Mass selection*
c) Pure line selection
d) *Combination breeding*

47. Chairman of CVRC will be
a) Director of Agriculture
b) Ministry of Agriculture
c) Deputy Director General (Crop Science)
d) Director of State Seeds Development Corporation

48. Self-incompatibility is exploited in hybrid seed production of which crop?
a) Tomato

d) Bignonia

35. Following plant stigmas bend in opposite direction to that of anthers to promote cross pollination
   a) Helianthus
   b) Gossypium
   c) Hibiscus
   d) *Gloriosa superba*

36. Which of the following is used in hybrid seed production?
   a) Homogamy
   b) Chasmogamy
   c) Male sterility
   d) Cleistogamy

37. The process of removal of male parts (stamens) from a bisexual flower selected as female parent is called
   a) Male sterility
   b) Emasculation
   c) Heterosis
   d) Hybrid vigour

38. Male sterile source for bajra crop is
   a) *Combined kafir*
   b) Dee Gee Woo Gen
   c) Tift 23A
   d) *S. spontaneum*

b) Cabbage
c) Maize
d) Bajra

49. Protandry (pollen shed before stigma is receptive) is seen in
   a) Wheat
   b) Maize
   c) Sunflower
   d) Bajra

50. *Herkogamy refers* to
   a) Separation of sexes to different flowers
   b) Maturity of pollen before stigma
   c) Physical barrier between pollen and stigma
   d) *Flower not* opening

Topic 3: Selection Methods & Hybridization

51. A progeny of a single homozygous plant of a self-pollinated species is
   a) *Mass selected* variety
   b) Hybrid
   c) Pure line
   d) Seedling

52. The concept of pure line was proposed by
   a) Shull
   b) Darwin
   c) Johansen
   d) Mendel

53. In crop improvement programmes, ________ hybridization is the most commonly used method
   a) *Intervarietal hybridization*
   b) *Intravarietal hybridization*
   c) *Interspecific hybridization*
   d) Both intervarietal and interspecific

54. *Aruna variety* of castor was developed through ________ method
   a) Pure line selection
   b) *Back cross* breeding
   c) Hybridization
   d) *Mutation breeding*

55. *Purity percentage* of Foundation Seed is
   a) 99.5%
   b) 99.0%
   c) 98.0%
   d) 97.0%

56. The part of graft combination which becomes the upper portion is termed as
   a) Root stock
   b) Under stock
   c) Scion
   d) Bud

57. *Air layering* is NOT practiced in
   a) Pomegranate

64. Which of the following is NOT a method for breeding vegetatively propagated crops?
   a) *Clonal selection*
   b) *Hybridization followed* by selection
   c) *Mutation breeding*
   d) *Pedigree selection*

65. The term 'Heterosis' was coined by
   a) Joseph Koelreuter
   b) C T Patel
   c) Shull
   d) Yuan Longping

66. Hybrid vigour is also known as
   a) *Inbreeding depression*
   b) Heterosis
   c) Mutation
   d) Polyploidy

67. *Inbreeding depression* is the opposite of
   a) Mutation
   b) Heterosis
   c) Polyploidy
   d) Selection

68. Hybrid seed is produced by crossing
   a) *Two varieties*
   b) *Two inbred* lines
   c) *Two pure* lines
   d) All of the above

69. Which crop was first successfully commercialized as a hybrid?
   a) Wheat
   b) Rice
   c) Maize
   d) Cotton

70. In India, commercial hybrid rice production uses which system?
   a) CMS system
   b) Self-incompatibility

b) Guava
c) Rose
d) Lemon

58. *Mass selection* is based on
a) *Phenotypic selection* of plants
b) *Genotypic selection*
c) DNA markers
d) *Tissue culture*

59. Pure line selection is applicable to
a) Cross-pollinated crops
b) Self-pollinated crops
c) *Vegetatively propagated* crops
d) *All crops*

60. *Pedigree method* of selection is used in ________ crop improvement
a) Self-pollinated crops after hybridization
b) Cross-pollinated crops
c) *Asexually reproduced* crops
d) All of the above

61. *Bulk method* of breeding differs from pedigree method in that
a) *Selection starts* early in bulk method
b) No selection is made in early generations in bulk method
c) Hybridization is not done in bulk method
d) *Individual plant* selection is done in F2 in bulk method

62. *Back cross* method is used for
a) *Developing hybrids*
b) *Transferring single* gene into an adapted variety
c) *Selecting polyploids*
d) *Mass selection*

63. *Recurrent selection* is used in improvement of ________ crops
a) Self-pollinated crops
b) Cross-pollinated crops
c) *Vegetatively propagated* crops
d) *All crops*

c) *Gametocide method*
d) *Manual emasculation*

71. Three-line system in hybrid seed production uses
a) A line, B line, C line
b) A line, B line, R line
c) CMS line, maintainer line, restorer line
d) All of the above

72. Which crop uses two-line system for hybrid seed production in India?
a) Maize
b) Bajra
c) Rice
d) Cotton

73. The male sterile line in three-line breeding is called
a) A line
b) B line
c) R line
d) C line

74. The restorer line in three-line breeding is called
a) A line
b) B line
c) R line
d) C line

75. Which organization releases varieties and hybrids in India?
a) ICAR
b) CVRC
c) State Agriculture Departments
d) All of the above

Topic 4: Mutation, Polyploidy & Seed Technology

76. The word 'Mutation' is derived from the *Latin word* 'MUTARE' which means
a) To disturb
b) To change
c) To collect
d) To split

77. Rate of occurrence of spontaneous mutations is
a) 10-7
b) 10-5
c) 10-8
d) 10-6

78. X-ray mutations were induced first time in barley by
a) Muller
b) Hugo de Vries
c) Stadler
d) Mendel

79. Which is commonly used for chromosome doubling?
a) *Malic hydrazide*
b) Formaldehyde
c) Colchicine
d) Ethane Methane Sulphonate

80. Ploidy of *Triticum aestivum* (bread wheat) is
a) Octaploid

89. Which of the following is the hierarchy of seed classes?
a) Certified → Foundation → Registered → Breeder
b) Breeder → Foundation → Certified → Truthfully Labelled
c) Breeder → Registered → Foundation → Certified
d) Foundation → Breeder → Certified → Registered

90. Failure of a viable seed to germinate even when given favourable environmental conditions is called as
a) Seed vigour
b) Seed viability
c) Seed dormancy
d) Heterosis

91. Seed viability refers to
a) Ability of seed to germinate
b) Percentage of pure seeds
c) *Freedom from* seed-borne diseases
d) *Storage life* of seeds

92. Seed vigour refers to
a) Ability to germinate under stress conditions
b) *Genetic purity*
c) Physical purity
d) *Moisture content*

93. Which of the following chemicals breaks seed dormancy?
a) *Gibberellic acid* (GA3)

b) Tetraploid
c) Hexaploid
d) Triploid

81. The basic chromosome number of an individual is known as
a) Haploid
b) Monoploid
c) Diploid
d) Tetraploid

82. *Polyploidy refers* to organisms having
a) More than two copies of chromosome sets
b) Only one chromosome set
c) *Exactly two* chromosome sets
d) *Structural chromosomal* abnormality

83. Which of the following is a mutagen?
a) Colchicine
b) EMS (*Ethyl methane* sulphonate)
c) *Gamma rays*
d) All of the above

84. *Mutation breeding* has been most successful in which type of crops?
a) Cross-pollinated crops
b) *Vegetatively propagated* crops and self-pollinated crops
c) Only self-pollinated crops
d) *All crops* equally

85. The largest programme of induced mutations for crop improvement is in which country?
a) USA
b) India
c) Japan
d) China

86. *Purity percentage* of Certified Seed is
a) 99.5%
b) 99.0%
c) 98.0%
d) 97.0%

87. *Purity percentage* of Breeder Seed is
a) 99.5%
b) 99.0%
c) 100%
d) 97.0%

88. Seed germination percentage required for certified seed of paddy is
a) 70%
b) 75%
c) 80%
d) 85%

b) *Abscisic acid*
c) Cytokinin
d) Ethylene

94. Seed certification is done by
a) ICAR
b) State Seed Certification Agencies
c) CVRC
d) *Private seed* companies

95. *Tag colour* for Breeder Seed is
a) White
b) *Azure blue*
c) *Golden yellow*
d) Green

96. *Tag colour* for Foundation Seed is
a) White
b) *Azure blue*
c) *Golden yellow*
d) Green

97. *Tag colour* for Certified Seed is
a) White
b) *Azure blue*
c) *Golden yellow*
d) Blue

98. *Nucleus seed* is produced by
a) ICAR/SAU Plant Breeders
b) State Seed Certification Agency
c) State Seed Corporations
d) Farmers

99. The science of improving hereditary qualities of a population is called
a) Eugenics
b) Plant breeding
c) Genetics
d) Biotechnology

100. Which of the following is the most important objective of plant breeding?
a) *Increased yield*
b) *Disease resistance*
c) *Better quality*
d) All of the above

ANSWER KEY — PLANT BREEDING & SEED TECHNOLOGY

1	2	3	4	5	6	7	8	9	10
C	C	D	A	B	C	D	B	A	D
11	12	13	14	15	16	17	18	19	20
C	C	D	C	C	D	D	B	C	D
21	22	23	24	25	26	27	28	29	30
D	A	C	C	C	C	B	D	D	D
31	32	33	34	35	36	37	38	39	40
C	B	C	B	C	C	B	C	C	C
41	42	43	44	45	46	47	48	49	50
C	D	B	C	B	D	C	B	B	C
51	52	53	54	55	56	57	58	59	60
C	C	D	D	A	C	C	A	B	A
61	62	63	64	65	66	67	68	69	70
B	B	B	D	C	B	B	D	C	A
71	72	73	74	75	76	77	78	79	80
D	C	A	C	B	B	D	C	C	C
81	82	83	84	85	86	87	88	89	90

B	A	D	B	C	B	C	D	C	C
91	92	93	94	95	96	97	98	99	100
A	A	A	B	C	B	D	A	B	D

SUBJECT 4: SOIL CHEMISTRY AND FERTILITY

Code: DA 121 | Total Questions: 150

Topic 1: Introduction to Soil Science & Soil Composition

1. *Three dimensional* body having length, breadth and depth is
   a) Colloid
   b) Soil
   c) *Bulk density*
   d) *Pore space*

2. The latin word 'solum' means
   a) Air
   b) Floor or ground
   c) Water
   d) Mineral

3. The person who is known as Father of Soil Science is
   a) Emil Traug
   b) Thomas Way
   c) Dokuchaev
   d) Jenny

4. The percentage of mineral matter in a representative loam soil is
   a) 25%
   b) 5%
   c) 80%
   d) 45%

5. The percentage of organic matter in a representative loam soil is
   a) 45%
   b) 25%
   c) 5%
   d) 80%

6. The percentage of water in a representative loam soil is
   a) 45%
   b) 25%
   c) 5%
   d) 80%

7. The percentage of air present in a representative loam soil is
   a) 45%
   b) 25%
   c) 80%
   d) 5%

8. Soil particle which can be visible only through an electron microscope is
   a) Sand
   b) Clay
   c) Silt
   d) *Coarse sand*

9. Size of the soil clay particle is
   a) <0.002 mm
   b) 2 mm
   c) <0.002 cm
   d) <0.02 cm

10. Size of silt particle ranges from
   a) 0.002-0.02 mm
   b) 0.02-2.0 mm
   c) <0.002 mm
   d) >2.0 mm

11. Size of fine sand particles ranges from
   a) 0.002-0.02 mm
   b) 0.1-0.25 mm
   c) 0.25-0.5 mm

14. *Inert sand* and silt fractions are referred as
   a) Flesh of the soil
   b) Skeleton of the soil
   c) Head of the soil
   d) Leg of the soil

15. *Nitrogen percentage* present in atmospheric air is
   a) 0.033
   b) 0.50
   c) 20.97
   d) 79.00

16. *Oxygen percentage* present in atmospheric air is
   a) 79.00
   b) 0.033
   c) 20.97
   d) 0.50

17. *Carbon dioxide* content in soil air is
   a) 0.033%
   b) 20.97%
   c) 0.50%
   d) 79.00%

18. Which method gives accurate results for estimation of soil texture?
   a) *Bouyoucos hydrometer* method
   b) *Robinson pipette* method
   c) *Particle density* method
   d) *Feel method*

19. By using which method can soil texture be estimated in field conditions?
   a) *Bouyoucos hydrometer* method
   b) *Robinson pipette* method
   c) *Particle density* method
   d) *Feel method*

20. Robinson's pipette method works on the principle of
   a) Continuous decrease in density of soil suspension
   b) Sedimentation
   c) Formation of gritters
   d) Deflocculation

21. Bouyoucos Hydrometer is used for estimation of the
   a) Soil texture
   b) Soil structure
   c) Soil colour
   d) Soil temperature

22. *Feel method* of soil texture estimation is also called as
   a) *Lab method*
   b) *Field method*
   c) *Pipette method*
   d) *Hydrometer method*

23. The most reactive soil particle is
   a) Sand
   b) Clay
   c) Silt
   d) *Coarse sand*

24. Permanent property of soil is
   a) Soil structure
   b) Soil plasticity
   c) Soil texture
   d) Soil colour

d) 0.5-1.0 mm

12. Range of size of gravel or pebble is
   a) 2.0-75.0 mm
   b) >250.0 mm
   c) 75.0-250.0 mm
   d) >800 mm

13. *Rock fragments* of size >250 mm are called
   a) Gravel
   b) Cobbles
   c) Stones or Boulders
   d) Pebbles

25. Which of the following soil type has the lowest total porosity?
   a) *Clay soil*
   b) Deep clay soil
   c) *Sandy soil*
   d) *Loamy soil*

Topic 2: Soil Physical Properties — Structure, Bulk Density & Porosity

26. Arrangement of sand, silt and clay under field conditions is called
   a) Soil texture
   b) Soil structure
   c) Soil colour
   d) Soil temperature

27. Natural aggregates present in soil are called
   a) Fragments
   b) Clods
   c) Peds
   d) Concretions

28. The broken peds are known as
   a) Concretions
   b) Clods
   c) Fragments
   d) Aggregates

29. In soil structural classification, the term 'Type' refers to the
   a) Size of the peds
   b) *Shape and* arrangement of aggregates
   c) *Distinctness and* durability
   d) Soil texture

30. In soil structural classification, 'Class' refers to
   a) Shape of aggregates
   b) Size of peds
   c) Durability of peds
   d) Soil texture

31. In soil structural classification, 'Grade' refers to
   a) Shape of aggregates
   b) Size of peds
   c) *Distinctness and* durability of peds
   d) Soil texture

32. *Predominant soil* structure found in arid and semi-arid regions of Rajasthan is
   a) *Block like*
   b) *Prism like*
   c) *Plate like*
   d) Spheroidal

33. Which of the following soil structure is most porous in nature?
   a) *Plate like*
   b) *Prism like*
   c) *Block like*
   d) Spheroidal

34. A coherent mass formed within the soil by precipitation of chemicals in percolating water is called
   a) Concretion
   b) Fragment
   c) Ped

39. Addition of organic matter to the soil leads to
   a) Increase in bulk density
   b) Decrease in bulk density
   c) No change in bulk density
   d) Decrease in porosity

40. *Particle density* of soils is always ________ than the bulk density
   a) Lower
   b) Equal
   c) Higher
   d) Variable

41. *Porosity percentage* seen in deep clay soils is
   a) 50
   b) 45
   c) 30
   d) 66

42. *Porosity percentage* of sandy soil is
   a) 66
   b) 45
   c) 30
   d) 50

43. *Porosity percentage* of deep black soil is
   a) 30
   b) 66
   c) 45
   d) 40

44. *Pores existing* between sand-sized granules that allow air and water movement are
   a) Micropores
   b) Macropores
   c) *Capillary pores*
   d) *Sandy pores*

45. At field capacity only ________ pores are filled with water
   a) Macropores
   b) Micropores
   c) Capillary
   d) Both macro and micro

46. *Macropores are* defined as pores with diameter greater than
   a) 0.002 mm
   b) 0.02 mm
   c) 0.06 mm
   d) 0.2 mm

47. Which of these soils has the lowest bulk density?
   a) *Sandy soils*
   b) *Loamy soil*
   c) *Clayey soils*
   d) *Sandy loams*

48. *Soils containing* more than 15% of colloidal clay exhibit

d) Clod

35. *Bulk density* of sand dominated soils is
   a) 0.5 Mgm-3
   b) 2.0 Mgm-3
   c) 1.0 Mgm-3
   d) 1.7 Mgm-3

36. *Bulk density* of organic peat soils is
   a) 2.0 Mgm-3
   b) 1.7 Mgm-3
   c) 0.5 Mgm-3
   d) 1.0 Mgm-3

37. *Bulk density* of compacted sub soils is
   a) 1.7 Mgm-3
   b) 0.5 Mgm-3
   c) 2.0 Mgm-3
   d) 1.0 Mgm-3

38. By long term intense tillage practice, bulk density of soil will
   a) Decrease
   b) Increase
   c) *Remain same*
   d) Decrease 2 times

a) Elasticity
b) Plasticity
c) Adhesion
d) Deflocculation

49. *Shrinkage and* swelling are more pronounced in
   a) *Montmorillonite clays*
   b) *Kaolinite clays*
   c) *Illite clays*
   d) *Laterite clays*

50. Total porosity is more in
   a) *Sandy soil*
   b) *Clayey soil*
   c) *Loamy soil*
   d) *Sandy loam*

Topic 3: Soil Water — Types, Constants & Movement

51. Soil moisture potential at field capacity is
   a) -31 bars
   b) -1/3 bars
   c) -1000 bars
   d) -10000 bars

52. Soil water held at a moisture potential of -31 bars is known as
   a) *Available water*
   b) Permanent wilting point
   c) *Hygroscopic coefficient*
   d) Dried soil

53. Permanent wilting point is expressed as
   a) -1/3 bars
   b) -15 bars
   c) -31 bars
   d) -1000 bars

54. The equilibrium tension of moisture in oven-dried soil is
   a) -1/3 bars
   b) -1000 bars
   c) -31 bars
   d) -10000 bars

55. *Water retained* between field capacity and permanent wilting point is called
   a) *Hygroscopic water*
   b) *Gravitational water*
   c) *Available water*
   d) *Capillary water*

56. The downward movement of water through soil is called
   a) Percolation
   b) Infiltration
   c) Permeability
   d) Runoff

57. *Downward movement* of water below the root zone depth is known as
   a) Permeability
   b) Percolation
   c) Seepage

64. *Wilting coefficient* is expressed as
   a) -1/3 bars
   b) -15 bars
   c) -31 bars
   d) -1000 bars

65. *Air dry* soil corresponds to moisture potential of
   a) -1/3 bars
   b) -15 bars
   c) -31 bars
   d) -1000 bars

66. Which type of soil water is not available to plants?
   a) *Capillary water*
   b) *Available water*
   c) *Gravitational water*
   d) *Hygroscopic water*

67. Broad *Bed and* Furrow (BBF) method is used in management of ________ soils
   a) Red soils
   b) *Alluvial soils*
   c) Black soils
   d) *Laterite soils*

68. *Reduced iron* imparts ________ colour to the soil
   a) Red
   b) White
   c) Blue green
   d) Grey

69. Soil colour in humid temperate regions is
   a) Red
   b) Yellow
   c) Grey
   d) Black

70. Red colour of soils is due to
   a) *Iron oxides*
   b) Quartz
   c) *Reduced iron*
   d) Humus

71. The presence of humus imparts ________ colour to the soil

d) Runoff

58. The ease with which water passes through a bulk mass of soil is called
   a) Infiltration
   b) Percolation
   c) Permeability
   d) Seepage

59. Loss of water from soil to atmosphere in the form of water vapour is
   a) Evaporation
   b) *Water holding* capacity
   c) Transpiration
   d) Mineralization

60. In deep black soil, porosity percentage is
   a) 40
   b) 66
   c) 50
   d) 45

61. Which soil type has the highest water holding capacity?
   a) *Sandy soil*
   b) *Sandy loam*
   c) *Clay soil*
   d) *Loamy soil*

62. *Doruvu technology* is practiced in ________ type of soils
   a) Red soils
   b) *Coastal sands*
   c) Black soils
   d) *Laterite soils*

63. *Saturation condition* of soil corresponds to soil moisture potential of
   a) -1/3 bars
   b) -15 bars
   c) 0 bars
   d) -1000 bars

   a) Red
   b) Dark brown
   c) Yellow
   d) White

72. Two or three colours occurring in patches in soil is known as
   a) Grade
   b) Mottling
   c) Type
   d) Soil colour

73. *Capillary pores* in soil are pores with diameter
   a) >0.06 mm
   b) 0.06-0.2 mm
   c) <0.06 mm
   d) 0.2-2.0 mm

74. Which instrument is used to measure soil moisture tension?
   a) Tensiometer
   b) Barometer
   c) Hygrometer
   d) Anemometer

75. The *Laterite type* of soil is present in Zaheerabad of Telangana
   a) Red soil
   b) Black soil
   c) *Alluvial soil*
   d) *Laterite soil*

Topic 4: Soil Colloids, CEC & Ion Exchange

76. Size of colloidal particle is
   a) <0.001 cm
   b) <0.002 cm
   c) <0.001 mm
   d) <0.002 mm

77. The continuous rapid zigzag movement of a colloidal particle in dispersion medium is called
   a) *Ion exchange*
   b) *Brownian movement*
   c) Flocculation
   d) Deflocculation

78. Aggregation or clumping together of individual tiny soil particles is called
   a) Deflocculation
   b) Flocculation
   c) Mineralization
   d) Immobilization

79. Which of the following is NOT a property of soil colloids?
   a) Plasticity
   b) *Swelling and* shrinkage
   c) *Brownian movement*
   d) Size of >0.001 mm

80. The concept of Cation Exchange Capacity of soils was given by
   a) Dokuchaev

89. AEC (me/100g of soil) of *Kaolinite clay* minerals is
   a) >5
   b) >10
   c) >20
   d) >43

90. Anionic Exchange Capacity (AEC) is greater in
   a) Montmorillonite
   b) Illite
   c) Kaolinite
   d) *All the* above

91. *Plasticity order* of clay minerals is
   a) Illite>Montmorillonite>Kaolinite
   b) Kaolinite>Montmorillonite>Illite
   c) Montmorillonite>Illite>Kaolinite
   d) Illite>Kaolinite>Montmorillonite

92. Which of the following statement is NOT correct regarding soil colloids?
   a) *All soil* colloids are clay soils
   b) Some clay soils may be soil colloids
   c) *All clay* soils are soil colloids
   d) *Colloidal soils* have less size than clay soils

93. Desorption is a phenomenon by which
   a) Adsorption of ions occurs
   b) Replacement or release of adsorbed ion species occurs
   c) *Flocculation occurs*

b) Thomas Way
c) Arnon
d) Mendel

81. The percentage of total CEC satisfied with basic cations is termed as
a) *Ion exchange*
b) *Buffering capacity*
c) *Base saturation*
d) CEC

82. *Base saturation* of highly fertile soils is
a) <50%
b) 50-80%
c) >80%
d) 30-50%

83. *Base saturation* of medium fertile soils is
a) 30-50%
b) 50-80%
c) 80-100%
d) <50%

84. *Correct order* of cations to flocculate soil colloids is
a) Na+>Ca2+>Al3+>Mg2+>K+>H+
b) K+>Ca2+>Al3+>Mg2+>Na+>H+
c) Al3+>Ca2+>H+>Mg2+>K+>Na+
d) Al3+>Ca2+>Na+>Mg2+>K+>H+

85. Presence of which cation causes deflocculation of soil particles?
a) Ca
b) Al
c) Na
d) Zn

86. Which element promotes flocculation of soil particles?
a) Magnesium
b) Potassium
c) Sodium
d) Calcium

87. In strong acid soils, which may act as dominant exchangeable ion?
a) Na+
b) Cl-
c) Al(OH)2+
d) Ca2+

88. CEC (me/100g of soil) of *Kaolinite clay* minerals is
a) 100-150
b) 10-33
c) 3-15
d) >43

d) *Brownian movement* stops

94. *Ion exchange* property may NOT be seen in
a) *Clay soils*
b) *Silty soils*
c) *Sandy soils*
d) *Soils with* high organic matter

95. The ability of soil to resist pH changes is called
a) *Base saturation*
b) *Buffering capacity*
c) Liming
d) *Ion exchange*

96. *Kaolinite crystals* are
a) *Rod shaped*
b) *Flake shaped*
c) Hexagonal
d) Tubular

97. *Montmorillonite crystals* are
a) Hexagonal
b) *Rod shaped*
c) Tubular
d) *Flake shaped*

98. Which clay mineral shows maximum shrink-swell property?
a) Kaolinite
b) Illite
c) Montmorillonite
d) Halloysite

99. The capacity of soil to resist pH change is called buffering capacity. This is highest in soils rich in
a) Sand
b) Silt
c) *Clay and* organic matter
d) Gravel

100. Soil pH is defined as
a) Positive logarithm of H+ concentration
b) Negative logarithm of H+ concentration
c) Ratio of H+ to OH-
d) Amount of lime in soil

Topic 5: Soil Organic Matter & C:N Ratio

101. The C:N ratio of humus is
a) 60-80:1
b) 150-500:1
c) 10-12:1
d) 13-25:1

102. C:N ratio of forest waste is
a) 13-25:1
b) 10-12:1
c) 150-500:1
d) 60-80:1

103. C:N ratio for cereal residues and straws is
a) 10-12:1
b) 13-25:1

114. Addition of humus to soils causes
a) Increase in bulk density
b) Increase in porosity
c) Decrease in bulk density
d) Both 2 and 3

115. *Example for* soil fauna is
a) Bacteria
b) Blue Green Algae
c) Actinomycetes
d) Snails

116. *Example for* soil flora is
a) Algae
b) Nematodes

c) 60-80:1
d) 150-500:1

104. C:N ratio for legume residues and green manures is
a) 10-12:1
b) 13-25:1
c) 60-80:1
d) 150-500:1

105. C:N ratio of sewage and sludge is
a) 5-14:1
b) 10-12:1
c) 60-80:1
d) 150-500:1

106. C:N ratio is very low in
a) *Cereal straw*
b) *Forest wastes*
c) *Legume residues*
d) *Sewage and* sludge

107. *Mineralization takes* place at C:N ratio of
a) >30:1
b) >20:1
c) <20:1
d) <10:1

108. *Immobilization takes* place at C:N ratio of
a) <20:1
b) >20:1
c) >30:1
d) <30:1

109. The conversion of organic forms of C, N, P and S to inorganic forms as a result of microbial decomposition is called
a) Mineralization
b) Immobilization
c) Hydrolysis
d) Decomposition

110. The conversion of element from inorganic to organic form in microbial tissues is called
a) Mineralization
b) Immobilization
c) Denitrification
d) Nitrification

111. Which of the following is a rapidly decomposed organic residue?
a) Lignins
b) Fats
c) Waxes
d) Sugars

112. Weight of organisms per unit weight of soil is known as
a) Soil fauna
b) Biomass
c) Soil flora
d) Microbes

113. *Microbial reduction* of nitrate to molecular nitrogen is called
a) Nitrification
b) Denitrification
c) Mineralization
d) Immobilization

c) Protozoa
d) Earthworms

117. The process of enrichment of surface water bodies with nutrients is called
a) Eutrophication
b) Denitrification
c) Leaching
d) Salinization

118. *Largest contributing* source of greenhouse gas is
a) Paddy fields
b) Deforestation
c) Burning of fossil fuels
d) Livestock

119. *Gas released* from rice fields that causes greenhouse effect is
a) CO2
b) N2O
c) Methane
d) SO2

120. *Gases responsible* for acid rain are
a) CO2 and CH4
b) NO2 and SO2
c) NH3 and CO2
d) CFC and O3

121. The natural radioactive material among the following is
a) Strontium-90
b) Cesium-137
c) Nitrogen-28
d) Potassium-40

122. Consumption of water with high nitrate levels causes ________ in infants
a) Fluorosis
b) Blue baby syndrome
c) Hypothyroidism
d) Arrhythmias

123. High concentration of ________ in groundwater causes Blue *Baby syndrome*
a) Fluoride
b) Phosphorous
c) Nitrate
d) Cadmium

124. WHO standard for drinking water nitrate-N level is
a) 23 mg/L
b) 32 mg/L
c) 10 mg/L
d) 22 mg/L

125. C:N ratio of microbial tissues is
a) 5-14:1
b) 6-12:1
c) 10-12:1
d) 13-25:1

Topic 6: Soil pH, Problem Soils & Classification

126. Presence of excess soluble salts of Ca, Mg and Na is characteristic of
a) *Alkali soils*

139. Hypomagnesemia in cattle is due to consumption of forage
a) Low in calcium
b) High in calcium

b) *Saline soils*
c) *Acid soils*
d) *Laterite soils*

127. Soil amendment used for reclamation of alkaline soils is
a) Lime
b) Gypsum
c) Dolomite
d) *Basic slag*

128. Removal of calcium carbonate present in soil can be achieved by treating with
a) Gypsum
b) NaOH
c) Diluted HCl
d) H2SO4

129. When soils have >15% ESP and pH >8.5, these soils are called
a) *Alkali soils*
b) *Saline soils*
c) *Acid soils*
d) *Solon chalks*

130. The electrical conductivity of saline-alkali soils is
a) >4 dSm-1
b) <4 dSm-1
c) =4 dSm-1
d) 0 dSm-1

131. *Critical value* for EC (dSm-1) is
a) 0-2
b) 2
c) 4
d) 6

132. Availability of phosphorus is high when soil pH is between
a) 4.0-5.5
b) 5.5-6.0
c) 6.0-7.5
d) 7.5-9.0

133. The availability of molybdenum increases under ________ condition
a) Acidic
b) Alkaline
c) Neutral
d) Waterlogged

134. Liming is applied to neutralize the effect of
a) *Alkaline soils*
b) *Saline soils*
c) *Acid soils*
d) *Laterite soils*

135. Which of the following is NOT a liming material?
a) *Calcium sulphate*
b) *Calcium carbonate*
c) *Calcium hydroxide*
d) *Calcium oxide*

136. Which is the source for both calcium and magnesium?
a) *Magnesium sulphate*
b) *Basic slag*
c) Dolomite
d) Lime

137. Quantity of lime required for strongly acidic soils is
a) 1-1.5 tonnes/acre
b) 2-4 tonnes/acre
c) 3-6 tonnes/acre
d) 4-6 tonnes/acre

138. *Phosphatic fertilizer* suitable for highly acidic soils is
a) SSP
b) DAP
c) Low in magnesium
d) High in magnesium

140. Which of the following is NOT a problematic soil?
a) *Acid soils*
b) *Waterlogged soils*
c) *Basic soils*
d) *Neutral soils*

141. Main reason for identifying saline soils as problematic is
a) High nitrogen
b) Hardness of soil
c) High salt content
d) High zinc content

142. The inherent capacity of soil to provide nutrients in adequate amounts is called
a) Soil fertility
b) Soil plasticity
c) Soil nutrition
d) Soil density

143. *Doruvu technology* can be followed in ________ soils
a) *Coastal sands*
b) *Laterite soils*
c) Red soils
d) Black soils

144. Deep black cotton soils with depth >120 cm occur in
a) *Karimnagar and* Adilabad
b) *Nizamabad and* Nalgonda
c) *Mahbubnagar and* Khammam
d) *Warangal and* Khammam

145. Which of the following Telangana soil type is suitable for BBF technology?
a) Red soils
b) *Laterite soils*
c) Black cotton soils
d) *Alluvial soils*

146. *Sustainable agriculture* is the practice of farming using principles of
a) *Intensive cultivation*
b) *Shifting cultivation*
c) Ecology
d) Economics

147. Practice of farming a small area with limited resources to produce just enough food for family needs is
a) *Sustainable agriculture*
b) *Subsistence farming*
c) *Intensive farming*
d) *Extensive farming*

148. *Shifting cultivation* is also called
a) *Jhum cultivation*
b) *Intensive farming*
c) *Terrace farming*
d) *Organic farming*

149. *Nitrification bacteria* are active when the soil pH is between
a) 5.0-5.5
b) <6.0
c) 6.5-7.5
d) <8.5

150. Which of these is an example of soil formation factor?
a) *Parent material*
b) *Organic matter*
c) Microorganisms
d) Fertilizers

c) *Rock phosphate*
d) TSP

ANSWER KEY — SOIL CHEMISTRY AND FERTILITY

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	**B**	**C**	**D**	**C**	**B**	**B**	**B**	**A**	**A**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	**A**	**C**	**B**	**D**	**C**	**C**	**B**	**D**	**B**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	**B**	**B**	**C**	**C**	**B**	**C**	**C**	**B**	**B**
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
C	**B**	**D**	**A**	**D**	**C**	**C**	**B**	**B**	**C**
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
D	**C**	**C**	**B**	**B**	**C**	**C**	**B**	**A**	**B**
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
B	**C**	**B**	**D**	**C**	**B**	**B**	**C**	**A**	**D**
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
C	**B**	**C**	**B**	**D**	**D**	**C**	**C**	**C**	**A**
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
B	**B**	**C**	**A**	**D**	**C**	**B**	**B**	**D**	**B**
81	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**
C	**C**	**B**	**C**	**C**	**D**	**C**	**C**	**D**	**C**
91	**92**	**93**	**94**	**95**	**96**	**97**	**98**	**99**	**100**
C	**C**	**B**	**C**	**B**	**C**	**D**	**C**	**C**	**B**
101	**102**	**103**	**104**	**105**	**106**	**107**	**108**	**109**	**110**
C	**C**	**C**	**B**	**A**	**D**	**C**	**C**	**A**	**B**
111	**112**	**113**	**114**	**115**	**116**	**117**	**118**	**119**	**120**
D	**B**	**B**	**D**	**D**	**A**	**A**	**C**	**C**	**B**
121	**122**	**123**	**124**	**125**	**126**	**127**	**128**	**129**	**130**
D	**B**	**C**	**C**	**B**	**B**	**B**	**C**	**A**	**A**
131	**132**	**133**	**134**	**135**	**136**	**137**	**138**	**139**	**140**
C	**C**	**B**	**C**	**A**	**C**	**C**	**C**	**C**	**D**
141	**142**	**143**	**144**	**145**	**146**	**147**	**148**	**149**	**150**
C	**A**	**A**	**C**	**C**	**C**	**B**	**A**	**C**	**A**

SUBJECT 5: MANURES AND FERTILIZERS

Code: DA 122 | Total Questions: 100

Topic 1: Essential Plant Nutrients

1. Number of elements known to be essential for plant growth is
   a) 20
   b) 16
   c) 25
   d) 18

2. Criteria of essentiality of nutrients was given by
   a) *Dokuchaev and* Jenny
   b) *Arnon and* Stout
   c) *Mendel and* Watson
   d) Thomas *Way and* Emil

3. C, H and O constitute ________ per cent of plant dry matter weight
   a) 80-90
   b) 90-95
   c) 70-80
   d) 60-70

4. Which of the following is NOT a secondary nutrient of plant?
   a) Sulphur
   b) Calcium
   c) Phosphorous
   d) Magnesium

5. Which of the following is an anionic micronutrient?
   a) Zn
   b) Cu
   c) Mn
   d) Mo

6. Which nutrient of plant is absorbed in both anionic and cationic forms?
   a) P
   b) N
   c) Cu
   d) Mo

7. Following nutrients are immobile within plants
   a) N & P
   b) S & Fe
   c) Zn & Mg
   d) Ca & B

8. *Highly mobile* nutrients in plants are
   a) Ca and B
   b) Zn
   c) N, P and K
   d) Fe and Mn

9. *Supply and* absorption of chemical compounds required for plant growth and metabolism is known as
   a) Plant breeding
   b) Plant nutrition
   c) Plant physiology
   d) Plant pathology

10. Which element plays a major role in protein improvement?
   a) Magnesium
   b) Sulphur
   c) Zinc
   d) Nitrogen

11. Which element plays an important role in root development and growth in plants?
   a) Nitrogen
   b) Calcium

14. Which nutrient is necessary for proper pollination and fruit setting?
   a) Boron
   b) Molybdenum
   c) Zinc
   d) Copper

15. Potassium is a unique element that plants can accumulate without exhibiting toxicity. This is called
   a) *Luxury consumption*
   b) *Ion exchange*
   c) *Base saturation*
   d) Buffering

16. MOP fertilizer is NOT used in ________ crops
   a) Tobacco
   b) *Oil palm*
   c) Grapes
   d) Both tobacco and grapes

17. Which of the following nutrient is immobile in soil?
   a) Ca
   b) B
   c) Zn
   d) K+

18. Which of the following nutrient deficiency symptoms are NOT seen in older leaves at initial stages?
   a) N
   b) P
   c) Ca
   d) K

19. *Element that* acts as 'electron carrier' in enzymes is
   a) Zn
   b) Mg
   c) Cu
   d) Fe

20. *Element involved* in charge balance of soil is
   a) Nitrogen
   b) Sulphur
   c) Potassium
   d) Phosphorous

21. Which element is essential in enzyme nitrate reductase in plants?
   a) Mo
   b) Mg
   c) Mn
   d) Cu

22. The role of manganese is closely associated with which nutrient?
   a) Copper
   b) Zinc
   c) Iron
   d) Molybdenum

23. Which of the following is NOT a heavy metal pollutant?
   a) Cadmium
   b) Lead
   c) Calcium
   d) Arsenic

24. Which of the following is a natural radioactive material?
   a) Strontium-90

c) Potassium
d) Phosphorous

12. Which element promotes disease resistance in plants?
a) Phosphorous
b) Nitrogen
c) Potassium
d) Calcium

13. The nutrient responsible for drought tolerance in plants is
a) K
b) N
c) P
d) Ca

b) Cesium-137
c) Nitrogen-28
d) Potassium-40

25. *Liming and* fertilization increases
a) CEC
b) pH
c) *Base saturation*
d) *All the* above

Topic 2: Nitrogen Cycle & Nitrogen Fertilizers

26. *Nitrogen losses* from soil due to volatilization occur in the form of
a) N2
b) NH3
c) NO3
d) NO2

27. Conversion of ammoniacal nitrogen to nitrite nitrogen is taken up by
a) Nitrosomonas
b) Nitrobacter
c) Bacillus
d) Xanthomonas

28. *Microbial reduction* of nitrate to molecular nitrogen is called
a) Nitrification
b) Denitrification
c) Mineralization
d) Immobilization

29. *Nitrification bacteria* are active at soil pH of
a) 5.0-5.5
b) <6.0
c) 6.5-7.5
d) <8.5

30. *Enzyme responsible* for conversion of ammonia to nitrite is
a) Urease
b) Aminase
c) *Nitrate reductase*
d) Nitrogenase

31. *Nitrogen percentage* in ammonium chloride fertilizer is
a) 21
b) 25-26
c) 30
d) 18

32. Which of the following is NOT a symptom of nitrogen toxicity?
a) *Excessive growth*
b) Increase in succulence
c) *Taller plants*
d) Early maturity

33. *Urea required* to correct nitrogen deficiency in dry situations (as foliar spray) is
a) 1%
b) 2%
c) 3%
d) 4%

34. Which of the following is NOT a symptom of nitrogen deficiency?

39. N% in urea is
a) 21%
b) 26%
c) 46%
d) 33%

40. N% in ammonium sulphate is
a) 20.6%
b) 46%
c) 21%
d) 33%

41. Which biofertilizer is used for paddy as nitrogen fixer?
a) Rhizobium
b) Azotobacter
c) Blue Green Algae (BGA)
d) PSB

42. *Rhizobium fixes* nitrogen in association with
a) Paddy
b) Legumes
c) Wheat
d) Cotton

43. Which of the following is a free-living nitrogen-fixing bacteria?
a) Rhizobium
b) Frankia
c) Azotobacter
d) Anabaena

44. *Azospirillum fixes* nitrogen in association with
a) Legumes
b) Cereals
c) Oilseeds
d) Vegetables

45. Which organism forms symbiotic association with non-legume trees for N fixation?
a) Rhizobium
b) Azotobacter
c) Frankia
d) Azospirillum

46. Which biofertilizer dissolves insoluble phosphates in soil?
a) Rhizobium
b) PSB
c) BGA
d) Azotobacter

47. *Nitrogen fixation* by *Rhizobium bacteria* per hectare ranges from
a) 50-90 kg
b) 50-70 kg
c) 50-100 kg

a) Yellowing of older leaves
b) *Stunted growth*
c) *Purple coloration* of leaves
d) Burning of leaf tips

35. Which nitrogen fertilizer has highest N content?
a) Urea
b) *Ammonium sulphate*
c) CAN
d) DAP

36. *Potassic fertilizer* applied to tobacco crop is
a) *Potassium chloride*
b) *Potassium nitrate*
c) *Potassium sulphate*
d) MOP

37. Which of the following is a slow-release nitrogen fertilizer?
a) Urea
b) *Ammonium sulphate*
c) Neem-coated urea
d) CAN

38. The nitrogen loses from soil through denitrification are enhanced under
a) Well-drained conditions
b) *Waterlogged conditions*
c) *Sandy soil*
d) *Acidic conditions*

d) 50-80 kg

48. Inoculation of ________ species of Rhizobium is done in soybean crop
a) *Rhizobium phaseoli*
b) *Rhizobium meliloti*
c) *Rhizobium leguminosarum*
d) *Rhizobium japonicum*

49. Inoculation of ________ species of Rhizobium is done in berseem crop
a) *Rhizobium phaseoli*
b) *Rhizobium meliloti*
c) *Rhizobium leguminosarum*
d) *Rhizobium japonicum*

50. Conversion of organic nitrogen to inorganic form is called
a) Immobilization
b) Mineralization
c) Decomposition
d) Leaching

Topic 3: Phosphorus, Potassium & Secondary Nutrient Fertilizers

51. P2O5 percentage in Single Super Phosphate (SSP) is
a) 16-22
b) 46
c) 30-38
d) 48

52. P2O5 percentage in DAP is
a) 16-22
b) 46
c) 30-38
d) 48

53. SSP contains 16-22% P2O5. Which other nutrient does SSP supply?
a) Nitrogen
b) Calcium
c) Sulphur
d) Magnesium

54. Which phosphatic fertilizer is suitable for highly acidic soils?
a) SSP
b) DAP
c) *Rock phosphate*
d) TSP

55. Which fertilizer reduces calcium deficiency in soil?
a) DAP
b) MOP
c) SSP
d) Urea

56. *Bone meal* is a good ________ fertilizer of organic origin
a) Nitrogen
b) Potassium
c) Phosphorous
d) Sulphur

57. *Horn meal* is a slow-acting fertilizer of
a) Nitrogen

64. Popping in groundnut is due to ________ deficiency
a) Zinc
b) Boron
c) Calcium
d) Manganese

65. *Heart rot* of sugar beet is due to deficiency of
a) Ca
b) B
c) Zn
d) Cu

66. *Top sickness* of tobacco is due to deficiency of
a) Molybdenum
b) Zinc
c) Boron
d) Iron

67. *Pahala blight* of sugarcane is due to deficiency of
a) Zn
b) Fe
c) Mn
d) Cu

68. *Marsh spot* of peas is due to deficiency of
a) Zinc
b) Phosphorous
c) Potassium
d) Manganese

69. White bud of maize is due to deficiency of
a) Iron
b) Zinc
c) Manganese
d) Copper

70. *Empty glumes* in wheat is a symptom of ________ deficiency
a) Mo
b) B

b) Potassium
c) Phosphorous
d) Sulphur

58. To avoid zinc deficiency, ________ of zinc sulphate should be applied per acre
a) 50 kg
b) 20 kg
c) 100 kg
d) 150 kg

59. Whiptail in cauliflower is due to deficiency of
a) Zn
b) Mn
c) Mo
d) Cu

60. *Khaira disease* of rice is due to deficiency of
a) Fe
b) Zn
c) Mn
d) Cu

61. *Mottle leaf* or Frenching of citrus is due to deficiency of
a) Iron
b) Zinc
c) Manganese
d) Copper

62. *Bitter pit* in apple is due to deficiency of
a) Ca
b) B
c) Zn
d) Cu

63. *Blossom end* rot of tomato is due to deficiency of
a) Boron
b) Zinc
c) Calcium
d) Magnesium

c) Cu
d) Ca

71. Young leaves turn white due to deficiency of ________ element
a) Iron
b) Nitrogen
c) Phosphorous
d) Sulphur

72. Discoloration (reddening) of leaves in cotton is due to deficiency of
a) Zinc
b) Sulphur
c) Iron
d) Magnesium

73. *Example for* chelated Fe fertilizer is
a) Fe-EDTA
b) FeSO4
c) FeS
d) FeCl3

74. *Critical deficiency* range of boron in *Gramineae plants* varies from
a) 5-10 ppm
b) 10-15 ppm
c) 20-70 ppm
d) 70-90 ppm

75. Toxicity of ________ element leads to delayed maturity
a) Nitrogen
b) Iron
c) Phosphorous
d) Zinc

Topic 4: Organic Manures, Biofertilizers & Vermicompost

76. Scientific method of breeding and raising earthworms in controlled conditions is
a) Vermicomposting
b) Vermiculture
c) Earthyculture
d) Wormiculture

77. pH range suitable for vermicompost preparation is
a) 6.5-7.5
b) 7.5-8.5
c) 8.5-9.5
d) 5.5-6.5

78. *Temperature range* suitable for vermicompost preparation is
a) 18-35°C
b) 19-36°C
c) 15-35°C
d) 13-32°C

79. 1000 earthworms may convert ________ kg waste material per day
a) 6
b) 8
c) 7
d) 5

80. The turnover of vermicompost is ________%
a) 65
b) 75

89. *Microbe living* freely without any association with plants and fixing atmospheric nitrogen is
a) Rhizobium
b) Frankia
c) Azolla
d) Azotobacter

90. Which of the following is a free-living nitrogen-fixing bacteria?
a) *Azorhizobium colidans*
b) Azospirillum
c) Clostridium
d) Frankia

91. Azolla is used as biofertilizer in ________ crop
a) Wheat
b) Maize
c) Rice
d) Cotton

92. Which biofertilizer is commonly used as a carrier material for Rhizobium?
a) Sand
b) Charcoal
c) Lignite
d) Peat

93. FYM (Farm Yard Manure) contains approximately ________ N

c) 55
d) 85

81. *Earthworm activity* is confined to ________ feet depth only in vermicompost
a) 5
b) 6
c) 3
d) 2

82. *Recommended dosage* of vermicompost for fruit trees is
a) 5-7 kg/tree
b) 5-8 kg/tree
c) 6-10 kg/tree
d) 5-10 kg/tree

83. Which of the following is NOT a characteristic of neem cake?
a) *Insect repellant*
b) *Insecticidal property*
c) Used as cattle feed
d) Used as coating material over urea

84. Which of the following is NOT a green leaf manuring crop?
a) *Pongamia glabra*
b) *Gliricidia maculata*
c) *Cassia siamea*
d) *Sesbania speciosa*

85. Which of the following is a Non-edible oil cake?
a) *Gingelly cake*
b) *Groundnut cake*
c) *Coconut cake*
d) Cotton cake

86. *Oil cakes* can be decomposed easily when
a) *Oil content* is more
b) Oil is added
c) *Oil content* is less
d) Cut in larger pieces

87. *Concentrated organic* manures are
a) Plant originated
b) *Animal originated*
c) Chemical originated
d) Both plant and animal originated

88. *Rhizobium fixes* ________ nitrogen/acre in root nodules
a) 20-35 kg
b) 25-30 kg
c) 25-50 kg
d) 25-55 kg

a) 0.5%
b) 1.5%
c) 2.5%
d) 3.5%

94. Green manuring improves soil by
a) *Adding organic* matter
b) *Fixing nitrogen*
c) *Improving soil* structure
d) All of the above

95. Which of the following is the best green manure crop for rice?
a) Sesbania
b) Sunnhemp
c) Dhaincha
d) All of the above

96. Compost is prepared from
a) *Rock minerals* only
b) Farm waste and crop residues
c) Chemical fertilizers
d) *Synthetic materials*

97. Which of the following biofertilizer is used for Azolla-*Anabaena association*?
a) *Phosphate solubilizing* bacteria
b) Cyanobacteria
c) Mycorrhiza
d) Trichoderma

98. *Mycorrhiza helps* plants mainly in absorption of
a) Nitrogen
b) Potassium
c) Phosphorus
d) Calcium

99. *Concentrated organic* manures have ________ nutrient content compared to bulky organic manures
a) Lower
b) Equal
c) Higher
d) Variable

100. *Blood meal* is a rich source of which nutrient?
a) Phosphorus
b) Potassium
c) Nitrogen
d) Calcium

ANSWER KEY — MANURES AND FERTILIZERS

1	2	3	4	5	6	7	8	9	10
A	B	B	C	D	B	D	C	B	D
11	12	13	14	15	16	17	18	19	20
D	C	A	A	A	D	C	C	C	C
21	22	23	24	25	26	27	28	29	30
A	C	C	D	D	B	A	B	C	D
31	32	33	34	35	36	37	38	39	40
B	D	B	D	A	C	C	B	C	A
41	42	43	44	45	46	47	48	49	50
C	B	C	B	C	B	C	D	B	B
51	52	53	54	55	56	57	58	59	60
A	B	C	C	C	C	A	A	C	B
61	62	63	64	65	66	67	68	69	70
B	A	C	C	B	C	C	D	B	C
71	72	73	74	75	76	77	78	79	80
A	D	A	A	A	B	A	A	C	B
81	82	83	84	85	86	87	88	89	90
C	D	C	D	D	C	D	C	D	C
91	92	93	94	95	96	97	98	99	100

C	D	A	D	D	B	B	C	C	C

SUBJECT 6: PRINCIPLES OF ENTOMOLOGY

Code: DA 131 | Total Questions: 100

Topic 1: Introduction to Insects & Classification

1. The largest phylum in the animal kingdom is
   a) Mollusca
   b) Chordata
   c) Echinodermata
   d) Arthropoda

2. Claw-bearing animals belong to the class
   a) Crustacea
   b) Onycophora
   c) Chilopoda
   d) Diplopoda

3. Number of orders in *Exopterygota are*
   a) 16
   b) 9
   c) 4
   d) 2

4. *Wings develop* externally in
   a) Hemimetabola
   b) Holometabola
   c) Endopterygota
   d) Apterygota

5. Which of the following order belongs to the division Endopterygota?
   a) Odonata
   b) Zoraptera
   c) Isoptera
   d) Diptera

6. True flies belongs to the order
   a) Odonata
   b) Diptera
   c) Hymenoptera
   d) Orthoptera

7. *Ants and* honey bees belong to the order
   a) Lepidoptera
   b) Dermaptera
   c) Hymenoptera
   d) Hemiptera

8. *Book lice* belong to the order
   a) Psocoptera
   b) Mallophaga
   c) Hemiptera
   d) Mecoptera

9. Larvae of order *Lepidoptera are* known as
   a) Maggots
   b) Grubs
   c) Naiads
   d) Caterpillars

10. *Parthenogenesis type* of reproduction is very common in
   a) Coleoptera
   b) Diptera
   c) Thysanoptera
   d) Thysanura

11. *Adectious type* of pupa is seen in order
   a) Hemiptera
   b) Hymenoptera
   c) Lepidoptera
   d) Coleoptera

12. Following insect is active during night time

14. *Bipectinate type* of antennae are seen in
   a) Butterflies
   b) Naiads
   c) Maggots
   d) Moths

15. The last segment in Proturan is called as
   a) Furcula
   b) Tenaculum
   c) Cornicles
   d) *Telson tail*

16. *Cornicles are* present in pregenital segments of
   a) Aphids
   b) Thysanura
   c) Collembola
   d) Lepidoptera

17. *Scent glands* are present in
   a) Leafhoppers
   b) Aphids
   c) Scales
   d) Bugs

18. *Wax glands* are present in
   a) *Scale insects*
   b) Whiteflies
   c) *Mealy bugs*
   d) *Scale insects* and mealy bugs

19. Which of the following is most advanced insect order?
   a) Thysanura
   b) Odonata
   c) Hymenoptera
   d) Collembola

20. *Dragonflies and* damselflies belong to order
   a) Orthoptera
   b) Hemiptera
   c) Odonata
   d) Neuroptera

21. *Cockroaches belong* to order
   a) Hemiptera
   b) Blattodea
   c) Isoptera
   d) Orthoptera

22. *Termites belong* to order
   a) Blattodea
   b) Hemiptera
   c) Isoptera
   d) Orthoptera

23. *Grasshoppers belong* to order
   a) Odonata
   b) Orthoptera
   c) Hemiptera
   d) Coleoptera

24. *Beetles belong* to order
   a) Hemiptera
   b) Lepidoptera
   c) Coleoptera
   d) Diptera

25. *Scale insects* and aphids belong to order
   a) Hemiptera

a) Moths
b) Butterflies
c) Cockroaches
d) Both moths and cockroaches

13. Type of head seen in Homoptera
a) Prognathous
b) Opisthognathous
c) Hypognathous
d) None

b) Homoptera
c) Both *Hemiptera and* Homoptera
d) Thysanoptera

Topic 2: External Morphology — Head, Thorax & Abdomen

26. Inverted Y-shaped suture in insect head is
a) *Occipital suture*
b) *Epicranial suture*
c) *Post occipital* suture
d) *Genal suture*

27. Inverted U-shaped suture in insect head is
a) *Occipital suture*
b) *Epicranial suture*
c) *Post occipital* suture
d) *Genal suture*

28. The area extending from below compound eyes to just above mandibles is
a) Occiput
b) Epicranium
c) Frons
d) Gena

29. Sclerite of ventral region of thorax is called as
a) Sternum
b) Tergum
c) Pleuron
d) Notum

30. *Forewings are* present in
a) Prothorax
b) Mesothorax
c) Metathorax
d) Pterothorax

31. Pterothorax is made up of
a) *Pro and* Mesothorax
b) *Meso and* Metathorax
c) *Metathorax and* 1st abdominal segment
d) None of the above

32. *Maximum number* of abdominal segments found in insects are
a) 10
b) 11
c) 13
d) 6

33. In abdomen, post-genital appendages are present in ________ segments
a) 7th and 9th
b) 9th and 10th
c) 8th and 9th
d) 10th and 11th

34. The number of spiracles found on thorax are
a) 3
b) 2
c) 4
d) 5

35. Following non-segmental legs on abdominal segments are NOT present in larvae of Lepidoptera

39. In Orthoptera, by rubbing hindwings against each other is ________ type of stridulation
a) Femoro-alary type
b) *Alary type*
c) *Air expulsion*
d) Vibration

40. *Scales known* as androconia seen on upper surface of wings of male *Lepidopterans are*
a) Gustatory in function
b) Odoriferous in function
c) Olfactory in function
d) *Sound producing*

41. Secondary maxillary segment is also called as
a) *Mandibular segment*
b) *Labral segment*
c) *Labial segment*
d) *Antennary segment*

42. *Insect leg* contains typically ________ segments
a) 5
b) 6
c) 4
d) 7

43. *Walking legs* of bugs are called as
a) Cursorial
b) Saltatorial
c) Ambulatorial
d) Fossorial

44. Following legs are used for jumping in grasshoppers
a) *Fore legs*
b) *Hind legs*
c) Both legs
d) *Metathoracic legs*

45. Percentage of oxygen in soil air in extreme conditions
a) 21%
b) 5-10%
c) 79%
d) 25%

46. *Hemielytra type* of forewings are present in
a) True bugs
b) Stem borers
c) Ants
d) Caterpillars

47. In Hemiptera, rostrum is formed by
a) Labium
b) Labrum
c) Maxillae
d) Galea

48. In siphoning type of mouthparts, galea is formed by
a) Labium
b) Labrum

a) 3rd abdominal segment
b) 4th abdominal segment
c) 7th abdominal segment
d) 10th abdominal segment

36. *Gustatory receptors* are associated with
a) Touch
b) Visual
c) Smell
d) Taste

37. Johnston's organs are present in ________ part of the antenna
a) Scape
b) Pedicel
c) Flagellum
d) Meriston

38. Which insect structure produces sound in crickets?
a) *Tymbal organ*
b) Femoro-alary type stridulation
c) *Hind legs* rubbing each other
d) *Alary type* stridulation

c) Mandible
d) Maxillae

49. *Upper lip* of the mouthparts of insects is
a) Labium
b) Mandibles
c) Maxillae
d) Labrum

50. Which of the following is NOT a type of insect mouthpart?
a) Chewing
b) Piercing-sucking
c) Siphoning
d) Grating

Topic 3: Wings, Metamorphosis & Sensory Organs

51. Example of insect which is having sound-making type of wings
a) Grasshopper
b) Bees
c) Wasps
d) Crickets

52. *Hemielytra type* of forewings are present in
a) True bugs
b) Stem borers
c) Ants
d) Caterpillars

53. *Scales known* as androconia on wings of male *Lepidoptera are* odoriferous. They are produced by
a) Hindwings
b) *Upper surface* of forewings
c) Underside of hindwings
d) *Wing margin*

54. In red gram plume moth, hindwing and forewing are divided into how many parts respectively?
a) 1 & 2
b) 2 & 3
c) 3 & 2
d) 3 & 4

55. Which of the following insects undergoes complete metamorphosis (holometabola)?
a) Grasshopper
b) Cockroach
c) Butterfly
d) Dragonfly

56. Which of the following is an example of hemimetabola?
a) Butterfly
b) Moth
c) Beetle
d) Grasshopper

57. *Nymph stage* is seen in which type of metamorphosis?
a) Holometabola
b) Hemimetabola
c) Ametabola
d) Paurometabola

64. Johnston's organ in insects detects
a) Light
b) *Sound vibrations*
c) Gravity
d) Humidity

65. Which sense organ helps insects detect pheromones?
a) *Compound eye*
b) *Simple eye* (ocelli)
c) Antennae
d) Tympanum

66. Tympanum (hearing organ) in grasshoppers is located on
a) Head
b) Thorax
c) 1st abdominal segment
d) Hindleg

67. *Insect wing* venation patterns are used for
a) *Flight only*
b) Identification/Classification
c) *Protection only*
d) Mating

68. *Elytra are* hardened forewings found in order
a) Lepidoptera
b) Hymenoptera
c) Coleoptera
d) Diptera

69. Halteres (balancers) are modified hindwings found in order
a) Lepidoptera
b) Diptera
c) Hymenoptera
d) Hemiptera

70. Number of instars in most holometabolous insects is
a) 2-3
b) 3-4
c) 4-6
d) 7-10

71. Moulting in insects is controlled by the hormone
a) *Juvenile hormone*
b) Ecdysone
c) Pheromone

58. *Pupa stage* is absent in which type of metamorphosis?
a) Holometabola
b) Hemimetabola
c) *Complete metamorphosis*
d) Endopterygota

59. *Maggots are* larvae of order
a) Lepidoptera
b) Coleoptera
c) Diptera
d) Hymenoptera

60. *Grubs are* larvae of order
a) Lepidoptera
b) Coleoptera
c) Diptera
d) Hymenoptera

61. *Naiads are* aquatic nymphs of order
a) Lepidoptera
b) Hemiptera
c) Odonata
d) Orthoptera

62. *Olfactory receptors* in insects are located on
a) Legs
b) Antennae
c) *Compound eyes*
d) Labrum

63. *Compound eyes* in insects detect
a) Sound
b) Temperature
c) *Movement and* colour
d) Smell

d) Sericin

72. Which hormone prevents metamorphosis and maintains larval characters?
a) Ecdysone
b) *Juvenile hormone*
c) Pheromone
d) *Brain hormone*

73. Diapause in insects is induced by
a) High temperature
b) *Unfavourable environmental* conditions
c) *Excess food*
d) Short day length only

74. Which of the following is a chemical used as sex pheromone trap in pest management?
a) CUE lure
b) Malathion
c) Chlorpyrifos
d) Endosulfan

75. Ocelli (simple eyes) in adult insects are used for
a) *Detecting movement*
b) *Detecting light* intensity
c) *Detecting colour*
d) *Detecting smell*

Topic 4: Insect Mouthparts & Feeding Habits

76. Which of the following mouthpart type does a butterfly have?
a) *Chewing type*
b) Piercing-sucking type
c) *Siphoning type*
d) Rasping-sucking type

77. Which of the following mouthpart type does a mosquito have?
a) *Chewing type*
b) Piercing-sucking type
c) *Siphoning type*
d) *Sponging type*

78. Which of the following mouthpart type does a housefly have?
a) *Chewing type*
b) Piercing-sucking type
c) *Siphoning type*
d) *Sponging type*

79. *Thrips have* ________ type of mouthparts
a) Chewing
b) Piercing-sucking
c) Rasping-sucking
d) Sponging

80. *Mandibles are* absent in which of the following mouthpart type?
a) *Chewing type*
b) Piercing-sucking type
c) *Siphoning type*
d) *Biting type*

89. *Sucking insects* cause damage by
a) Making holes in leaves
b) Feeding on plant sap and causing indirect damage
c) *Boring into* stem
d) *Cutting roots*

90. The proboscis in Lepidoptera is formed by modification of
a) Labium
b) Labrum
c) Galea of maxillae
d) Mandibles

91. Which of the following is an indicator of piercing-sucking pest damage?
a) Large irregular holes in leaves
b) Yellowing, wilting, sooty mould, honeydew
c) Webbing on plant
d) Linear mines on leaves

92. Honeydew is secreted by
a) Butterflies
b) Bees
c) *Aphids and* scale insects
d) Beetles

93. *Sooty mould* grows on the honeydew secreted by
a) Caterpillars
b) *Aphids and* whiteflies
c) Beetles
d) Moths

94. *Lacinia and* galea are parts of
a) Labrum
b) Mandibles

c) Maxillae
d) Labium

95. *Stipes and* cardo are parts of
a) Labrum
b) Mandibles
c) Maxillae
d) Labium

96. The hypopharynx in insects is associated with
a) Vision
b) Taste
c) Hearing
d) Balance

97. Which mouthpart carries salivary duct opening?
a) Labrum
b) Mandibles
c) Maxillae
d) Hypopharynx

98. *Insects with* chewing mouthparts are pests of which crop part?
a) *Roots and* stem only
b) Leaves, stems, fruits and roots
c) *Sap only*
d) *Flowers only*

99. Which insect order has the most economically important chewing pests?
a) Hemiptera
b) Diptera
c) Lepidoptera
d) Hymenoptera

100. The epiglottis in insects is equivalent to the
a) Labrum
b) Clypeus
c) Hypopharynx
d) Galea

81. Labrum (upper lip) in chewing mouthparts is
a) Unpaired
b) Paired
c) Absent
d) Modified into proboscis

82. Which is the lower lip in insect mouthparts?
a) Labrum
b) Mandibles
c) Maxillae
d) Labium

83. Stylets in piercing-sucking mouthparts are formed by modified
a) *Labrum and* labium
b) *Mandibles and* maxillae
c) *Mandibles only*
d) *Maxillae only*

84. The chewing and lapping type mouthparts are found in
a) Housefly
b) Butterfly
c) *Honey bee*
d) Aphid

85. Which insect pest causes damage by scraping the plant surface before sucking sap?
a) Aphid
b) Whitefly
c) Thrips
d) *Scale insect*

86. *Insects with* siphoning type of mouthparts feed on
a) *Solid food*
b) Plant sap
c) *Flower nectar*
d) Blood

87. Which mouthpart type causes maximum visible injury to leaves?
a) Piercing-sucking
b) Siphoning
c) Chewing
d) Sponging

88. In piercing-sucking insects, which structure is inserted into the plant?
a) Labrum
b) Stylets
c) Galea
d) Mentum

ANSWER KEY — PRINCIPLES OF ENTOMOLOGY

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	B	A	A	D	B	C	A	D	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	D	B	D	D	A	D	D	C	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	C	B	C	B	B	A	D	A	B
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	B	D	B	A	D	B	C	B	B
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
C	A	A	D	B	A	A	D	D	D
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
D	A	B	C	C	D	B	B	C	B
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
C	B	C	B	C	C	B	C	B	C
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
B	B	B	A	B	C	B	D	C	C
81	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**
A	D	B	C	C	C	C	B	B	C
91	**92**	**93**	**94**	**95**	**96**	**97**	**98**	**99**	**100**

B	C	B	C	C	B	D	B	C	C

SUBJECT 7: PESTS OF CROPS & THEIR MANAGEMENT

Code: DA 132 | Total Questions: 100

Topic 1: Pests of Pulse Crops

1. The holes of about 1 mm in diameter covered with a thin membrane readily seen on infested pods is noticed in ________ pest of pulses
   a) *Pod bug*
   b) *Pod fly*
   c) Gram caterpillar
   d) Leaf miner

2. *Visual symptoms* are NOT appeared due to the incidence of
   a) *Spotted pod* borer
   b) *Pod fly*
   c) Gram pod borer
   d) *Plume moth*

3. Which crop is recommended as guard crop in IPM practices of pulse crops?
   a) Rice
   b) Bhendi
   c) Jowar
   d) Cotton

4. Which insect in red gram has its forewing and hindwing divided into 3 and 2 parts respectively?
   a) *Pod bug*
   b) Gram pod borer
   c) *Plume moth*
   d) *Pod fly*

5. Presence of red band on head and red streak along costal margin of forewings is seen in
   a) *Amsacta albistriga*
   b) *A. moorei*
   c) Both 1 & 2
   d) None of the above

6. In gram pod borer (*Helicoverpa armigera*), the damage is caused by
   a) *Adults only*
   b) *Larvae only*
   c) Both adults and larvae
   d) Nymphs

7. The caterpillar of gram pod borer bores into pods and feeds on
   a) Leaves
   b) *Developing seeds*
   c) Stems
   d) Roots

8. Inter-cropping of pulse crops with ________ to attract Helicoverpa is recommended
   a) Maize
   b) Sorghum
   c) Bhendi/Marigold
   d) Castor

9. Which of the following is NOT a pest of pulse crops?
   a) *Spotted pod* borer
   b) *Pod fly*
   c) *American bollworm*
   d) Stem fly

10. Leaf miner damage in pulses appears as
   a) Yellowing of leaves
   b) *Serpentine mines* on leaves
   c) Holes in pods
   d) Webbing on shoots

14. *Flooding the* field for 24 hours kills grub population is a management practice for ________ pest of groundnut
   a) *Pod bug*
   b) White grub
   c) Root grub
   d) Both white grub and root grub

15. The cut end of the attacked stem of a dead groundnut plant is swollen due to
   a) Leaf miner
   b) *Pod bug*
   c) White grub
   d) *Jewel beetle*

16. Which pest of groundnut is a vector of peanut clump virus?
   a) Leaf miner
   b) Thrips
   c) White grub
   d) Aphid

17. Groundnut Bud Necrosis Virus is transmitted by
   a) Aphids
   b) Whiteflies
   c) Thrips
   d) Leafhoppers

18. The pod bug of groundnut sucks sap from the pods and causes
   a) Shrivelling of seeds
   b) *Pod drop*
   c) *Pod borer* damage
   d) Leaf yellowing

19. Which of the following is an important biological control agent for *Helicoverpa armigera*?
   a) NPV (Nuclear Polyhedrosis Virus)
   b) *Bacillus thuringiensis*
   c) *Trichogramma chilonis*
   d) All of the above

20. *Spotted pod* borer of redgram has scientific name
   a) *Maruca vitrata*
   b) *Melanagromyza obtusa*
   c) *Callosobruchus chinensis*
   d) *Exelastis atomosa*

21. Which of the following is an example of a pulse storage pest?
   a) *Callosobruchus chinensis*
   b) *Helicoverpa armigera*
   c) *Spodoptera litura*
   d) *Agrotis ipsilon*

22. Which of the following practices reduces pod fly infestation?
   a) Early sowing
   b) Late sowing
   c) *Intercropping with* maize
   d) *Flood irrigation*

23. In blackgram and greengram, which pest causes sucking damage and spreads yellow mosaic virus?
   a) Thrips
   b) Whitefly
   c) Aphid
   d) Jassid

11. *Castor pest* that attacks unripened citrus fruits is
a) *Hairy caterpillar*
b) *Semi looper*
c) *Slug caterpillar*
d) *Tobacco caterpillar*

12. The scientific name of castor shoot and capsule borer is
a) *Achaea janata*
b) *Microplitis ophiusae*
c) *Conogethes punctiferalis*
d) *Trialeurodes ricini*

13. Terminal leaves turn crinkled and silvery white in ________ pest of castor
a) *Castor whitefly*
b) *Castor mites*
c) *Castor thrips*
d) *Castor semilooper*

24. Jassids in pulses cause damage by
a) *Boring into* pods
b) *Sucking sap* from leaves causing 'hopperburn'
c) *Mining leaves*
d) *Cutting roots*

25. Which of the following management practices is used against stored grain pests of pulses?
a) *Mixing with* neem seed kernel extract
b) Soil drenching with insecticide
c) Spraying at harvesting
d) *Border trap* cropping

Topic 2: Pests of Cotton

26. ________ cotton pest is vector of leaf curl virus in cotton crop
a) Cotton aphid
b) Cotton leafhopper
c) Cotton whitefly
d) Cotton leaf miner

27. 'Rosette' flowers in cotton are due to the infestation of
a) *Spotted bollworm*
b) Pink bollworm
c) *Tobacco caterpillar*
d) *American bollworm*

28. 'Flared or open' squares are symptoms noticed in ________ cotton pest
a) Pink bollworm
b) *Spotted bollworm*
c) *American bollworm*
d) *Tobacco caterpillar*

29. ________ varieties of cotton are tolerant/resistant to whiteflies
a) Saritha
b) MCU-5
c) Narasimha
d) Kanchana

30. Inter-cropping of cotton with other legume crops should be grown in ________ ratio
a) 1:2
b) 1:5
c) 1:6
d) 1:3

31. Which of the following is NOT a chilli pod borer?
a) *American bollworm*
b) *Spotted bollworm*
c) *Tobacco caterpillar*
d) *Utethesia pulchella*

32. Which pest causes 'dead heart' in cotton seedling stage?
a) Cotton aphid
b) Whitefly
c) Cutworm
d) Jassid

33. Which cotton pest causes silvering of leaves due to rasping?
a) Whitefly
b) Jassid (leafhopper)
c) Thrips

39. Which cotton pest lays eggs inside cotton squares?
a) *American bollworm*
b) Pink bollworm
c) *Spotted bollworm*
d) *Tobacco caterpillar*

40. Red cotton bug (*Dysdercus cingulatus*) causes damage to cotton by
a) *Sucking sap* from leaves
b) *Sucking sap* from bolls causing staining of lint
c) *Boring into* bolls
d) Feeding on roots

41. Which insecticide is recommended as seed treatment for cotton against sucking pests?
a) Chlorpyrifos
b) Imidacloprid
c) Malathion
d) Endosulfan

42. *Jassid damage* in cotton is called
a) *Witches broom*
b) Leaf curl
c) Tile-like symptom
d) Yellowing

43. The predator commonly used in biological control of cotton bollworms is
a) Trichogramma
b) *Chrysoperla carnea*
c) NPV
d) All of the above

44. Which of the following measures helps in reducing pink bollworm in cotton?
a) Early sowing
b) Destruction of crop residue after harvest
c) Both 1 and 2
d) Neither 1 nor 2

45. Aphids in cotton excrete ________ which promotes sooty mould growth
a) Wax
b) Honeydew
c) Latex
d) Resin

46. *Tobacco caterpillar* (*Spodoptera litura*) is a ________ pest
a) Specific to cotton only
b) *Polyphagous pest*

d) Aphid

34. Which cotton pest causes 'Hopperburn' symptom?
a) Thrips
b) Aphid
c) Jassid (leafhopper)
d) Whitefly

35. Pink bollworm infests cotton mainly during
a) *Seedling stage*
b) *Boll formation* stage
c) *Flowering stage*
d) *Vegetative stage*

36. Scientific name of *American bollworm* is
a) *Pectinophora gossypiella*
b) *Earias vittella*
c) *Helicoverpa armigera*
d) *Spodoptera litura*

37. Which of the following is a systemic insecticide recommended for cotton sucking pests?
a) Monocrotophos
b) Chlorpyrifos
c) Imidacloprid
d) Endosulfan

38. Bt cotton is resistant to which of the following pests?
a) *Sucking pests*
b) Bollworms (lepidopteran pests)
c) Both bollworms and sucking pests
d) Soil pests

c) *Monophagous pest*
d) *Oligophagous pest*

47. Which of the following sucking pests of cotton has been increasing due to Bt cotton cultivation?
a) Bollworms
b) *Sucking pests* (whitefly, aphid, jassid)
c) Stem borers
d) Cutworms

48. *Spotted bollworm* (*Earias vittella*) damage in cotton results in
a) Dead heart in early stage
b) *Rosetted flowers*
c) Both dead heart and damaged squares
d) Leaf curl

49. Which trap is used to monitor adult moths of cotton bollworms?
a) Yellow sticky trap
b) Light trap and pheromone trap
c) Blue sticky trap
d) *Pitfall trap*

50. Economic Threshold Level (ETL) concept in pest management was introduced to
a) *Eliminate all* pests
b) *Apply pesticides* only when pest population reaches economic damage level
c) *Use more* pesticides
d) *Ban chemical* pesticides

Topic 3: Pests of Sugarcane & Oilseed Crops

51. Which pest of sugarcane may act as internode borer during cane formation stages?
a) Early shoot borer
b) *Top shoot* borer
c) Both 1 & 2
d) *Internode borer*

52. Due to attack by which pest does cane juice become more watery and jaggery quality is adversely affected?
a) *Aleurolobus barodensis*
b) *Chilo sacchariphagus*
c) *Odontotermes obesus*
d) *Chilo infuscatellus*

53. Earthing up operation is recommended against
a) Early shoot borer
b) *Internode borer*
c) *Top shoot* borer
d) *Fruit borer*

54. *Scale infestation* in sugarcane reduces juice sucrose content up to ________%
a) 13
b) 47
c) 28
d) 26

55. The scientific name of gingelly (sesame) leaf and pod borer is
a) *Asphondylia sesami*
b) *Acherontia styx*
c) *Antigastra catalaunalis*
d) *Elasmolomus sordidus*

56. *Webbed leaves* at top with young caterpillars is a symptom noticed in ________ pest of sesame
a) Gingelly Leaf and Pod Borer

64. *Top shoot* borer in sugarcane causes damage in
a) *Germination stage*
b) Early tillering stage
c) *Grand growth* period
d) *Maturity stage*

65. White grub in sugarcane causes damage by feeding on
a) Leaves
b) Shoots
c) Roots
d) *Cane juice*

66. Which insect pest of sugarcane lives in soil and cuts roots?
a) Early shoot borer
b) Termite
c) *Top borer*
d) *Scale insect*

67. *Castor hairy* caterpillar (Amsacta) causes skeletonization of
a) Roots
b) Stems
c) Leaves
d) Pods

68. Which biopesticide is recommended against *Spodoptera litura* in groundnut?
a) NPV
b) Bt
c) *Beauveria bassiana*
d) All of the above

69. *Biopesticide recommended* for tobacco caterpillar control is
a) SINPV (Spodoptera NPV)
b) Bt var. kurstaki
c) *Neem oil*
d) All of the above

b) *Gall fly*
c) *Hawk moth*
d) Leaf hopper

57. ________ is recommended as ID crop to escape gingelly leaf and pod borer pest
a) Rabi crop
b) Summer crop
c) Kharif crop
d) Both rabi and summer crop

58. The variety of sesame which is resistant to gall fly is
a) N 166-3
b) N 166-6
c) N 166-5
d) N 166-7

59. ________ pest infestation in sesame results in malformation of pod without proper setting of seeds
a) Gingelly Leaf and Pod Borer
b) *Gall fly*
c) *Hawk moth*
d) Leaf hopper

60. *Galled flowers* and buds resulting in malformation of pod in sesamum is due to attack of
a) Gall Fly
b) Gall Midge
c) Pod Borer
d) Hawk Moth

61. Which species of thrips infests lower surface of leaves which curl in sunflower?
a) *Scirtothrips dorsalis*
b) *Caliothrips indicus*
c) *Frankliniella schultzei*
d) *Stenotarsonemus spinki*

62. *Sunflower capitulum* borer causes damage by boring into
a) Stem
b) Capitulum (flower head)
c) Leaves
d) Roots

63. Which pest of sugarcane causes 'Dead Heart' in early stage?
a) *Internode borer*
b) Early shoot borer
c) *Top shoot* borer
d) *Scale insect*

70. Which pest causes 'Tikka' disease by transmitting pathogen in groundnut?
a) Leaf miner
b) *Pod fly*
c) White grub
d) Thrips

71. The pyrethroid insecticide recommended for bollworm management in cotton is
a) Chlorpyrifos
b) Lambda-cyhalothrin
c) Monocrotophos
d) Phosalone

72. Systemic insecticides that are absorbed by plants and kill sucking pests are called
a) *Contact insecticides*
b) *Stomach poisons*
c) Systemic insecticides
d) Fumigants

73. *Neonicotinoid insecticides* act on which insect system?
a) *Digestive system*
b) *Nicotinic acetylcholine* receptors in nervous system
c) *Muscular system*
d) *Reproductive system*

74. Which of the following is an organophosphate insecticide?
a) Endosulfan
b) Malathion
c) Cypermethrin
d) Imidacloprid

75. Biological control using *Trichogramma egg* parasitoid is recommended against
a) *Sucking pests*
b) Stem borers and bollworms (Lepidoptera)
c) Soil pests
d) *Storage pests*

Topic 4: Pests of Mango & Vegetable Crops

76. Double dip method is used to control ________ pest in mango
a) Mango Stem Borer
b) Mango Fruit Fly
c) Mango Nut Weevil
d) Mango Leaf Hopper

77. Masses of frass and sap exuding from bore holes is due to attack of ________ pest in mango
a) Mango Fruit Fly
b) Mango Leaf Hopper
c) Mango Stem Borer
d) Mango Nut Weevil

78. CUE LURE is an effective attractant being used to trap
a) Mango Stem Borer
b) Cucurbit Fruit Fly
c) Mango Leaf Hopper
d) Mango Fruit Borer

89. Mango Red Tree *Ant does* NOT cause direct damage but acts as carrier for
a) *Fruit fly* and stem borer
b) *Scale insects*, mealy bugs and causes nuisance to workers
c) Leaf hopper and nut weevil
d) Leaf webber and fruit borer

90. Damage of ________ pest results in withering and drying of new terminal shoots in mango
a) Mango Leaf Hopper
b) Mango Shoot Borer
c) Mango Stem Borer
d) Mango Leaf Webber

91. Caterpillar of ________ pest in mango bores into fruits at beak region and feeds inside reaching the kernel
a) Mango Shoot Borer
b) Mango Leaf Webber
c) Mango Fruit Borer

79. Scientific name of Mango Leaf Hopper is
   a) *Batocera rufomaculata*
   b) *Amritodus atkinsoni*
   c) *Sternochetus mangifera*
   d) *Idioscopus niveosparsus*

80. Scientific name of Mango Stem Borer is
   a) *Batocera rufomaculata*
   b) *Amritodus atkinsoni*
   c) *Sternochetus mangifera*
   d) *Idioscopus niveosparsus*

81. Scientific name of Mango Nut Weevil (Stone Weevil) is
   a) *Batocera rufomaculata*
   b) *Amritodus atkinsoni*
   c) *Sternochetus mangifera*
   d) *Idioscopus niveosparsus*

82. *Hot water* treatment of fruit at ________°C kills mango weevil inside the stone
   a) 50
   b) 60
   c) 70
   d) 80

83. *Wrapping polythene* sheets and pasting grease over stems is done to prevent migration of ________ pest in mango
   a) Mango Leaf Hopper
   b) Mango Stem Borer
   c) Mealy Bug
   d) Mango Fruit Fly

84. Caterpillar of ________ pest in mango webs terminal leaves and feeds by scraping green portion
   a) Mango Leaf Webber
   b) Mango Fruit Borer
   c) Mango Shoot Borer
   d) Mango Stem Borer

85. Raking of soil after harvest reduces ________ pest incidence in cucurbits
   a) Mango Fruit Fly
   b) Cucurbit Fruit Fly
   c) Mango Weevil
   d) Stem Borer

86. Infestation of ________ pest results in premature drop and decaying of fruits due to secondary bacterial infection in cucurbits
   a) Cucurbit Fruit Fly
   b) Mango Fruit Fly
   c) Stem Borer
   d) Leaf miner

87. 'Arka Tinda' and 'Arka Suryamukhi' are resistant varieties to ________ pest in cucurbits
   a) Aphid
   b) Whitefly
   c) Cucurbit Fruit Fly
   d) Thrips

88. *Silken webs* on undersurface of leaves which turn yellow is an indication of ________ pest in mango
   a) Mango Leaf Hopper
   b) Mango Stem Borer
   c) Red Spider Mite
   d) Mealy Bug

   d) Mango Stem Borer

92. Scientific name of Diamond Back Moth is
   a) *Plutella xylostella*
   b) *Bagrada cruciferarum*
   c) *Pieris brassicae*
   d) *Athalia proxima*

93. Transplanting 2 rows of mustard as trap crop for every 25 rows of cabbage is done to attract ________ pest
   a) Painted Bug
   b) Diamond Back Moth
   c) Cabbage Borer
   d) Mustard Sawfly

94. *Webbed leaves* and holes in cabbage head with faecal matter is a symptom of ________ pest
   a) Painted Bug
   b) Diamond Back Moth
   c) Cabbage Borer
   d) Mustard Sawfly

95. Scientific name of painted bug is
   a) *Plutella xylostella*
   b) *Bagrada cruciferarum*
   c) *Pieris brassicae*
   d) *Athalia proxima*

96. The female of ________ pest possesses a saw-like ovipositor
   a) Painted Bug
   b) Diamond Back Moth
   c) Cabbage Borer
   d) Mustard Sawfly

97. On slightest touch, larva of ________ pest falls to ground feigning death
   a) Painted Bug
   b) Diamond Back Moth
   c) Cabbage Borer
   d) Mustard Sawfly

98. Scientific name of Snake Gourd Semilooper is
   a) *Anadevidia peponis*
   b) *Plutella xylostella*
   c) *Bagrada cruciferarum*
   d) *Pieris brassicae*

99. *R. foveicollis* has reddish brown elytra, *A. intermedia* has ________ elytra and *A. cincta* has grey with black border elytra
   a) Green
   b) Blue black
   c) Red
   d) Yellow

100. *Avoid growing* of castor in mango orchards to eliminate *C. punctiferalis* which comes as
   a) Leaf webber
   b) *Fruit borer*
   c) Stem borer
   d) *Shoot borer*

ANSWER KEY — PESTS OF CROPS & THEIR MANAGEMENT

1	2	3	4	5	6	7	8	9	10
B	B	B	C	A	B	B	C	C	B
11	12	13	14	15	16	17	18	19	20

A	C	C	D	D	B	C	A	D	A
21	22	23	24	25	26	27	28	29	30
A	A	B	B	A	C	B	C	A	D
31	32	33	34	35	36	37	38	39	40
D	C	C	C	B	C	C	B	B	B
41	42	43	44	45	46	47	48	49	50
B	C	D	C	B	B	B	C	B	B
51	52	53	54	55	56	57	58	59	60
C	B	A	D	C	A	D	C	B	A
61	62	63	64	65	66	67	68	69	70
B	B	B	C	C	B	C	A	D	A
71	72	73	74	75	76	77	78	79	80
B	C	B	B	B	B	C	B	B	A
81	82	83	84	85	86	87	88	89	90
C	B	C	A	B	A	C	C	B	B
91	92	93	94	95	96	97	98	99	100
C	A	B	C	B	D	D	A	B	B

SUBJECT 8: FARM POWER AND MACHINERY

Code: DA 151 | Total Questions: 100

Topic 1: Sources of Farm Power & IC Engines

1. The most important source of farm power in India is
   a) animal power
   b) mechanical power
   c) human power
   d) solar power

2. India has approximately how many cattle making it highest in world
   a) 5 crore
   b) 10 crore
   c) 22.68 crore
   d) 50 crore

3. The average force a bullock can exert is approximately one tenth of its
   a) age
   b) height
   c) leg length
   d) body weight

4. *Power developed* by an average pair of bullocks is about
   a) 1 hp
   b) 5 hp
   c) 10 hp
   d) 0.5 hp

5. Thermal efficiency of diesel engine varies from
   a) 10 to 20 percent
   b) 50 to 60 percent
   c) 32 to 38 percent
   d) 5 to 10 percent

6. Thermal efficiency of petrol engine varies from
   a) 10 to 15 percent
   b) 25 to 32 percent
   c) 50 to 60 percent
   d) 40 to 50 percent

7. *Spark ignition* engines use
   a) diesel as fuel
   b) LPG only
   c) petrol or kerosene as fuel
   d) hydrogen only

8. *Compression ignition* engines use
   a) diesel as fuel
   b) petrol as fuel
   c) kerosene only
   d) natural gas only

9. In modern days almost all tractors and power tillers are operated by
   a) petrol engines
   b) kerosene engines
   c) diesel engines
   d) electric motors

10. *Diesel engines* are used for operating irrigation pumps flour mills and
   a) only transport
   b) only lighting
   c) only cooling
   d) chaff cutters threshers and winnowers

11. IC engine stands for
   a) Irrigation *Control engine*
   b) Intercrop *Control engine*

14. *Power transmission* system transmits power from engine to
   a) fuel tank
   b) working implement or wheels
   c) cooling system
   d) lubrication system

15. Advantages of mechanical power over animal power include
   a) low initial cost
   b) supplies manure
   c) higher efficiency and not affected by weather
   d) low maintenance

16. Disadvantage of mechanical power is
   a) slow work
   b) affected by seasons
   c) lives on farm produce
   d) high initial capital investment and requires technical knowledge

17. Advantages of animal power include
   a) high efficiency
   b) easily available low initial investment and supplies manure
   c) not affected by weather
   d) can work continuously

18. The process of converting chemical energy of fuel to mechanical energy in IC engine is called
   a) combustion
   b) lubrication
   c) working cycle
   d) cooling

19. *Four stroke* engine completes one working cycle in
   a) two rotations of crankshaft
   b) one rotation of crankshaft
   c) four rotations
   d) half rotation

20. *Two stroke* engine completes one working cycle in
   a) two rotations
   b) one rotation of crankshaft
   c) four rotations
   d) half rotation

21. The compression ratio of diesel engine is generally
   a) 4 to 6
   b) 8 to 10
   c) 2 to 4
   d) 14 to 22

22. Horsepower HP is a unit used to measure
   a) power output of engine
   b) torque
   c) fuel consumption
   d) temperature

23. The quality of diesel fuel is expressed by
   a) octane number
   b) API gravity
   c) cetane number
   d) RON value

24. Camshaft in IC engine controls
   a) fuel injection pressure
   b) opening and closing of valves
   c) cooling water flow
   d) lubrication

c) Indian *Combustion engine*
d) Internal *Combustion engine*

12. The fuel system of IC engine includes
a) fuel tank pump injector and filters
b) cooling and lubrication only
c) only the tank
d) only the injector

13. *Ignition system* of IC engine provides
a) cooling
b) lubrication
c) spark to ignite fuel-air mixture at right time
d) fuel supply

25. The function of flywheel in IC engine is to
a) maintain uniform rotation by storing energy
b) cool the engine
c) filter fuel
d) pump water

Topic 2: Lubrication Cooling & Tractor Systems

26. The main objective of lubrication in IC engine is to
a) cool the engine
b) filter fuel
c) reduce friction between moving parts
d) provide power

27. *Engine oil* viscosity is measured in
a) Pascal
b) Newton
c) bar
d) SAE units

28. *Water cooled* engines use a
a) air blast
b) water jacket around cylinder with radiator
c) oil for cooling
d) no cooling

29. *Air cooled* engines are cooled by
a) water
b) fins on cylinder and airflow
c) oil
d) refrigerant

30. The radiator in water-cooled engine removes heat by
a) radiation only
b) conduction
c) radiation and convection through airflow
d) evaporation

31. Thermostat in cooling system maintains
a) fuel pressure
b) optimum engine temperature
c) oil pressure
d) air pressure

32. A tractor is a self-propelled power unit used mainly for
a) transport only
b) carrying passengers
c) irrigation only
d) pulling implements and driving stationary machinery

33. The main types of tractors are
a) walking and riding types
b) wheeled and crawler tractors
c) small and large tractors
d) diesel and petrol tractors

34. *Power tiller* is a
a) two-wheel walk-behind type tractor for small farms
b) large tractor
c) four-wheel tractor
d) rotary cultivator only

35. PTO stands for
a) Power Take-*Off shaft* to drive implements

39. *Draft control* in tractor hydraulic system maintains constant
a) engine speed
b) wheel speed
c) fuel flow
d) implement depth by adjusting to soil resistance

40. Clutch in a tractor
a) connects and disconnects engine from transmission
b) steers the tractor
c) controls hydraulics
d) controls PTO speed

41. *Gear box* in tractor provides
a) different speed and torque combinations for various field conditions
b) cooling
c) fuel injection
d) lighting

42. *Field capacity* of a machine is the
a) maximum speed
b) fuel consumption per hour
c) area covered per unit time under field conditions
d) implement width

43. Maintenance of tractor includes checking
a) only tyres
b) only fuel level
c) engine oil coolant air filter tyres and battery regularly
d) only brakes

44. The function of air cleaner in tractor is to
a) clean air before it enters engine to prevent wear
b) filter fuel
c) cool the engine
d) reduce smoke

45. *Disc plough* is preferred over mould board plough in
a) hard dry rocky or sticky soils
b) light loamy soils
c) waterlogged soils
d) nursery beds

46. Rotavator is used for
a) pulverizing and mixing soil in one operation
b) harvesting
c) threshing
d) transport

47. Seed drill sows seeds at
a) uniform depth and spacing with fertilizer placement
b) random depths
c) only on surface
d) only in furrows manually

48. Sprayer is used for applying

b) Power Transfer Output
c) Pressure Temperature Output
d) Primary Traction Operation

36. *Three point* linkage on tractor is used for
a) attaching and controlling implements at rear
b) steering only
c) fuel storage
d) driver safety

37. The differential of a tractor allows
a) engine to stop
b) fuel injection
c) wheels to rotate at different speeds during turning
d) cooling

38. *Hydraulic system* of tractor is used for
a) fuel supply
b) steering only
c) braking only
d) lifting and controlling mounted implements

a) liquid pesticides fertilizers and herbicides uniformly
b) seeds
c) solid fertilizers
d) compost

49. *Combine harvester* performs operations of
a) only cutting
b) reaping threshing and cleaning grain in one pass
c) only threshing
d) only cleaning

50. *Preventive maintenance* of tractor is done to
a) prevent breakdowns and extend machine life
b) increase fuel consumption
c) reduce power
d) increase downtime

Topic 3: Tillage Implements & Sowing Equipment

51. *Mould board* plough inverts the soil and
a) only cuts weeds
b) only loosens
c) only breaks clods
d) buries weeds and crop residues while turning soil

52. *Disc plough* is suitable for
a) hard soils with heavy crop residues
b) light soils only
c) flooded paddy fields
d) greenhouse soils

53. *Chisel plough* is used for
a) making ridges
b) primary tillage with minimal soil inversion
c) seed sowing
d) fertilizer application

54. Subsoiler is used to break the
a) topsoil
b) hardpan or plough pan at depth below normal tillage
c) surface crust
d) weed roots only

55. *Disc harrow* is used for
a) deep ploughing
b) secondary tillage to break clods and cut crop residues
c) seed sowing
d) fertilizer application

56. Cultivator is used for
a) deep ploughing
b) inter-row cultivation and weed control after crop emergence
c) threshing
d) harvesting

57. *Blade harrow* is used for
a) deep tillage
b) seed sowing
c) surface tillage and weed control in dry soils
d) irrigation

58. Ridger is used to make
a) flat seedbeds
b) ridges and furrows for crops like groundnut sugarcane
c) only furrows
d) only basins

64. *Boom sprayer* sprays pesticides over
a) one row only
b) large field area through multiple nozzles on a boom
c) only soil
d) only seeds

65. *Knapsack sprayer* is a
a) manually carried portable sprayer on back
b) large tractor-mounted sprayer
c) engine-powered sprayer
d) motorized vehicle sprayer

66. *Thresher separates* grain from
a) weeds
b) straw and chaff after harvesting
c) soil
d) roots

67. *Winnower separates* grain from
a) chaff and lighter material by blowing air
b) roots
c) soil
d) stones

68. *Chaff cutter* cuts
a) crop into furrows
b) fodder crops into small pieces for animal feeding
c) soil
d) seeds

69. *Reaper cuts* and gathers
a) only soil
b) weeds only
c) seeds only
d) standing crops at harvest time

70. *Groundnut digger* lifts
a) groundnut pods from soil without damage
b) soil for ploughing
c) weeds
d) seeds from storage

71. Transplanting of vegetables is done by
a) seed drill
b) sprayer
c) thresher
d) manual or mechanical vegetable transplanters

59. A seed drill places seeds at
- a) surface only
- b) random depths
- c) specific depth and spacing in rows
- d) only broadcasting

60. Ferti-seed drill simultaneously sows
- a) only seeds
- b) seeds and fertilizer together in adjacent bands
- c) only fertilizer
- d) seeds at random

61. *Zero till* drill sows seeds into
- a) prepared seedbed
- b) only flooded soil
- c) only loosened soil
- d) undisturbed soil without prior tillage

62. Paddy transplanter is used to transplant paddy seedlings from
- a) seed to field directly
- b) nursery mat to main field at uniform spacing
- c) farmer to dealer
- d) lab to field

63. Duster is used to apply
- a) liquid pesticides
- b) liquid fertilizers
- c) dust formulation of pesticides
- d) irrigation water

72. *Potato planter* opens furrow places tuber covers and firms soil in
- a) 4 separate operations
- b) one continuous operation
- c) 2 operations
- d) 3 separate operations

73. The working width of an implement determines its
- a) field coverage per pass
- b) fuel consumption
- c) depth of operation
- d) seed rate

74. Calibration of seed drill means
- a) adjusting to deliver correct seed rate per unit area
- b) cleaning the drill
- c) oiling moving parts
- d) checking for damage

75. Draft of an implement is the
- a) horizontal pull force required to operate it
- b) speed of operation
- c) depth of working
- d) weight of implement

Topic 4: Irrigation Pumps Sprinkler & Drip Systems

76. *Centrifugal pump* operates by
- a) rotating impeller creating centrifugal force to lift water
- b) reciprocating pistons
- c) gravity only
- d) air pressure

77. *Turbine pump* is used to lift water from
- a) surface water bodies only
- b) deep bore wells and deep water sources
- c) only shallow wells
- d) only rivers

78. *Submersible pump* is installed
- a) inside the well or borewell submerged in water
- b) above ground level
- c) at surface level
- d) in overhead tank

79. *Diesel pumpset* is used where
- a) electricity is available
- b) electricity is not available or unreliable
- c) solar power is present
- d) wind power is available

80. *Electric pumpset* is preferred where
- a) reliable electricity supply is available
- b) power supply is unavailable
- c) diesel is cheap
- d) no water source is present

81. The efficiency of a pump is ratio of
- a) cost to output
- b) speed to pressure
- c) discharge to suction
- d) water output energy to input energy

82. *Sprinkler irrigation* distributes water through
- a) furrows
- b) basins

89. *Pressure required* for drip system is generally
- a) low 1 to 3 kg/cm2
- b) very high above 10 kg/cm2
- c) medium 5 to 8 kg/cm2
- d) extremely low below 0.5 kg/cm2

90. Lateral pipes in drip system are laid
- a) vertically down
- b) across rows only
- c) randomly in field
- d) along each crop row

91. Clogging of drippers is prevented by using
- a) filters at pump inlet and sand separator
- b) higher pressure
- c) more water
- d) acid treatment only

92. Automation in irrigation is achieved through
- a) manual labour only
- b) only timers
- c) only sensors
- d) solenoid valves timers and sensors

93. *Check valve* in irrigation system prevents
- a) water flow
- b) pressure increase
- c) backflow of water
- d) filtration

94. Discharge of a pump is expressed in
- a) kg per hour
- b) metres
- c) litres per second or cubic metres per hour
- d) percentage

95. Head of a pump is the
- a) weight of water
- b) discharge rate

c) nozzles that spray water like rain
d) underground pipes directly to roots

83. *Sprinkler irrigation* is suitable for
a) only flat terrain
b) uneven terrain sandy soils and crops not tolerating waterlogging
c) paddy only
d) waterlogged areas

84. *Drip irrigation* delivers water
a) by flooding
b) through sprinklers
c) by furrows
d) drop by drop directly to root zone of each plant

85. Main components of drip system are pump main line sub main lateral pipes and
a) sprinkler heads
b) valves only
c) drippers or emitters at each plant
d) nothing else

86. *Water use* efficiency is highest in
a) flood irrigation
b) furrow irrigation
c) sprinkler irrigation
d) drip irrigation

87. *Fertilizer injection* through drip system is called
a) fertigation
b) chemigation
c) irrigation
d) foliar spray

88. *Rain gun* sprinkler covers
a) large area by projecting water in rotating jet
b) small area near the nozzle
c) only one row
d) only greenhouse crops

c) vertical height water is lifted expressed in metres
d) power consumed

96. *Water lifting* devices like *Persian wheel* and chain pump are examples of
a) modern irrigation equipment
b) pumping stations
c) overhead tanks
d) traditional indigenous water lifting devices

97. *Irrigation scheduling* means deciding
a) when and how much water to apply to crop
b) crop variety
c) fertilizer dose
d) pesticide application

98. Soil moisture sensor helps in
a) measuring soil pH
b) applying fertilizers
c) detecting soil moisture to automate irrigation
d) controlling pests

99. *Hydraulic ram* pump operates using
a) electricity
b) energy of flowing water itself without external power
c) diesel engine
d) solar energy

100. *Surge irrigation* in furrows improves
a) fertilizer uptake
b) water advance and distribution uniformity
c) drainage
d) aeration

ANSWER KEY — FARM POWER AND MACHINERY

1	2	3	4	5	6	7	8	9	10
A	C	D	A	C	B	C	A	C	D
11	12	13	14	15	16	17	18	19	20
D	A	C	B	C	D	B	C	A	B
21	22	23	24	25	26	27	28	29	30
D	A	C	B	A	C	D	B	B	C
31	32	33	34	35	36	37	38	39	40
B	D	B	A	A	A	C	D	D	A
41	42	43	44	45	46	47	48	49	50
A	C	C	A	A	A	A	A	B	A
51	52	53	54	55	56	57	58	59	60
D	A	A	B	B	B	C	B	C	B
61	62	63	64	65	66	67	68	69	70
D	B	C	B	A	B	A	B	D	A
71	72	73	74	75	76	77	78	79	80
D	B	C	A	A	A	B	A	B	A
81	82	83	84	85	86	87	88	89	90
D	C	B	D	C	D	A	A	A	D
91	92	93	94	95	96	97	98	99	100
A	D	C	C	C	D	A	C	B	B

SUBJECT 9: PLANT PATHOLOGY & CROP DISEASES

Code: DA 171 | Total Questions: 100

Topic 1: Introduction to Plant Pathology & Disease Concepts

1. Plant pathology is derived from *Greek words* Phyton *Pathos and Logos meaning*
   a) plant study of soil
   b) plant growth and study
   c) plant air and study
   d) plant suffering and study of diseases

2. Plant Pathology is defined as the branch of science dealing with
   a) insect pests
   b) soil fertility
   c) crop production
   d) cause etiology resulting losses and management of plant diseases

3. *Fungi are* eukaryotic spore bearing and
   a) achlorophyllous heterotrophic organisms
   b) photosynthetic organisms
   c) nitrogen-fixing organisms
   d) autotrophic organisms

4. *Bacteria are* extremely minute prokaryotic and
   a) eukaryotic
   b) unicellular organisms lacking chlorophyll
   c) photosynthetic
   d) multicellular organisms

5. *Phytoplasmas are* found in the sieve tube cells of plant
   a) xylem tissue
   b) phloem tissue
   c) mesophyll
   d) epidermis

6. *Phytoplasmas are* transmitted by phloem-feeding
   a) beetles
   b) leafhoppers and planthoppers
   c) aphids only
   d) fungi

7. A virus is a sub-microscopic obligate parasite consisting of
   a) cells with nucleus
   b) cellulose walls
   c) lipid only
   d) nucleic acid and protein only

8. The pathogen or its parts or products seen on a host plant is known as
   a) Symptom
   b) Disease
   c) Sign
   d) Syndrome

9. *Disease that* occurs at very irregular intervals in few instances is known as
   a) Endemic
   b) Epidemic
   c) Pandemic
   d) Sporadic

10. *Abnormal increase* in size of plant part due to increase in cell size is
   a) Hyperplasia
   b) Hypertrophy
   c) Necrosis
   d) Chlorosis

14. *Restricted blackening* of vascular bundles in maize is characteristic symptom of
   a) *Turcicum leaf* blight
   b) *Banded leaf* and sheath blight
   c) Downy mildew
   d) *Charcoal rot*

15. Principles of plant disease management include Avoidance Exclusion *Eradication and*
   a) Protection
   b) Mutation
   c) Hybridization
   d) Domestication

16. Classification of fungicides includes protectant and
   a) insecticide
   b) bactericide
   c) nematicide
   d) systemic fungicide

17. *Host plant* resistance is the most economical method of
   a) fertilizer application
   b) irrigation scheduling
   c) weed control
   d) plant disease management

18. *Saprophytic survival* of pathogens occurs in
   a) dead organic matter in soil
   b) living host tissue only
   c) water only
   d) atmosphere

19. *Infection process* in plant disease management includes pre-penetration penetration and
   a) post-penetration establishment
   b) sporulation only
   c) dispersal only
   d) dormancy only

20. Dormant spores help pathogens to survive in soil for
   a) long periods without host
   b) few hours
   c) one season only
   d) rainy season only

21. *Nematodes are* microscopic roundworms that attack plant
   a) leaves only
   b) roots and cause various symptoms including galls
   c) stems only
   d) flowers only

22. Roguing is a disease management practice of
   a) applying fungicide
   b) adding fertilizer
   c) irrigating field
   d) removing and destroying infected plants from field

23. Crop sanitation helps in reducing
   a) fertilizer cost
   b) irrigation requirement
   c) primary inoculum source of pathogens
   d) weed population

24. Koch's postulates are used to
   a) prove that a specific organism causes a specific disease
   b) classify fungi
   c) measure rainfall

11. *Abnormal increase* in number of cells causing abnormal growth is
a) Hypertrophy
b) Wilting
c) Necrosis
d) Hyperplasia

12. Passive dispersal of plant pathogens can be done through
a) All of the above
b) Soil
c) Seed
d) *Planting material*

13. The temperature at which nematodes are killed is
a) 72 degrees C
b) 82 degrees C
c) 50 degrees C
d) 60 degrees C

d) test soil pH

25. Biocontrol of plant diseases uses organisms like
a) chemical pesticides
b) only fertilizers
c) synthetic compounds
d) Trichoderma *Pseudomonas and Bacillus species*

Topic 2: Diseases of Rice Wheat and Sorghum

26. Blast disease of paddy is caused by
a) *Xanthomonas oryzae*
b) *Helminthosporium oryzae*
c) *Magnaporthe oryzae Pyricularia oryzae*
d) *Rhizoctonia solani*

27. *Bacterial leaf* blight of paddy is caused by
a) *Pyricularia oryzae*
b) *Rhizoctonia solani*
c) *Xanthomonas oryzae* pv oryzae
d) Helminthosporium

28. *Sheath blight* of paddy is caused by
a) *Magnaporthe oryzae*
b) *Rhizoctonia solani*
c) *Xanthomonas oryzae*
d) Pyricularia

29. Brown spot of paddy is caused by
a) *Magnaporthe oryzae*
b) *Helminthosporium oryzae Bipolaris oryzae*
c) *Rhizoctonia solani*
d) Xanthomonas

30. *False smut* of paddy is caused by
a) *Ustilago oryzae*
b) *Claviceps oryzae*-sativae *Ustilaginoidea virens*
c) *Tilletia oryzae*
d) *Puccinia oryzae*

31. *Grassy stunt* virus of paddy is transmitted by
a) aphids
b) whitefly
c) brown planthopper *Nilaparvata lugens*
d) leafhopper

32. *Tungro disease* of paddy is transmitted by
a) brown planthopper
b) aphids
c) whitefly
d) green leafhopper *Nephotettix virescens*

33. Rust diseases of wheat are caused by species of
a) Puccinia
b) Helminthosporium
c) Fusarium
d) Alternaria

34. Black rust stem rust of wheat is caused by
a) *Puccinia graminis* tritici
b) *Puccinia striiformis*

39. Ergot of sorghum is caused by
a) *Helminthosporium turcicum*
b) *Rhizoctonia solani*
c) *Claviceps sorghi*
d) *Fusarium spp*

40. *Grain mold* of sorghum is caused by
a) single fungus
b) *Puccinia only*
c) complex of Curvularia Fusarium *Alternaria and* Helminthosporium
d) *Ustilago only*

41. Anthracnose of sorghum is caused by
a) *Puccinia purpurea*
b) *Colletotrichum graminicola*
c) Helminthosporium
d) Sclerospora

42. *Turcicum leaf* blight of maize is caused by
a) *Fusarium moniliforme*
b) *Pythium spp*
c) Rhizoctonia
d) *Helminthosporium turcicum Setosphaeria turcica*

43. *Charcoal rot* of sorghum and maize is caused by
a) *Puccinia sorghi*
b) *Macrophomina phaseolina*
c) Colletotrichum
d) Fusarium

44. Downy mildew of pearl millet is caused by
a) *Helminthosporium spp*
b) *Sclerospora graminicola*
c) *Puccinia penniseti*
d) Peronospora

45. *Covered smut* of sorghum is caused by
a) *Ustilago tritici*
b) *Puccinia sorghi*
c) *Sporisorium sorghi Sphacelotheca sorghi*
d) Claviceps

46. Seed treatment with Vitavax Carboxin is recommended against
a) leaf blast
b) bacterial diseases
c) virus diseases
d) smut and bunt diseases

c) *Puccinia recondita*
d) *Tilletia spp*

35. Yellow rust stripe rust of wheat is caused by
a) *Puccinia graminis*
b) *Puccinia recondita*
c) *Puccinia striiformis*
d) *Puccinia sorghi*

36. Brown rust leaf rust of wheat is caused by
a) *Puccinia graminis*
b) *Puccinia striiformis*
c) *Puccinia sorghi*
d) *Puccinia recondita* tritici

37. Loose smut of wheat is caused by
a) *Tilletia species*
b) *Puccinia species*
c) *Ustilago tritici*
d) Alternaria

38. *Karnal bunt* of wheat is caused by
a) *Neovossia indica Tilletia indica*
b) *Ustilago tritici*
c) *Puccinia species*
d) Helminthosporium

47. Spray of Propiconazole or Tebuconazole is recommended for management of
a) bacterial blight
b) nematode attack
c) rust diseases of wheat
d) virus diseases

48. Spray of *Copper oxychloride* or Streptomycin is used for
a) fungal diseases
b) bacterial diseases of crops
c) virus diseases
d) nematodes

49. Resistant varieties are the most economical way to manage
a) only soil problems
b) only irrigation problems
c) only fertilizer deficiency
d) plant diseases

50. *Detached leaf* or seedling test is used to screen varieties for
a) drought tolerance
b) disease resistance
c) insect resistance
d) salt tolerance

Topic 3: Diseases of Oilseeds Pulses and Cotton

51. Tikka disease early and late leaf spot of groundnut is caused by
a) *Puccinia arachidis*
b) *Cercospora arachidicola* and *Phaeoisariopsis personata*
c) *Rhizoctonia solani*
d) *Sclerotium rolfsii*

52. Stem rot of groundnut is caused by
a) *Fusarium oxysporum*
b) *Sclerotium rolfsii*
c) *Alternaria arachidis*
d) Cercospora

53. Rust of groundnut is caused by
a) *Cercospora arachidicola*
b) *Puccinia arachidis*
c) *Sclerotium rolfsii*
d) *Rhizoctonia solani*

54. *Aflatoxin contamination* in groundnut is produced by
a) *Aspergillus flavus* and A parasiticus
b) *Fusarium moniliforme*
c) *Penicillium spp*
d) *Rhizopus spp*

55. Wilt of castor is caused by
a) *Fusarium oxysporum* f sp ricini
b) *Alternaria casticola*
c) *Cercospora ricinella*
d) Botrytis

56. *Alternaria blight* of sunflower is caused by
a) *Puccinia helianthi*
b) *Sclerotinia sclerotiorum*
c) *Fusarium spp*
d) *Alternaria helianthi*

57. Downy mildew of sunflower is caused by
a) *Alternaria helianthi*
b) *Plasmopara halstedii*
c) *Sclerotinia sclerotiorum*
d) *Fusarium spp*

58. Yellow mosaic virus of soybean is transmitted by

64. *Bacterial blight* angular leaf spot of cotton is caused by
a) *Alternaria spp*
b) *Fusarium oxysporum*
c) *Xanthomonas axonopodis* pv malvacearum
d) Rhizoctonia

65. Leaf curl virus disease of cotton is transmitted by
a) aphids
b) thrips
c) whitefly *Bemisia tabaci*
d) leafhoppers

66. *Damping off* disease in nursery is caused by
a) *Alternaria spp*
b) *Fusarium only*
c) bacteria only
d) *Pythium spp* and *Rhizoctonia solani*

67. Seed treatment with Metalaxyl or *Captan prevents*
a) leaf rust
b) bacterial blight
c) virus diseases
d) damping off and seed rot

68. *Aspergillus crown* rot of groundnut is caused by
a) *Puccinia arachidis*
b) *Aspergillus niger*
c) *Rhizoctonia solani*
d) Cercospora

69. *Cercospora leaf* spot of chilli is caused by
a) *Pythium spp*
b) *Colletotrichum spp*
c) *Cercospora capsici*
d) *Fusarium oxysporum*

70. Anthracnose of chilli is caused by
a) *Alternaria spp*
b) *Cercospora capsici*
c) *Colletotrichum capsici*
d) *Pythium spp*

71. Biological control of soil borne diseases uses
a) chemical pesticides

a) whitefly *Bemisia tabaci*
b) aphids
c) thrips
d) leafhoppers

59. Root rot of soybean caused by *Phytophthora sojae* is favored by
a) dry conditions
b) waterlogged and poorly drained soils
c) sandy soils
d) saline soils

60. Wilt of pigeonpea is caused by
a) *Alternaria spp*
b) *Fusarium udum*
c) *Rhizoctonia solani*
d) Phytophthora

61. *Sterility mosaic* of pigeonpea is transmitted by
a) aphids
b) whitefly
c) eriophyid mite *Aceria cajani*
d) leafhoppers

62. Yellow vein mosaic of okra is transmitted by
a) aphids
b) whitefly *Bemisia tabaci*
c) thrips
d) leaf miner

63. *Fusarium wilt* of cotton is caused by
a) *Verticillium dahliae*
b) *Alternaria macrospora*
c) Xanthomonas
d) *Fusarium oxysporum* f sp vasinfectum

b) synthetic fungicides
c) organic acids
d) *Trichoderma viride* and T harzianum

72. Collar rot of groundnut is caused by
a) *Fusarium oxysporum*
b) *Aspergillus niger*
c) *Sclerotium rolfsii*
d) Alternaria

73. The disease triangle concept involves interaction of
a) soil water and air
b) temperature rainfall and humidity
c) insect plant and fungus
d) host pathogen and favourable environment

74. *Bordeaux mixture* contains
a) lime and sulphur
b) copper sulphate and urea
c) copper sulphate and lime
d) lime and nitrogen

75. Spray of Mancozeb or Chlorothalonil is recommended for management of
a) fungal diseases like blights and spots
b) bacterial diseases
c) virus diseases
d) nematode diseases

Topic 4: Wilt Virus Nematode and Post-harvest Diseases

76. *Fusarium wilt* symptoms include yellowing wilting and
a) browning or darkening of vascular tissue
b) swelling of stem
c) leaf spots
d) fruit rot

77. *Verticillium wilt* is caused by
a) *Pythium spp*
b) *Verticillium dahliae* and V albo-atrum
c) *Fusarium oxysporum*
d) *Sclerotium rolfsii*

78. *Panama wilt* of banana is caused by
a) *Alternaria spp*
b) *Phytophthora spp*
c) *Fusarium oxysporum* f sp cubense
d) *Sclerotinia spp*

79. *Tobacco mosaic* virus TMV is transmitted by
a) aphids
b) mechanical sap inoculation and contact
c) whitefly
d) leafhoppers

80. Mosaic symptoms in plants appear as
a) irregular light and dark green patches on leaves
b) uniform yellowing
c) brown necrotic spots
d) water soaked lesions

81. Leaf curl disease is characterized by
a) yellowing only
b) upward or downward curling of leaves
c) stem rot

89. Soil drenching with Metalaxyl Mancozeb is done to manage
a) leaf spot diseases
b) soil borne diseases like *Pythium and* Phytophthora
c) virus diseases
d) nematodes

90. Collar rot of chilli is caused by
a) *Cercospora capsici*
b) *Alternaria spp*
c) *Phytophthora capsici*
d) Fusarium

91. *Cucumber mosaic* virus CMV has a very wide host range and is transmitted by
a) whitefly
b) aphids
c) thrips
d) beetles

92. *Potato virus* Y PVY is transmitted by
a) whitefly
b) soil
c) water
d) aphids

93. Post-harvest diseases of fruits and vegetables are caused mainly by
a) nematodes
b) fungi like Botrytis *Penicillium and* Colletotrichum
c) bacteria only
d) viruses only

94. *Botrytis grey* mold of grapes and strawberries is caused by
a) *Colletotrichum spp*

d) root rot

82. *Nematodes that* cause root knot disease belong to genus
a) Heterodera
b) Pratylenchus
c) Tylenchulus
d) Meloidogyne

83. Root knot nematode *Meloidogyne causes* characteristic
a) yellowing of leaves
b) galls or knots on roots
c) stem cankers
d) fruit blight

84. *Cyst nematodes* belong to genus
a) Meloidogyne
b) Pratylenchus
c) Criconemella
d) *Heterodera and* Globodera

85. Biological control of nematodes uses
a) *Paecilomyces lilacinus* and *Trichoderma spp*
b) chemical nematicides
c) fungicides
d) bactericides

86. Seed treatment with Carbofuran or Phorate is recommended against
a) fungal diseases
b) bacteria
c) nematodes and soil insects
d) viruses

87. *Phytophthora foot* rot of black pepper is caused by
a) *Fusarium spp*
b) *Alternaria spp*
c) *Cercospora spp*
d) *Phytophthora capsici*

88. *Damping off* disease affects seedlings in
a) mature stage
b) fruiting stage
c) harvesting stage
d) nursery stage causing pre or post emergence death

b) *Alternaria spp*
c) *Botrytis cinerea*
d) Fusarium

95. Blue mold rot of stored fruits is caused by
a) *Botrytis cinerea*
b) *Penicillium expansum* and P italicum
c) *Colletotrichum spp*
d) *Fusarium spp*

96. *Sooty mold* on plants is a secondary fungal growth on
a) honeydew secreted by sucking insects
b) roots
c) leaves without insects
d) soil surface

97. *Phytoplasma diseases* are managed mainly by
a) fungicide sprays
b) controlling insect vectors and roguing infected plants
c) bactericide
d) nematicide

98. *Witches broom* symptom is caused by
a) fungi
b) nematodes
c) phytoplasma
d) bacteria

99. *Little leaf* disease of brinjal is caused by
a) phytoplasma
b) virus
c) fungi
d) bacteria

100. *Hot water* seed treatment at 50 degrees C for 30 minutes is done to manage
a) nematodes only
b) virus diseases
c) soil borne diseases
d) seed borne fungi and bacteria

ANSWER KEY — PLANT PATHOLOGY & CROP DISEASES

1	2	3	4	5	6	7	8	9	10
D	D	A	B	B	B	D	C	D	B
11	12	13	14	15	16	17	18	19	20
D	A	C	D	A	D	D	A	A	A
21	22	23	24	25	26	27	28	29	30
B	D	C	A	D	C	C	B	B	B
31	32	33	34	35	36	37	38	39	40
C	D	A	A	C	D	C	A	C	C
41	42	43	44	45	46	47	48	49	50
B	D	B	B	C	D	C	B	D	B
51	52	53	54	55	56	57	58	59	60
B	B	B	A	A	D	B	A	B	B
61	62	63	64	65	66	67	68	69	70
C	B	D	C	C	D	D	B	C	C
71	72	73	74	75	76	77	78	79	80
D	C	D	C	A	A	B	C	B	A
81	82	83	84	85	86	87	88	89	90
A	D	B	D	A	C	D	D	B	C
91	92	93	94	95	96	97	98	99	100
A	D	B	C	B	A	B	C	A	D

SUBJECT 10: CROP PRODUCTION II – RABI AND COMMERCIAL CROPS

Code: DA 201 | Total Questions: 100

Topic 1: Oilseed Crops – Groundnut and Castor

1. *Crops cultivated* for production of oils are known as
   a) pulse crops
   b) oilseed crops
   c) cereal crops
   d) fibre crops

2. India ranks first in world production of
   a) soybean
   b) sunflower
   c) wheat
   d) castor safflower sesame and niger

3. India accounts for about what percent of world oilseed production
   a) 20 percent
   b) 35 percent
   c) 50 percent
   d) 7 percent

4. Technology Mission on Oilseeds TMO was initiated in
   a) 1975
   b) 1986 May
   c) 2001
   d) 1991

5. Directorate of Oilseed Research DOR now named IIOR is located at
   a) Rajendranagar Hyderabad
   b) Bhopal
   c) Coimbatore
   d) Kanpur

6. *Groundnut botanical* name is
   a) *Arachis hypogaea*
   b) *Glycine max*
   c) *Vigna radiata*
   d) *Cajanus cajan*

7. In *Greek language Arachis means* legume and hypogaea means
   a) above ground
   b) below ground referring to pod formation in soil
   c) spreading plant
   d) annual plant

8. Groundnut is a member of family
   a) Poaceae
   b) Asteraceae
   c) Leguminosae Fabaceae
   d) Malvaceae

9. *Groundnut pods* develop
   a) above ground
   b) on stem
   c) below ground due to geocarpy
   d) on branches above soil

10. Seed rate for groundnut in shelling basis is
   a) 150 to 180 kg per ha
   b) 5 kg per ha
   c) 50 kg per ha
   d) 300 kg per ha

11. *Castor botanical* name is
   a) *Ricinus communis*
   b) *Arachis hypogaea*
   c) *Helianthus annuus*

14. Castor is a non-edible oilseed crop because
   a) oil has low yield
   b) it tastes bitter
   c) it has low oil content
   d) it contains toxic ricin compound

15. Wilt of castor is caused by
   a) *Alternaria ricini*
   b) *Cercospora ricinella*
   c) Botrytis
   d) *Fusarium oxysporum* f sp ricini

16. Earthing up in groundnut is done to
   a) apply fertilizer
   b) facilitate peg penetration and pod development underground
   c) remove weeds
   d) apply irrigation

17. Maturity in groundnut is indicated by
   a) yellowing of leaves
   b) stem turning red
   c) all flowers falling
   d) inner wall of pod darkening and seed filling

18. *Reasons for* low yields in groundnut include
   a) excess water supply
   b) too much fertilizer
   c) use of non-certified seed poor spacing and drought stress
   d) too dense plant population

19. *Groundnut requires* well drained soils because
   a) it is a C4 plant
   b) it needs saline soil
   c) it is a deep-rooted crop only
   d) waterlogging causes peg rot and poor pod development

20. *Rugose mosaic* virus of groundnut is spread by
   a) soil
   b) whitefly
   c) water
   d) aphid *Aphis craccivora*

21. Crop of groundnut is harvested at which growth stage
   a) vegetative stage
   b) pod maturity when outer pod surface shows venation and inner wall darkens
   c) during flowering
   d) early germination

22. *Spreading type* varieties of groundnut have duration of
   a) 90 to 100 days
   b) 120 to 150 days
   c) 60 to 70 days
   d) 200 days

23. Erect or bunch type groundnut varieties have duration of
   a) 150 to 200 days
   b) 60 to 70 days
   c) 90 to 110 days
   d) 200 to 250 days

24. *Castor hybrid* DCH 177 was developed at
   a) IARI New Delhi
   b) CRRI Cuttack
   c) IIMR Hyderabad
   d) DOR Hyderabad

d) *Sesamum indicum*

12. *Castor belongs* to family
    a) Fabaceae
    b) Asteraceae
    c) Euphorbiaceae
    d) Malvaceae

13. *Castor oil* is used industrially for
    a) edible purposes
    b) only lighting
    c) lubricants cosmetics and biofuel
    d) only medicine

25. India productivity of oilseeds is about 935 kg per ha while world average is
    a) 1632 kg per ha
    b) 500 kg per ha
    c) 2500 kg per ha
    d) 800 kg per ha

Topic 2: Rabi Crops – Wheat and Chickpea

26. Botanical name of wheat is
    a) *Oryza sativa*
    b) *Hordeum vulgare*
    c) *Triticum aestivum*
    d) *Zea mays*

27. Wheat belongs to family
    a) Fabaceae
    b) Solanaceae
    c) Poaceae
    d) Asteraceae

28. Wheat is a
    a) Kharif crop
    b) Zaid crop
    c) Rabi crop sown in winter
    d) perennial crop

29. The dwarf gene in wheat varieties introduced during Green Revolution is
    a) *Rht gene Reduced height* gene
    b) Sd1 gene
    c) d1 gene
    d) dwf1 gene

30. *Critical stages* of irrigation in wheat are CRI and
    a) germination only
    b) harvesting only
    c) tillering jointing flowering and grain filling
    d) after harvesting

31. *Recommended seed* rate for wheat is
    a) 5 kg per ha
    b) 100 kg per ha
    c) 200 kg per ha
    d) 50 kg per ha

32. *Chickpea gram* botanical name is
    a) *Vigna radiata*
    b) *Cajanus cajan*
    c) *Cicer arietinum*
    d) *Phaseolus vulgaris*

33. *Chickpea belongs* to family
    a) Leguminosae Fabaceae
    b) Poaceae
    c) Solanaceae
    d) Asteraceae

34. Gram caterpillar *Helicoverpa armigera* is a major pest of
    a) paddy
    b) wheat
    c) sunflower
    d) chickpea

35. Wilt of chickpea is caused by
    a) *Alternaria spp*

39. Wheat sowing is done during
    a) October to December
    b) June to July
    c) March to April
    d) January to February

40. Late sown wheat has
    a) higher yield due to cool temperatures
    b) same yield as timely sown
    c) reduced yield due to heat stress at grain filling
    d) zero yield

41. The *Karnal bunt* of wheat is caused by
    a) *Neovossia indica*
    b) *Ustilago tritici*
    c) *Tilletia foetida*
    d) *Puccinia graminis*

42. *Lentil masoor* botanical name is
    a) *Pisum sativum*
    b) *Cicer arietinum*
    c) *Vigna radiata*
    d) *Lens culinaris*

43. *Mustard botanical* name is
    a) *Helianthus annuus*
    b) *Brassica juncea*
    c) *Sesamum indicum*
    d) *Linum usitatissimum*

44. *Mustard belongs* to family
    a) Fabaceae
    b) Asteraceae
    c) Brassicaceae Cruciferae
    d) Solanaceae

45. *Safflower botanical* name is
    a) *Helianthus annuus*
    b) *Sesamum indicum*
    c) *Brassica juncea*
    d) *Carthamus tinctorius*

46. *Sesamum gingelly* botanical name is
    a) *Helianthus annuus*
    b) *Carthamus tinctorius*
    c) *Sesamum indicum*
    d) *Linum usitatissimum*

47. *Linseed botanical* name is
    a) *Helianthus annuus*
    b) *Sesamum indicum*
    c) *Linum usitatissimum*
    d) *Brassica juncea*

48. *Soybean belongs* to family
    a) Poaceae
    b) Asteraceae

b) *Rhizoctonia solani*
c) Colletotrichum
d) *Fusarium oxysporum* f sp ciceri

36. Chickpea is mainly a
a) Kharif crop
b) Zaid crop
c) Rabi crop
d) perennial crop

37. *Boron deficiency* in wheat causes
a) yellow stripes
b) grain sterility and reduced tillering
c) root rot
d) leaf blight

38. Seed treatment in wheat with Bavistin or *Vitavax prevents*
a) leaf rust
b) bacterial blight
c) virus diseases
d) seed-borne smut diseases

c) Brassicaceae
d) Leguminosae Fabaceae

49. Chickpea is mainly grown in which state in India
a) Madhya Pradesh *Rajasthan and* Maharashtra
b) West Bengal
c) Kerala
d) Assam

50. India is the largest producer of chickpea in the world accounting for about
a) 10 percent
b) 60 to 65 percent
c) 30 percent
d) 80 percent

Topic 3: Commercial Crops – Sugarcane and Tobacco

51. Botanical name of sugarcane is
a) *Saccharum sinense*
b) *Saccharum robustum*
c) *Saccharum officinarum*
d) *Saccharum spontaneum*

52. *Sugarcane belongs* to family
a) Poaceae
b) Fabaceae
c) Solanaceae
d) Malvaceae

53. Sugarcane is propagated by
a) seeds
b) setts stem cuttings
c) tubers
d) root cuttings

54. Sugarcane is mainly grown for
a) fibre
b) grain
c) vegetable use
d) sucrose production for sugar and jaggery industries

55. Red rot of sugarcane is caused by
a) *Ustilago scitaminea*
b) *Colletotrichum falcatum*
c) *Xanthomonas albilineans*
d) *Leifsonia xyli*

56. Smut of sugarcane is caused by
a) *Colletotrichum falcatum*
b) *Rhizoctonia solani*
c) *Puccinia melanocephala*
d) *Ustilago scitaminea Sporisorium scitamineum*

57. *Grassy shoot* disease of sugarcane is caused by
a) phytoplasma
b) fungi
c) bacteria
d) virus

58. Botanical name of tobacco is
a) *Datura stramonium*
b) *Nicotiana tabacum*
c) *Capsicum annuum*
d) *Solanum lycopersicum*

59. *Tobacco belongs* to family

64. *Ratoon crop* in sugarcane is the crop grown from
a) fresh setts
b) stubble remaining after main crop harvest
c) seeds
d) aerial roots

65. *Grand growth* period of sugarcane is the stage of
a) germination
b) ripening
c) maximum vegetative growth tillering and elongation
d) planting

66. Ripening of sugarcane refers to
a) harvesting
b) germination
c) ratoon growth
d) accumulation of sucrose in stem before harvest

67. *Trash mulching* in sugarcane is done to
a) increase disease
b) increase evaporation
c) conserve moisture suppress weeds and supply organic matter
d) compact soil

68. Cotton is a
a) food crop
b) oil crop only
c) commercial fibre crop belonging to Malvaceae
d) spice crop

69. *Jute botanical* name is
a) *Corchorus capsularis* and C olitorius
b) *Gossypium hirsutum*
c) *Cannabis sativa*
d) *Hibiscus cannabinus*

70. *Mesta botanical* name is
a) *Corchorus olitorius*
b) *Cannabis sativa*
c) *Jute only*
d) *Hibiscus cannabinus* and H sabdariffa

71. *Sugarcane harvest* is done when juice purity reaches
a) 60 percent
b) 100 percent
c) 85 to 90 percent
d) 50 percent

a) Fabaceae
b) Solanaceae
c) Poaceae
d) Asteraceae

60. Central Tobacco Research Institute CTRI is located at
a) Rajahmundry Andhra Pradesh
b) Mysore
c) Hyderabad
d) Coimbatore

61. Flue-cured tobacco is used for
a) chewing
b) cigarette manufacture
c) hookah
d) bidis

62. *Bidi tobacco* Natu is sun-cured and used mainly for
a) cigarettes
b) bidis
c) chewing
d) snuff

63. Nitrogen is applied in sugarcane in splits because
a) crop duration is long and to reduce losses
b) it is cheap
c) it improves colour
d) it prevents lodging

72. *Sett treatment* in sugarcane before planting is done with
a) Aretan or *Agallol solution* to prevent diseases
b) only water
c) only FYM
d) only salt water

73. *Sugarcane red* rot symptom is
a) yellowing of leaves
b) stem blackening
c) root rot
d) internal red discolouration of stalk with white patches

74. Mosaic virus of sugarcane is transmitted by
a) aphid *Aphis maidis*
b) whitefly
c) thrips
d) mealybug

75. Planting of sugarcane is mainly done in
a) Rabi early spring October to February
b) Kharif season only
c) summer only
d) any time equally

Topic 4: Vegetables and Pulses

76. *Pigeon pea* redgram botanical name is
a) *Cicer arietinum*
b) *Cajanus cajan*
c) *Vigna mungo*
d) *Phaseolus vulgaris*

77. *Pigeon pea* belongs to family
a) Poaceae
b) Solanaceae
c) Leguminosae Fabaceae
d) Asteraceae

78. *Greengram mung* bean botanical name is
a) *Cicer arietinum*
b) *Cajanus cajan*
c) *Phaseolus vulgaris*
d) *Vigna radiata*

79. Black gram urad botanical name is
a) *Vigna radiata*
b) *Cicer arietinum*
c) *Cajanus cajan*
d) *Vigna mungo*

80. *Cowpea botanical* name is
a) *Cajanus cajan*
b) *Phaseolus vulgaris*
c) *Vigna unguiculata*
d) *Cicer arietinum*

81. Botanical name of tomato is
a) *Solanum melongena*
b) *Capsicum annuum*
c) *Solanum lycopersicum Lycopersicon esculentum*
d) *Solanum tuberosum*

82. *Tomato belongs* to family
a) Cucurbitaceae
b) Solanaceae
c) Asteraceae
d) Fabaceae

89. Pulse crops are important because they fix atmospheric nitrogen through
a) industrial process
b) chemical reaction
c) physical absorption
d) symbiosis with *Rhizobium bacteria* in root nodules

90. *Okra lady* finger botanical name is
a) *Hibiscus sabdariffa*
b) *Hibiscus cannabinus*
c) *Abelmoschus esculentus*
d) *Capsicum annuum*

91. *Okra belongs* to family
a) Malvaceae
b) Solanaceae
c) Cucurbitaceae
d) Fabaceae

92. Yellow vein mosaic disease of okra is a serious
a) viral disease transmitted by whitefly
b) fungal disease
c) bacterial disease
d) nematode problem

93. *Enation leaf* curl of chilli is caused by a
a) fungus
b) bacterium
c) nematode
d) virus transmitted by aphids

94. Nursery is raised for vegetables like tomato brinjal and
a) cowpea and groundnut
b) paddy and wheat
c) chilli for transplanting
d) maize and sorghum

95. Hardening of seedlings before transplanting is done by
a) gradually reducing irrigation and exposing to sun
b) applying more fertilizer
c) increasing shade

83. Botanical name of brinjal eggplant is
a) *Solanum lycopersicum*
b) *Capsicum annuum*
c) *Cucumis sativus*
d) *Solanum melongena*

84. Botanical name of chilli is
a) *Solanum melongena*
b) *Capsicum annuum*
c) *Solanum lycopersicum*
d) *Cucumis sativus*

85. *Fruit borer* of tomato is
a) *Earias vittella*
b) *Helicoverpa armigera*
c) *Leucinodes orbonalis*
d) *Spodoptera litura*

86. *Shoot and* fruit borer of brinjal is
a) *Helicoverpa armigera*
b) *Spodoptera litura*
c) *Leucinodes orbonalis*
d) *Agrotis ipsilon*

87. Yellow mosaic virus of mung bean is transmitted by
a) aphids
b) thrips
c) whitefly *Bemisia tabaci*
d) leafhopper

88. Wilt of pigeon pea is caused by
a) *Alternaria spp*
b) *Rhizoctonia solani*
c) Phytophthora
d) *Fusarium udum*

d) applying fungicide

96. *Bitter gourd* botanical name is
a) *Cucumis sativus*
b) *Cucurbita maxima*
c) *Luffa cylindrica*
d) *Momordica charantia*

97. *Pumpkin ash* gourd and bitter gourd belong to family
a) Cucurbitaceae
b) Solanaceae
c) Fabaceae
d) Malvaceae

98. *Potato botanical* name is
a) *Solanum lycopersicum*
b) *Dioscorea alata*
c) *Ipomoea batatas*
d) *Solanum tuberosum*

99. *Potato late* blight caused by *Phytophthora infestans* is historically famous for causing
a) *Irish potato* famine of 1845
b) Indian famine
c) *American famine*
d) *European famine* of 1900

100. *Sweet potato* botanical name is
a) *Solanum tuberosum*
b) *Dioscorea alata*
c) *Colocasia esculenta*
d) *Ipomoea batatas*

ANSWER KEY — CROP PRODUCTION II – RABI AND COMMERCIAL CROPS

1	2	3	4	5	6	7	8	9	10
B	D	D	B	A	A	B	C	C	A
11	12	13	14	15	16	17	18	19	20
A	C	C	D	D	B	D	C	D	D
21	22	23	24	25	26	27	28	29	30
B	B	C	D	A	C	C	C	A	C
31	32	33	34	35	36	37	38	39	40
B	C	A	D	D	C	B	D	A	C
41	42	43	44	45	46	47	48	49	50
A	D	B	C	D	C	C	D	A	B
51	52	53	54	55	56	57	58	59	60
C	A	B	D	B	D	A	B	B	A
61	62	63	64	65	66	67	68	69	70
B	B	A	B	C	D	C	C	A	D
71	72	73	74	75	76	77	78	79	80
C	A	D	A	A	B	C	D	D	C
81	82	83	84	85	86	87	88	89	90
C	B	D	B	C	C	C	D	D	C
91	92	93	94	95	96	97	98	99	100
A	A	D	C	A	D	A	D	A	D

SUBJECT 11: AGRICULTURAL EXTENSION & RURAL DEVELOPMENT

Code: DA 291 | Total Questions: 100

Topic 1: Education – Formal Informal and Non-formal

1. Education is the process of bringing desirable change into the
   a) economy
   b) environment
   c) crops
   d) behavior of human beings

2. *Formal education* refers to hierarchically structured education system from
   a) only universities
   b) kindergarten through university including vocational training
   c) only schools
   d) only colleges

3. *Informal education* is a
   a) planned curriculum-based education
   b) structured institutional education
   c) only classroom learning
   d) lifelong process in which persons acquire knowledge from daily experiences

4. *Informal education* is
   a) pre-planned and systematic
   b) incidental spontaneous and not pre-planned
   c) structured and curriculum based
   d) compulsory for all

5. Non-formal education is organized systematic education activity carried on
   a) inside formal school system
   b) only through mass media
   c) only in homes
   d) outside the framework of formal system for specific groups

6. In formal education attendance is
   a) optional
   b) voluntary
   c) compulsory
   d) not required

7. In non-formal education the learners are mainly
   a) school children
   b) only teachers
   c) adults and youth in actual life situations
   d) university students

8. Which of the following is a characteristic of informal education
   a) incidental and spontaneous learning
   b) prescribed timetable
   c) pre-planned curriculum
   d) compulsory attendance

9. *Extension education* is concerned mainly with
   a) formal school system
   b) urban education
   c) adult education for agricultural and rural development
   d) university education

10. *Formal education* is concerned with growth of children preparing them for
   a) present situations
   b) agricultural work only
   c) future life and employment
   d) social events

14. According to Rodney *Stark education* is the cheapest most rapid and most reliable path to
   a) entertainment
   b) economic advancement
   c) political power
   d) religious knowledge

15. Non-formal education is derived from the expression
   a) informal education
   b) formal education
   c) vocational education
   d) extension education

16. *Mass media* like radio television and newspapers are tools of
   a) formal education only
   b) only government communication
   c) only agricultural research
   d) informal and non-formal education

17. E-learning or online education is a modern form of
   a) only formal education
   b) a blend of formal non-formal and informal education
   c) only informal education
   d) only agricultural training

18. *Practical training* in agricultural polytechnic is a component of
   a) only informal education
   b) formal education with practical elements
   c) only non-formal education
   d) only extension education

19. *Field visits* and demonstrations are primarily methods of
   a) formal school education
   b) informal family education
   c) non-formal extension education for farmers
   d) urban education

20. National Literacy *Mission aims* at reducing illiteracy through
   a) formal schooling only
   b) non-formal adult literacy programmes
   c) online education only
   d) university programmes

21. Agricultural polytechnic provides education that is
   a) purely theoretical
   b) only informal
   c) only extension education
   d) formal technical vocational education in agriculture

22. Farmers Field School FFS is a participatory learning approach under
   a) formal education system
   b) informal learning only
   c) non-formal extension education
   d) university system

23. ATMA stands for
   a) Agronomy Technology Management Association
   b) Agricultural *Training and* Management Authority
   c) Agricultural Technology Management Agency
   d) Agro-Technical Modern Agriculture

24. Krishi Vigyan Kendra KVK is an agricultural

11. Non-formal education serves the needs of
a) all age groups without distinction
b) identified specific group with particular learning needs
c) only urban population
d) only educated people

12. The difference between formal and non-formal education includes that non-formal is
a) more structured
b) outside the formal education realm and more flexible
c) more expensive
d) compulsory

13. Education as defined by Webster is the process of teaching to develop
a) only memory
b) knowledge skill or character of student
c) only physical fitness
d) only farming skills

a) farm science centre for technology demonstration training and frontline demonstrations
b) university
c) market centre
d) storage facility

25. The primary goal of agricultural extension education is
a) to increase university enrollment
b) to promote commercial farming only
c) to improve farmers knowledge skills and attitude for better farm income and living
d) to teach formal subjects

Topic 2: Extension Education – Meaning Principles and Scope

26. *Extension education* meaning is
a) out-of-school non-formal education helping people help themselves
b) formal school education
c) university education
d) only distance learning

27. The word *Extension was* first used in agricultural context in
a) Britain Cambridge University in 1867
b) France
c) USA
d) India

28. Agricultural Extension in India was started during
a) pre-independence era
b) 2001
c) 1947 only
d) post-independence with Community Development Programme in 1952

29. The principle of extension education is to
a) help people help themselves by building their capacity
b) provide subsidies only
c) give free inputs
d) do all farming for farmers

30. T and V system of extension stands for
a) *Training and Visit system* introduced by World Bank
b) *Technology and* Visit
c) *Transfer and* Validation
d) *Teaching and* Verification

31. The scope of agricultural extension includes
a) technology transfer education communication and rural development
b) crop production only
c) only livestock
d) only irrigation

32. *Extension worker* should first understand the
a) government policies only
b) needs problems and resources of farmers and community
c) market prices only
d) weather forecast only

33. Participation of farmers in extension activities ensures
a) more government control
b) reduced costs only
c) ownership relevance and sustainability of programmes

39. NABARD stands for
a) National Agriculture *Bank and* Rural Development
b) National *Bank for Agriculture and* Rural Development
c) National *Board for* Agricultural *Research and* Development
d) National Advisory *Bureau for Agriculture and* Rural Districts

40. Kisan Credit Card KCC scheme provides
a) revolving credit to farmers for agricultural inputs and needs
b) only seeds to farmers
c) only insurance
d) only subsidies

41. Self Help Groups SHGs are formed to
a) only grow crops
b) sell only government products
c) provide only government employment
d) promote savings and credit among rural poor especially women

42. *Nine steps* of programme planning in extension are used to
a) plan crop varieties
b) plan irrigation only
c) plan crop harvest
d) systematically plan agricultural extension programmes

43. *Front line* demonstration FLD shows new technology on farmers fields under
a) farmers fields with scientists guidance to demonstrate technology potential
b) scientists complete control
c) only laboratory conditions
d) greenhouse only

44. Trainers Training Programme ToT is conducted to
a) train farmers directly
b) train extension workers who then train farmers
c) train scientists only
d) train government officials only

45. IFFCO stands for
a) Indian Food and Fertilizer Corporation
b) International Federation of Farmers Cooperatives
c) Indian Federation of *Farmers and* Cooperatives Organization
d) Indian Farmers Fertiliser Cooperative

d) faster technology adoption

34. The concept of need-based extension means
a) providing all technologies
b) standardized package for all
c) addressing specific felt needs of farmers in their local context
d) government-decided priorities only

35. *Block level* extension worker in India is called
a) DDA Deputy Director of Agriculture
b) State Agriculture Minister
c) Agricultural Extension Officer AEO or VAS
d) ICAR scientist

36. Gram Panchayat is the lowest unit of
a) central government
b) district administration
c) local self-government in rural areas Panchayati Raj
d) state government

37. Mandal Parishad is the intermediate level of Panchayati Raj at
a) village level
b) district level
c) mandal or block level
d) state level

38. Zilla Parishad is the
a) village level body
b) mandal level body
c) state level body
d) district level Panchayati *Raj institution*

46. Agricultural cooperative society helps farmers by
a) only providing credit
b) only buying produce
c) only providing seeds
d) collective purchase of inputs marketing of produce and credit services

47. Adoption of innovation by farmers follows stages of awareness interest evaluation
a) trial and adoption
b) and buying
c) and rejection
d) and forgetting

48. Early adopters are farmers who adopt innovations
a) last after all others
b) first before anyone
c) shortly after innovators and are respected opinion leaders
d) never adopt innovations

49. *Mass media* are useful in extension for
a) reaching large number of people quickly with information
b) interpersonal communication only
c) only entertainment
d) one-to-one communication

50. *Village contact* programme involves extension worker making
a) regular visits to specific villages following schedule
b) monthly visits to state capital
c) random visits to any village
d) visits only during harvest

Topic 3: Extension Teaching Methods

51. *Extension teaching* methods are classified as individual group and
a) mass media methods
b) only individual methods
c) only group methods
d) only demonstration methods

52. *Method demonstration* shows farmers
a) only final result
b) only written instructions
c) only videos
d) HOW to perform a specific practice step by step

53. *Result demonstration* shows farmers
a) how to do a practice
b) only lab results
c) WHAT results can be achieved by using recommended practices on their own farm
d) only statistics

54. Farm visit by extension worker to individual farmer home or field is
a) group method
b) individual contact method
c) mass media method
d) written method

55. *Office call* or office visit by farmer to extension office is
a) group method
b) mass media method
c) printed method
d) individual contact method

56. *Field day* or kisan mela is a
a) group method for farmers to observe results at demonstration site

64. *Flip chart* is used in
a) mass media
b) group meetings to present information page by page using illustrations
c) individual visits only
d) radio

65. *Puppetry and* folk media are used to
a) entertain only
b) train scientists
c) teach in universities
d) communicate agricultural messages using traditional art forms in local language

66. ICT *Information and* Communication Technology in extension includes
a) internet mobile apps kisan call centres and *Agri portals*
b) only radio
c) only television
d) only newspapers

67. Kisan Call *Centre provides*
a) free inputs
b) toll-free helpline 1800-180-1551 where farmers can call and get agricultural advice
c) free seeds
d) free irrigation

68. mKisan portal sends
a) SMS based agricultural advisories to farmers mobile phones
b) only market prices
c) only weather forecast
d) only scheme information

b) individual method
c) mass media method
d) only radio programme

57. *Farmer field* school is a
a) formal school
b) individual method
c) mass media method
d) participatory group method where farmers learn by doing in field

58. Radio is a
a) individual contact method
b) group method
c) only agricultural method
d) mass media method for reaching large audience

59. Television is used in extension as
a) individual method
b) mass media audio-visual method
c) group method only
d) only print media

60. Leaflet is a
a) audio visual aid
b) printed single sheet communication material for farmers
c) group method
d) individual visit

61. Folder is
a) same as leaflet
b) audio visual aid
c) mass media
d) multi-page printed material with more detailed information than leaflet

62. Bulletin is a
a) daily newspaper
b) radio programme
c) mass media method
d) periodic publication of department on specific agricultural topic

63. Poster is used in extension to
a) provide detailed instructions
b) teach complex techniques
c) attract attention and convey simple message quickly
d) record data

69. Advantages of group methods over individual methods include
a) reaching more farmers at one time with less cost
b) more personalized attention
c) better for illiterate farmers
d) less effective overall

70. Disadvantages of mass media methods include
a) high reach
b) low cost
c) quick dissemination
d) no feedback from audience and message may not be understood by all

71. Storytelling as extension method is effective because it
a) is scientific
b) is culturally familiar engaging and easily remembered by farmers
c) gives exact data
d) is very expensive

72. *Mela agricultural* fair is used for
a) only selling produce
b) demonstrating new varieties technologies and educating farmers through exhibitions
c) only entertainment
d) only government display

73. *Exposure visit* for farmers to progressive farms or research stations helps in
a) seeing successful adoption which motivates them to adopt
b) increasing government control
c) only tourism
d) entertainment

74. *Campaign method* in extension creates
a) mass awareness by intensive focused effort on single important issue over short period
b) individual awareness
c) no awareness
d) only confusion

75. Participatory Rural Appraisal PRA is a method of
a) only data collection
b) crop cutting
c) understanding village situation by villagers and extension workers together using participatory tools
d) market analysis

Topic 4: Agricultural Information Materials & Rural Development

76. Agricultural information materials include leaflets folders pamphlets circular letters and
a) tractors
b) pesticides
c) newsletters bulletins and news stories
d) fertilizers

77. The purpose of agricultural information materials is to
a) sell products
b) replace extension workers
c) communicate agricultural technology information to farmers
d) conduct research

78. *News story* in extension is a
a) fictional story
b) radio drama
c) television show

89. Soil Health *Card scheme* provides farmers with
a) free fertilizer
b) free seeds
c) crop insurance
d) card containing soil nutrient status and fertilizer recommendations for their field

90. e-NAM National Agriculture Market is a
a) physical market
b) bank
c) online trading platform for agricultural commodities
d) insurance scheme

91. FPO Farmer Producer Organization is a form of
a) government department
b) collective enterprise owned and managed by farmers for input purchase marketing and services
c) individual farm
d) research station

d) news article about agricultural achievements or innovations for newspapers

79. Community Development Programme CDP in India was launched in
a) 1947
b) 1952
c) 1966
d) 1985

80. National Extension Service NES was established in
a) 1947
b) 1966
c) 1985
d) 1953

81. Intensive Agricultural Development Programme IADP was launched in
a) 1947
b) 1952
c) 1960-61
d) 1985

82. Green Revolution in India was achieved during
a) 1960s and 1970s with HYV wheat and rice
b) 1947 to 1950
c) 1980s
d) 2000s

83. White *Revolution refers* to increase in production of
a) wheat
b) cotton
c) milk through Operation Flood
d) sugar

84. Operation *Flood was* launched by
a) NABARD
b) National Dairy Development Board NDDB
c) ICAR
d) Ministry of Agriculture

85. Blue *Revolution refers* to increase in
a) water harvesting
b) milk production
c) poultry production
d) fisheries and aquaculture production

86. Mahatma Gandhi National Rural Employment Guarantee Act MGNREGA provides
a) 100 days of guaranteed wage employment to rural households
b) free seeds
c) free irrigation
d) free fertilizers

87. Pradhan Mantri Fasal Bima Yojana PMFBY is a scheme for
a) crop insurance to protect farmers from losses due to natural calamities
b) rural employment
c) input subsidy
d) direct income support

88. PM Kisan Samman *Nidhi provides* direct income support of
a) Rs 1000 per year
b) Rs 6000 per year in three equal instalments
c) Rs 10000 per year
d) Rs 500 per month

92. PMKSY Pradhan Mantri Krishi Sinchayee *Yojana aims* at
a) seed distribution
b) fertilizer supply
c) crop insurance
d) expanding irrigation coverage and improving water use efficiency

93. Rashtriya Krishi Vikas Yojana RKVY is a scheme to
a) provide employment
b) incentivize states to achieve increased public investment in agriculture
c) distribute seeds
d) provide crop insurance

94. Agricultural credit in India is provided by
a) only commercial banks
b) only NABARD
c) cooperative banks Regional Rural Banks NABARD and commercial banks
d) only government

95. *Rural cooperative* credit structure provides short term credit through
a) Primary Agricultural Credit Societies PACS at village level
b) RBI directly
c) NABARD directly
d) commercial banks only

96. The first Agricultural Polytechnic in Telangana PJTSAU system was established at
a) Rajendranagar Hyderabad
b) various locations under PJTSAU
c) Warangal
d) Adilabad

97. District Agriculture Officer DAO supervises extension activities at
a) village level
b) state level
c) national level
d) district level

98. *Lead farmer* or contact farmer approach uses
a) only scientists
b) progressive farmers as local change agents to spread technology among neighbours
c) only extension workers
d) only government officials

99. *Participatory technology* development PTD involves
a) only scientists conducting research
b) only government planning
c) only market research
d) farmers and scientists jointly developing and testing technologies

100. Food Security Act 2013 in India aims to provide
a) only agricultural credit
b) only crop insurance
c) subsidized food grain to about two thirds of population
d) only employment

ANSWER KEY — AGRICULTURAL EXTENSION & RURAL DEVELOPMENT

1	2	3	4	5	6	7	8	9	10
D	B	D	B	D	C	C	A	C	C
11	12	13	14	15	16	17	18	19	20
B	B	B	B	B	D	B	B	C	B

21	22	23	24	25	26	27	28	29	30
D	C	C	A	C	A	A	D	A	A
31	32	33	34	35	36	37	38	39	40
A	B	C	C	C	C	C	D	B	A
41	42	43	44	45	46	47	48	49	50
D	D	A	B	D	D	A	C	A	A
51	52	53	54	55	56	57	58	59	60
A	D	C	B	D	A	D	D	B	B
61	62	63	64	65	66	67	68	69	70
D	D	C	B	D	A	B	A	A	D
71	72	73	74	75	76	77	78	79	80
B	B	A	A	C	C	C	D	B	D
81	82	83	84	85	86	87	88	89	90
C	A	C	B	D	A	A	B	D	C
91	92	93	94	95	96	97	98	99	100
B	D	B	C	A	B	D	B	D	C

SUBJECT 12: FARM MANAGEMENT AND AGRICULTURAL ECONOMICS

Code: DA 241 | Total Questions: 100

Topic 1: Resources Production Function and Law of Variable Proportions

1. *Anything that* aids in production is called a
   a) product
   b) resource
   c) capital
   d) labour

2. Fixed resources remain unchanged irrespective of
   a) crop season
   b) market price
   c) government policy
   d) the level of production

3. Examples of fixed resources are land building and
   a) seeds
   b) fertilizers
   c) labour
   d) machinery

4. Variable resources vary with the level of production and examples include seeds fertilizers and
   a) land
   b) chemicals
   c) building
   d) machinery

5. *Flow resources* cannot be stored and should be used as and when available example is
   a) services of a labourer on a particular day
   b) seeds
   c) tractor
   d) land

6. *Stock resources* can be stored for later use example is
   a) labour services
   b) electricity
   c) seeds
   d) water flow

7. *Production function* is written as
   a) y = x + c
   b) y = x times c
   c) y = f(c) only
   d) y = f(x) where y is output and x is input

8. Short run production period is when one or more resources are fixed and written as
   a) Y=f(X1,X2,X3) all variable
   b) Y=constant
   c) Y=f(market price)
   d) Y=f(X1,X2/X3...Xn) some fixed

9. Long run production period is when
   a) all resources can be varied
   b) all resources are fixed
   c) only land is fixed
   d) only labour is fixed

10. Law of Variable Proportions is also known as
   a) Law of Proportionality
   b) Law of Returns to Scale
   c) Law of Demand
   d) Law of Supply

11. Total Product TP is the amount of product resulting from
   a) single fixed input only
   b) different quantities of variable input
   c) only labour input

14. When Total Product is maximum Marginal Product is
   a) maximum
   b) negative
   c) zero
   d) equal to average product

15. When Marginal Product is more than Average Product
   a) Average *Product decreases*
   b) Average *Product increases*
   c) Average Product is constant
   d) Total *Product decreases*

16. Stage II of production is the rational zone because
   a) MVP greater than MFC
   b) MPP is negative
   c) TPP is declining
   d) MVP equals MFC and profit is maximized

17. In Stage I of production elasticity of production is
   a) less than zero
   b) equal to zero
   c) greater than one
   d) less than one

18. In Stage III of production Marginal Physical Product is
   a) positive and increasing
   b) zero
   c) negative
   d) at maximum

19. Elasticity of Production Ep equals
   a) MPP divided by APP
   b) APP divided by MPP
   c) TPP divided by APP
   d) MPP times APP

20. Law of Increasing *Returns means* each successive unit of variable input adds
   a) less and less to total product
   b) equal amounts
   c) more and more to total product
   d) zero to total product

21. Law of Decreasing *Returns means* Marginal Physical Product is
   a) increasing
   b) constant
   c) declining
   d) zero always

22. Agricultural Production *Economics deals* with choice of production patterns and
   a) political decisions
   b) marketing only
   c) resource use to maximize objective within limited resources
   d) storage only

23. Production Economics is concerned with productivity which means use and incomes from productive inputs such as
   a) seeds only
   b) market prices only
   c) government subsidies
   d) land labour capital and management

24. *Transformation period* is the time required for a resource to be completely transformed into a

d) only capital input

12. Average Product AP is calculated as
a) Total *Product divided* by quantity of input
b) Total *Product divided* by price
c) Marginal *Product divided* by input
d) Total *Product plus* input

13. Marginal Product MP is
a) total output divided by total input
b) additional quantity of output from an additional unit of input
c) average output
d) minimum output

a) product
b) by-product
c) waste
d) raw material

25. Continuous production function arises for inputs that can be divided into smaller doses like
a) ploughings
b) weedings
c) seeds and fertilizers
d) harvestings

Topic 2: Farm Management Cost Concepts and Farm Records

26. Farm Management is the science of organizing and managing a farm for maximum
a) area coverage
b) continuous profit
c) crop diversity
d) labour employment

27. Fixed costs are costs that do not vary with the level of production example is
a) seed cost
b) fertilizer cost
c) rent of land and depreciation
d) labour wages

28. Variable costs vary with the level of production examples are
a) seed fertilizer labour and irrigation costs
b) land rent
c) building depreciation
d) machinery purchase cost

29. Total Cost TC equals
a) Fixed *Cost plus* Variable Cost
b) Fixed *Cost minus* Variable Cost
c) Variable *Cost only*
d) Fixed *Cost only*

30. Average Total Cost ATC equals
a) Total *Cost times* output
b) Marginal *Cost plus* Fixed Cost
c) Total *Cost divided* by output
d) Variable *Cost divided* by Fixed Cost

31. Marginal Cost MC is the additional cost incurred to produce
a) total output
b) average output
c) minimum output
d) one additional unit of output

32. Break-Even *Analysis determines* the level of output at which
a) profit is maximum
b) cost is minimum
c) total revenue equals total cost and profit is zero
d) loss is maximum

33. Farm records include Field Book Cash Book Stock *Register and*
a) weather records only
b) Crop Production *Record and* Labour Register
c) market price only
d) government scheme records

34. Balance Sheet is a statement showing
a) income only
b) assets and liabilities of a farm at a point in time

39. Farm planning involves deciding
a) only which crop to grow
b) what to produce how to produce and how much to produce for maximum profit
c) only when to harvest
d) only marketing

40. *Owned resources* in farming include land building machinery and
a) farmer family labour and capital
b) hired labour
c) rented land
d) custom hiring services

41. Depreciation is the decline in value of assets due to
a) inflation
b) market fluctuation
c) use and passage of time
d) crop failure

42. Net Farm *Income equals*
a) *Gross income* only
b) Total costs minus revenue
c) Variable cost only
d) Gross *Income minus* Total Costs

43. Gross Farm *Income includes* value of all farm products sold consumed and
a) none
b) donated
c) stored plus changes in inventory
d) wasted

44. Self Help Groups SHGs help farmers by promoting
a) individual loans only
b) collective savings credit and group marketing
c) only crop insurance
d) only seed distribution

45. Kisan Credit Card KCC scheme provides revolving credit to farmers for
a) agricultural inputs working capital and allied activities
b) only seeds
c) only irrigation
d) only machinery

46. Crop insurance protects farmers against losses due to
a) market price fall only
b) input shortage
c) natural calamities like drought flood and pests
d) labour shortage

47. Farm size classification in India: Small farm is
a) 1 to 2 hectares
b) below 0.5 ha

c) expenses only
d) profit only

35. *Opportunity cost* is the
a) cost of next best alternative forgone
b) actual cost paid
c) market price
d) selling price

36. *Economic efficiency* in farming means producing maximum output from
a) maximum inputs
b) given inputs at minimum cost
c) imported inputs
d) subsidized inputs

37. Diversification in farming means
a) specializing in one crop
b) using more fertilizers
c) growing multiple crops and enterprises to reduce risk
d) irrigating more area

38. Specialization in farming means
a) growing all crops
b) using all resources equally
c) concentrating on one or few crops where comparative advantage exists
d) diversifying always

c) above 10 ha
d) above 25 ha

48. *Marginal farm* in India has an area of
a) 1 to 2 ha
b) 2 to 4 ha
c) less than 1 hectare
d) above 5 ha

49. *Custom hiring* centre CHC provides
a) credit to farmers
b) seeds on loan
c) irrigation on hire
d) farm machinery on hire to small farmers at affordable cost

50. FPO Farmer Producer Organisation is owned and managed by
a) government only
b) banks only
c) NGOs only
d) farmers collectively for input purchase and marketing

Topic 3: Agricultural Marketing and Price Determination

51. Market is defined as any arrangement whereby buyers and sellers come into
a) conflict
b) contact for exchange of goods and services
c) competition
d) confusion

52. Agricultural marketing includes all activities from
a) only selling
b) only transport
c) production to final consumption including assembling processing storage transport and sale
d) only storage

53. Classification of markets based on time span includes spot market and
a) futures market
b) local market
c) village market
d) national market

54. *Perfect market* characteristics include many buyers and sellers homogeneous product perfect knowledge and
a) monopoly pricing
b) government control
c) single seller
d) free entry and exit

55. *Imperfect market* types include monopoly oligopoly and
a) perfect competition
b) free market
c) barter market
d) monopsony and monopolistic competition

56. *Marketing functions* include buying selling transportation storage grading and
a) nothing else
b) standardization packaging financing risk bearing and market information
c) only advertising

64. e-NAM National Agriculture Market is a
a) physical market
b) online trading platform linking mandis across India
c) bank
d) insurance company

65. *Regulated markets* APMC markets ensure
a) fair trade practices transparent weighing and proper facilities for farmers
b) free unregulated trade
c) monopoly by traders
d) no government intervention

66. *Warehousing provides* farmers the benefit of
a) free storage
b) transport
c) insurance
d) time utility allowing them to store and sell at better price

67. *Cold chain* infrastructure is essential for marketing of
a) cereals
b) fertilizers
c) perishable horticultural produce
d) equipment

68. *Price spread* is the difference between
a) input cost and output cost
b) wholesale and retail
c) two seasons
d) price paid by consumer and price received by producer

69. Margin of middlemen in agricultural marketing represents their profit for
a) no service
b) government tax
c) farmer loss
d) services rendered in marketing chain

70. *Cooperative marketing* societies help farmers by
a) buying their land

d) only distribution

57. Technical efficiency in marketing means
a) reducing losses during handling transport and storage
b) maximum profit
c) maximum price
d) minimum quality

58. *Economic efficiency* in marketing means
a) high transport cost
b) maximum production at minimum marketing cost
c) minimum output
d) maximum loss

59. *Price determination* under perfect competition is done by
a) government fixing price
b) single seller
c) intersection of demand and supply curves
d) single buyer

60. *Grading and* standardization in marketing ensures
a) uniform quality and enables better price
b) lower price
c) more wastage
d) less marketing

61. *Marketing risks* arise from price fluctuation quality deterioration and
a) uncertainty in quantity and timing of production
b) soil erosion
c) water shortage
d) input cost

62. Minimum Support Price MSP is announced by
a) State government only
b) Government of India based on CACP recommendations
c) RBI
d) NABARD

63. CACP stands for
a) Central Agriculture Credit Programme
b) Central Agricultural Cooperative Production
c) Crop Agriculture Control Policy
d) *Commission for* Agricultural *Costs and* Prices

b) collectively marketing produce to eliminate middlemen and get better price
c) providing only credit
d) providing only seeds

71. *Forward market* allows farmers to
a) sell produce after harvest only
b) borrow money
c) get insurance
d) sell at pre-determined future price reducing price risk

72. *Buffer stock* is maintained by government to
a) reduce production
b) increase market monopoly
c) reduce farmer income
d) stabilize prices by releasing stock when prices rise

73. Primary market in agricultural marketing is the
a) consumer market
b) export market
c) village level market where farmers first sell their produce
d) international market

74. *Wholesale market* assembles produce from
a) one village only
b) only from government
c) only exports
d) many primary markets and sells to retailers and processors

75. *Retail market* is where
a) farmers sell directly
b) wholesalers buy
c) consumers buy small quantities for final consumption
d) export buyers purchase

Topic 4: Loans Credit and Rural Development Schemes

76. Agricultural credit can be classified based on time into short-term medium-term and
a) seasonal credit
b) long-term credit
c) weekly credit
d) daily credit

77. Short-term agricultural credit is for a period of
a) more than 5 years
b) 10 to 15 years
c) up to 15 months for seasonal operations
d) 2 to 5 years

78. Medium-term agricultural credit is for
a) up to 15 months
b) 15 months to 5 years for equipment land improvement
c) more than 10 years
d) daily needs

79. Long-term credit in agriculture is for periods exceeding
a) 15 months
b) 1 year
c) 2 years
d) 5 years for land purchase irrigation wells and permanent improvements

89. Rashtriya Krishi Vikas Yojana RKVY incentivizes states to increase
a) food imports
b) public investment in agriculture and allied sectors
c) urban development
d) industrial output

90. National Food Security Mission NFSM aims to increase production of
a) rice wheat and pulses through area expansion and productivity enhancement
b) only oilseeds
c) only cotton
d) only sugarcane

91. Paramparagat Krishi Vikas Yojana PKVY promotes
a) organic farming in cluster mode
b) chemical farming
c) drip irrigation only
d) seed distribution

92. PM *Kisan scheme* provides annual income support of Rs 6000 to
a) urban workers
b) only small farmers

80. *Institutional sources* of agricultural credit include cooperative banks commercial banks and
a) moneylenders
b) Regional Rural *Banks and* NABARD
c) friends and relatives
d) shopkeepers

81. NABARD stands for
a) National Agriculture *Bank and* Rural Development
b) National Advisory *Bureau for* Agricultural *Research and* Development
c) National *Board for* Agri-Rural Districts
d) National *Bank for Agriculture and* Rural Development

82. Primary Agricultural Credit Societies PACS operate at
a) district level
b) state level
c) national level
d) village or primary level

83. District Central Cooperative Bank DCCB operates at
a) village level
b) district level
c) state level
d) national level

84. State Cooperative Bank SCB operates at
a) village level
b) district level
c) national level
d) state level providing credit to DCCBs

85. Problems in getting loans from institutions include lack of collateral security complex procedures and
a) low interest rates
b) illiteracy corruption and delays
c) easy access
d) no documentation

86. Non-institutional sources of credit include moneylenders traders landlords and
a) relatives and friends
b) NABARD
c) cooperative banks
d) commercial banks

87. *Moneylenders charge* very high interest rates and are known for
a) transparent dealing
b) low interest
c) exploitative practices and debt traps
d) quick service

88. PM Fasal Bima *Yojana provides* crop insurance with
a) no premium
b) very high premium
c) low premium for kharif and rabi crops
d) only for wheat

c) all landholder farmer families in three equal instalments
d) only marginal farmers

93. Soil Health *Card provides* farmers with information on
a) soil nutrient status and recommended doses of fertilizers for their specific soil
b) market prices
c) weather forecast
d) crop insurance

94. Sub-Mission on Agricultural Mechanization SMAM promotes
a) farm mechanization especially for small and marginal farmers through subsidies
b) manual farming
c) only large farms
d) only export farms

95. National Horticulture Mission NHM focuses on holistic growth of
a) only cereals
b) only oilseeds
c) only pulses
d) horticulture sector including fruits vegetables flowers and spices

96. *Drip and* sprinkler irrigation subsidy is provided under
a) MGNREGA
b) RKVY
c) PMKSY Pradhan Mantri Krishi Sinchayee Yojana
d) PKVY

97. Agri-clinics and Agri-Business Centres ACABC scheme is for
a) agricultural graduates to set up agri-enterprises with bank loan subsidy
b) government officers
c) NGOs only
d) scientists only

98. Gramin Bhandaran Yojana Rural Storage *Scheme provides* subsidy for
a) irrigation
b) construction of scientific storage godowns in rural areas
c) crop insurance
d) seed production

99. *Interest subvention* scheme provides short-term crop loans to farmers at reduced interest rate of
a) 12 percent
b) 7 percent per annum through banks
c) 3 percent
d) 15 percent

100. Agricultural debt waiver scheme announced in 2008 by government provided relief to farmers from
a) outstanding crop loans from scheduled commercial banks
b) input costs
c) irrigation charges
d) land revenue

ANSWER KEY — FARM MANAGEMENT AND AGRICULTURAL ECONOMICS

1	2	3	4	5	6	7	8	9	10
B	D	D	B	A	C	D	D	A	A
11	12	13	14	15	16	17	18	19	20
B	A	B	C	B	D	C	C	A	C
21	22	23	24	25	26	27	28	29	30
C	C	D	A	C	B	C	A	A	C
31	32	33	34	35	36	37	38	39	40
D	C	B	B	A	B	C	C	B	A
41	42	43	44	45	46	47	48	49	50
C	D	C	B	A	C	A	C	D	D

51	52	53	54	55	56	57	58	59	60
B	C	A	D	D	B	A	B	C	A
61	62	63	64	65	66	67	68	69	70
A	B	D	B	A	D	C	D	D	B
71	72	73	74	75	76	77	78	79	80
D	D	C	D	C	B	C	B	D	B
81	82	83	84	85	86	87	88	89	90
D	D	B	D	B	A	C	C	B	A
91	92	93	94	95	96	97	98	99	100
A	C	A	A	D	C	A	B	B	A

SUBJECT 13: LAND SURVEYING AND WATER ENGINEERING

Code: DA 252 | Total Questions: 100

Topic 1: Surveying Fundamentals and Classification

1. Surveying is the art of determination of horizontal distances differences in elevation directions angles and
   a) soil types
   b) locations areas and volumes on or near earth surface
   c) crop yields
   d) weather data

2. The primary object of survey is the preparation of
   a) a soil report
   b) a plan or map
   c) an agricultural report
   d) an irrigation schedule

3. *Geodetic surveying* is also called
   a) plane surveying
   b) topographic surveying
   c) trigonometrical surveying accounting for earth curvature
   d) agricultural surveying

4. *Plane surveying* considers the earth surface as a plane and does not account for
   a) curvature of the earth
   b) distances
   c) angles
   d) directions

5. Agricultural surveying is a simple plane surveying used for laying out contour lines terrace lines and
   a) geodetic triangulation
   b) drainage lines irrigation lines and computing field areas
   c) satellite surveys
   d) deep sea surveys

6. Classification of surveys based on instrument employed includes chain survey theodolite survey tacheometric survey compass survey and
   a) soil survey only
   b) rainfall survey
   c) plane table survey and photographic survey
   d) temperature survey

7. Land survey includes determination of
   a) only distances
   b) only elevations
   c) only water levels
   d) distances areas elevations and directions on land

8. *Marine surveys* are conducted for
   a) land measurement
   b) crop area estimation
   c) soil testing
   d) navigation purposes on water bodies

9. *Traverse survey* is a method where
   a) triangles are formed
   b) only distances are measured
   c) only heights are measured
   d) connected survey lines are measured with angles between them

10. Scale of a map is the fixed relation that every distance on map bears to
    a) corresponding distance on the ground
    b) the map size
    c) the elevation
    d) the area

14. *Contour line* connects points of
    a) different elevations
    b) same soil type
    c) same elevation above a datum
    d) same crop yield

15. *Contour interval* is the
    a) horizontal distance between contours
    b) vertical distance between successive contour lines
    c) length of survey line
    d) width of field

16. *Pacing technique* in surveying is used for
    a) precise measurement
    b) rough approximate distance measurement by counting steps
    c) theodolite work
    d) levelling

17. Chaining is the method of measuring distance with a chain or tape and is the most
    a) approximate
    b) expensive
    c) accurate and common method of distance measurement
    d) difficult

18. *Metric chain* IS 1492 is made in lengths of
    a) 10 and 20 meters
    b) 50 and 100 meters
    c) 20 and 30 meters
    d) 5 and 10 meters

19. *Gunters chain* is 66 feet long and divided into
    a) 50 links
    b) 150 links
    c) 200 links
    d) 100 links

20. *Engineers chain* is 100 feet long and divided into 100 links with each link equal to
    a) 0.66 ft
    b) 2 ft
    c) 0.5 ft
    d) 1 ft

21. *Invar tape* is made of alloy of steel 64 percent and nickel 36 percent with very low coefficient of
    a) thermal expansion
    b) electrical resistance
    c) magnetic permeability
    d) optical refraction

22. *Ranging rods* are used for
    a) measuring distance
    b) measuring height
    c) digging pits
    d) ranging lines and marking distant stations

23. Arrows or chain pins are used to
    a) measure angles
    b) set right angles
    c) level the ground
    d) mark the end of each chain length during chaining

24. *Plumb bob* is used to
    a) locate points directly below or above another point to check verticality

11. If one cm on map represents 5 m on ground the scale of map is
a) 1 to 50
b) 1 to 5
c) 5 m to 1 cm or 1 to 500
d) 1 to 5000

12. *Triangulation survey* is used for
a) small area surveys
b) field surveys only
c) building surveys only
d) precise determination of positions over large areas forming network of triangles

13. Reconnaissance is the
a) preliminary survey to collect general information about area
b) final detailed survey
c) soil sampling
d) weather data collection

b) measure rainfall
c) measure wind speed
d) level the instrument

25. Cross staff is used for
a) measuring distance
b) measuring height
c) chaining
d) finding foot of perpendicular and setting right angles on a survey line

Topic 2: Compass Survey and Levelling

26. Bearing of a line is the
a) length of the line
b) horizontal angle between the line and a reference direction
c) elevation of the line
d) area of the plot

27. *Whole circle* bearing WCB is measured from
a) magnetic south
b) east
c) west
d) north in clockwise direction from 0 to 360 degrees

28. *Quadrant bearing* QB system measures angles from
a) north or south towards east or west in four quadrants 0 to 90 degrees
b) north only
c) east to west
d) west to east

29. *Magnetic declination* is the angle between
a) true north and south
b) true meridian and magnetic meridian at a place
c) east and west
d) two survey lines

30. *Prismatic compass* gives readings in
a) quadrant bearing system
b) only rough directions
c) only cardinal directions
d) whole circle bearing WCB read at the prism

31. Local attraction causes errors in compass survey due to presence of
a) rainfall
b) magnetic materials like iron rails and electric cables near the compass
c) vegetation
d) soil type

32. Dip of magnetic needle is the angle between the horizontal plane and
a) true north
b) the direction in which freely suspended magnetic needle points
c) survey line
d) contour line

33. Levelling is the operation of determining difference in

39. Height of Instrument HI method in levelling calculates HI as
a) RL minus BS
b) RL of BM plus BS reading
c) FS minus BS
d) IS minus RL

40. *Rise and Fall method* checks arithmetic by comparing sum of rise and fall with
a) total distance
b) total readings
c) difference between first and last RL and sum of BS and FS
d) total area

41. *Contour map* is useful for
a) planning roads canals irrigation drainage and estimating earthwork
b) measuring soil pH
c) crop variety selection
d) fertilizer recommendation

42. *Profile levelling* determines elevations along a
a) area
b) line or route to design road or canal gradient
c) contour
d) boundary

43. Cross-section levelling is done at right angles to the
a) contour
b) boundary
c) survey line
d) centerline of a road or canal to design earthwork

44. *Fly levelling* is a rapid method of levelling done to
a) precise levelling
b) transfer benchmark or check elevation quickly over long distance
c) contour survey
d) area calculation

45. Land levelling for agriculture improves
a) uniform distribution of irrigation water and farm operations efficiency
b) soil pH
c) crop variety
d) seed rate

46. Grade in land levelling refers to
a) slope given to field for drainage usually 0.1 to 0.2 percent

a) horizontal distance
b) elevation between two or more points on the earth surface
c) bearing
d) magnetic north

34. Datum in levelling is the reference surface from which
a) all horizontal distances
b) all bearings
c) all elevations are measured usually mean sea level
d) all areas

35. Reduced Level RL of a point is its
a) height above the assumed datum level
b) horizontal distance from datum
c) magnetic bearing
d) horizontal angle

36. Bench Mark BM is a
a) fixed reference point of known elevation marked permanently
b) survey line
c) temporary station
d) traverse point

37. Backsight BS in levelling is the first staff reading taken on a
a) change point
b) foresight point
c) known point or bench mark to start levelling
d) turning point

38. Foresight FS in levelling is the last reading taken on a
a) bench mark
b) change point or closing point to end levelling
c) backsight point
d) intermediate point

b) soil texture
c) crop grade
d) fertilizer grade

47. *Cut and* fill in land levelling means
a) ploughing and harrowing
b) cutting crops and filling silos
c) cutting high spots and filling low spots to achieve uniform slope
d) nothing related to agriculture

48. *Laser land* leveller uses laser beam guided equipment to achieve
a) precise field levelling with high accuracy and efficiency
b) soil testing
c) contour ploughing
d) irrigation scheduling

49. Theodolite is used for measuring
a) rainfall
b) soil pH
c) wind speed
d) horizontal and vertical angles precisely in surveying

50. *Plane table* surveying plots the survey directly on paper in the field using
a) chain and tape
b) a drawing board mounted on tripod with alidade
c) theodolite only
d) compass only

Topic 3: Water Engineering Irrigation and Drainage

51. Watershed is an area of land that drains to a common
a) field
b) reservoir wall
c) outlet stream or point
d) road

52. *Watershed management* aims to conserve soil and water for
a) urban use only
b) industrial use
c) mining
d) sustainable agricultural production and reducing runoff and erosion

53. Runoff is the portion of rainfall that
a) infiltrates into soil
b) evaporates directly
c) is absorbed by crops
d) flows over land surface to streams

54. Infiltration is the process of water entering
a) soil through the soil surface
b) atmosphere
c) streams
d) plant roots from atmosphere

55. *Field channel* is used to convey irrigation water from distribution channel to
a) reservoir
b) river
c) canal head
d) individual fields

56. *Canal irrigation* conveys water by gravity from reservoirs through

64. Sub-surface drainage removes excess water from
a) surface only
b) below the soil surface using underground pipe drains or mole drains
c) atmosphere
d) reservoirs

65. *Water table* control is essential in irrigated areas to prevent
a) waterlogging and soil salinization
b) drought
c) wind erosion
d) crop burning

66. Soil erosion by water involves detachment transport and
a) infiltration
b) percolation
c) evaporation
d) deposition of soil particles

67. *Gully erosion* forms
a) sheet erosion only
b) only rill erosion
c) no visible damage
d) deep channels on sloping land caused by concentrated runoff

68. *Check dams* are built across gullies or streams to
a) slow runoff harvest water and reduce soil erosion
b) increase runoff
c) increase erosion
d) increase stream velocity

69. *Graded bunds* are earthen embankments constructed along
a) contour lines with zero slope

a) pipes under pressure
b) open channels to fields
c) drip pipes
d) sprinklers

57. *Tank irrigation* uses small surface water storage tanks which are common in
a) north India
b) peninsular India especially Telangana *Andhra and* Tamil Nadu
c) *Himalayan regions*
d) desert areas

58. *Lift irrigation* pumps water from
a) rivers tanks wells or reservoirs using mechanical pumps
b) rainfall directly
c) clouds
d) underground only

59. Evapotranspiration ET is the combined loss of water through
a) runoff and seepage
b) evaporation from soil surface and transpiration from plants
c) rainfall and irrigation
d) drainage and percolation

60. Crop water requirement CWR equals evapotranspiration minus
a) irrigation amount
b) effective rainfall
c) total rainfall
d) runoff

61. *Irrigation efficiency* is the ratio of water used by crop to
a) rainfall received
b) water stored in reservoir
c) water in river
d) total water applied at field inlet

62. *Conveyance efficiency* is the ratio of water delivered to field to
a) water evaporated
b) water stored in soil
c) crop water use
d) water diverted at the head of canal

63. *Surface drainage* removes excess surface water from fields through
a) open drains field channels and main drains
b) drip pipes
c) sprinklers
d) deep wells

b) field boundaries only
c) road sides only
d) contour with slight slope to lead runoff to outlets safely

70. Farm pond collects and stores runoff water for
a) household use only
b) supplemental irrigation during dry spells
c) cattle washing only
d) industrial use

71. *Water harvesting* structures include farm ponds check dams percolation tanks and
a) irrigation canals only
b) only wells
c) only reservoirs
d) nala bunds and contour trenches

72. *Percolation tank* is designed to
a) store water for direct use
b) supply domestic water
c) enhance groundwater recharge through percolation
d) irrigate directly

73. *Stream flow* measurement uses current meter or
a) weir and flume to measure discharge
b) rain gauge
c) barometer
d) thermometer

74. *Drip irrigation* has water use efficiency of about
a) 40 to 50 percent
b) 20 to 30 percent
c) 80 to 95 percent
d) 60 to 70 percent

75. *Sprinkler irrigation* has water use efficiency of about
a) 70 to 80 percent
b) 30 to 40 percent
c) 90 to 95 percent
d) 20 to 30 percent

Topic 4: Greenhouse Technology and Farm Structures

76. Greenhouse is a structure covered with transparent material allowing
a) only rain to enter
b) sunlight to pass through while controlling inside temperature humidity and CO_2
c) only wind to pass
d) only insects to enter

77. Main advantage of greenhouse cultivation is
a) year-round production of crops regardless of outside climate
b) lower cost always
c) no need for inputs
d) no irrigation needed

89. *Mole drainage* is created by pulling a solid cylinder through soil to form
a) visible open drains
b) concrete pipes
c) unlined underground drainage channels in clay soil
d) plastic pipes

90. Terrace is a broad nearly level bench constructed along
a) hill slope to reduce slope length and control erosion
b) valley bottom
c) flat land
d) paddy field boundary

91. *Bench terracing* converts steep slope into series of
a) gullies

78. *Materials used* for greenhouse covering include glass polythene film and
a) fiberglass reinforced plastic FRP and polycarbonate sheets
b) cement
c) metal sheets
d) wood

79. Natural ventilated greenhouse uses
a) fans and blowers
b) air conditioners
c) refrigeration
d) ridge and sidewall vents to allow natural air movement

80. *Evaporative cooling* pad and fan system in greenhouse reduces inside temperature by
a) heating
b) compression
c) evaporative cooling through wet pads with outside air drawn in by fans
d) radiation

81. *Hydroponic cultivation* grows plants without
a) water
b) soil using nutrient solution in water
c) light
d) temperature control

82. *Drip irrigation* is the most suitable irrigation method for
a) flooded paddy
b) greenhouse crops and drip-fertigated vegetables
c) broadcasting crops
d) forage crops

83. *Shade net* house provides
a) full sunlight exposure
b) partial shading reducing light and temperature for shade loving plants
c) heating
d) no climate control

84. Farm road should be
a) very wide and expensive
b) temporary only
c) hard surfaced with proper gradient and drainage for all-weather use
d) unpaved always

85. Farm building site selection should consider
a) only cost
b) only appearance
c) topography drainage wind direction and proximity to water and fields
d) only size

86. Orientation of farm buildings in India should preferably be
a) east-west for long axis to reduce heat gain
b) any direction
c) north-south for long axis to reduce heat gain
d) facing only west

87. Roof of farm building should be designed to
a) retain heat
b) increase heat inside
c) shed rainwater and provide insulation with adequate slope
d) be waterproof only

88. Silo is a structure used for
a) storing silage which is fermented green fodder
b) storing grain in bags
c) storing seeds in cold storage
d) storing machinery

b) nearly level steps reducing erosion and enabling cultivation
c) ravines
d) drainage channels

92. Broad base terrace is suitable for
a) very steep slopes only
b) very flat land
c) waterlogged areas
d) gentle to moderate slopes allowing machinery operation

93. Agrostological or vegetation measures to control erosion include
a) concrete structures only
b) deep ploughing only
c) grass waterways contour hedgerows and ground cover crops
d) deep drainage only

94. *Wind erosion* is controlled by
a) increasing wind speed
b) removing all vegetation
c) deep tillage
d) shelter belts windbreaks and stubble mulching

95. *Gabion structure* is a wire mesh basket filled with
a) stones or gravel placed across gullies to check erosion
b) sand only
c) concrete
d) soil

96. *Irrigation scheduling* means deciding when and how much to irrigate based on
a) convenience only
b) government order
c) soil moisture crop water demand and weather
d) market price

97. *Tensiometer measures*
a) soil pH
b) wind speed
c) rainfall
d) soil moisture tension to indicate when to irrigate

98. Gypsum is applied to reclaim
a) sodic alkali soils by replacing excess sodium with calcium
b) acidic soils
c) sandy soils
d) waterlogged soils

99. *Saline soils* have high concentration of soluble salts and are reclaimed by
a) adding more salt
b) adding gypsum
c) deep ploughing only
d) leaching with good quality water and providing drainage

100. *Acid soils* are reclaimed by application of
a) sulphur
b) gypsum
c) salt
d) lime calcium carbonate or dolomite to raise pH

ANSWER KEY — LAND SURVEYING AND WATER ENGINEERING

1	2	3	4	5	6	7	8	9	10
B	B	C	A	B	C	D	D	D	A
11	12	13	14	15	16	17	18	19	20
C	D	A	C	B	B	C	C	D	D
21	22	23	24	25	26	27	28	29	30
A	D	D	A	D	B	D	A	B	D
31	32	33	34	35	36	37	38	39	40
B	B	B	C	A	A	C	B	B	C
41	42	43	44	45	46	47	48	49	50
A	B	D	B	A	A	C	A	D	B
51	52	53	54	55	56	57	58	59	60
C	D	D	A	D	B	B	A	B	B
61	62	63	64	65	66	67	68	69	70
D	D	A	B	A	D	D	A	D	B
71	72	73	74	75	76	77	78	79	80
D	C	A	C	A	B	A	A	D	C
81	82	83	84	85	86	87	88	89	90
B	B	B	C	C	C	C	A	C	A
91	92	93	94	95	96	97	98	99	100
B	D	C	D	A	C	D	A	D	D

SUBJECT 14: COMPUTER APPLICATIONS IN AGRICULTURE

Code: DA 262 | Total Questions: 100

Topic 1: Introduction to Computers and Generations

1. A computer is an electronic device that can perform mathematical logical and
   a) biological operations
   b) weather forecasting only
   c) soil testing only
   d) graphical manipulations following program instructions

2. Data is defined as
   a) collection of raw facts figures and symbols
   b) organized information
   c) processed results
   d) computer programs

3. Information is data that has been
   a) collected randomly
   b) processed and presented in organized and meaningful manner
   c) stored as raw facts
   d) entered into computer

4. A program is a set of instructions that enables a computer to perform a given
   a) weather forecast
   b) agricultural activity manually
   c) task or operation
   d) soil test

5. First generation computers 1946 to 1954 used
   a) vacuum tubes
   b) transistors
   c) integrated circuits
   d) microprocessors

6. Second generation computers 1955 to 1965 used
   a) transistors
   b) vacuum tubes
   c) integrated circuits VLSI
   d) microprocessors

7. Third generation computers 1968 to 1975 used
   a) vacuum tubes
   b) transistors
   c) Integrated Circuits IC
   d) VLSI circuits

8. Fourth generation computers 1976 to 1980 used
   a) Very Large Scale Integrated Circuits VLSI
   b) vacuum tubes
   c) transistors
   d) IC

9. Fifth generation computers use
   a) vacuum tubes
   b) transistors
   c) Ultra Scale Integrated Circuits ULSI and *Silicon chips* with AI
   d) IC

10. Advantages of computers include high speed accuracy storage automation and
   a) high cost always
   b) inability to store data
   c) diligence versatility and cost effectiveness
   d) slow processing

11. Limitation of computers is that they cannot

14. 1024 bytes equals
   a) 1 Megabyte
   b) 1 Gigabyte
   c) 1 Kilobyte KB
   d) 1 Terabyte

15. 1024 KB equals
   a) 1 Megabyte MB
   b) 1 Kilobyte
   c) 1 Gigabyte
   d) 1 Terabyte

16. ASCII stands for
   a) Agricultural Standard *Code for* Information Interchange
   b) Applied Standard Computer Interface
   c) American Standard *Code for* Information Interchange
   d) Automated Standard *Code for* Information Input

17. *Analog computers* operate by
   a) counting discrete numbers
   b) measuring continuously varying quantities
   c) using binary digits
   d) running programs

18. *Digital computers* operate by
   a) measuring analog signals
   b) analog circuits
   c) vacuum tubes only
   d) counting using discrete digits or numbers

19. Hybrid computers combine features of
   a) analog and digital computers
   b) only digital computers
   c) only mainframe computers
   d) only supercomputers

20. *Super computers* have extremely large storage and are used for
   a) large scale numerical problems like weather forecasting and nuclear simulation
   b) home use
   c) office work
   d) small calculations

21. PARAM is India's indigenous
   a) satellite
   b) nuclear reactor
   c) agricultural robot
   d) supercomputer

22. *Micro computers* or personal computers PC are the smallest general purpose systems designed for use by
   a) large organizations only
   b) only scientists
   c) only government
   d) one person at a time

23. PC-XT means Personal *Computer with*
   a) Extended Technology
   b) Expanded Technology
   c) Extra Technology
   d) External Technology

24. The first microprocessor in *Intel series* was
   a) Pentium
   b) 8088
   c) 80286

a) think learn by experience or work without clear instructions
b) store data
c) perform arithmetic
d) process data

12. *Computer stores* information using binary digits which are
a) decimal 0 to 9
b) hexadecimal 0 to F
c) binary 0 and 1
d) octal 0 to 7

13. 8 bits equals
a) 1 byte
b) 1 kilobyte
c) 1 megabyte
d) 1 nibble

d) 8080 introduced in 1974

25. *Clock speed* of computer processor is measured in
a) Megahertz MHz or Gigahertz GHz
b) *Bytes per* second
c) Kilobytes
d) Pixels

Topic 2: Computer Hardware Input Output and Memory

26. *Hardware refers* to the
a) software programs
b) data stored in computer
c) internet connection
d) physical components of computer that can be touched

27. CPU stands for
a) Central Processing *Unit which* is the brain of computer
b) Central Power Unit
c) Computer Processing Unit
d) Central Program Unit

28. *Three main* parts of CPU are Memory Unit Arithmetic Logic Unit ALU and
a) Monitor
b) Keyboard
c) Control Unit
d) Printer

29. ALU performs
a) memory storage
b) input operations
c) arithmetic operations and logical functions like true or false
d) output display

30. Control *Unit acts* as central nervous system ensuring
a) program instructions are followed in proper sequence and coordinates all devices
b) only arithmetic
c) only output
d) only input

31. Keyboard is the standard input device with most common keyboards having
a) 50 keys
b) 104 keys
c) 200 keys
d) 26 keys

32. *Mouse controls* the movement of cursor on monitor and has buttons and a
a) light beam only
b) scroll wheel between buttons
c) built-in speaker
d) built-in camera

33. *Scanner reads* text or illustrations printed on paper and translates into
a) printed output
b) audio

39. *Plotter produces* drawings or graphs through pens and is used for
a) technical drawings maps and engineering plans
b) text printing
c) scanning documents
d) data entry

40. RAM Random Access Memory is also known as read write memory and its contents are
a) temporary volatile lost when power is off
b) permanent non-volatile
c) read only
d) stored permanently

41. ROM Read Only *Memory contains* built-in instructions by manufacturer and is
a) non-volatile permanent memory
b) volatile lost when power off
c) temporary
d) erasable easily

42. Secondary memory is also called
a) main memory
b) auxiliary or backup storage used for large permanent data storage
c) cache memory
d) register

43. *Floppy disk* is made of flexible vinyl material and high density 3.5 inch floppy stores
a) 1.44 MB
b) 360 KB
c) 700 MB
d) 4.7 GB

44. Hard disk capacity ranges from
a) 1 KB to 1 MB
b) 100 bytes to 1 KB
c) 1 byte to 100 bytes
d) 20 MB to several Gigabytes

45. CD-ROM stands for Compact Disk Read Only *Memory with* storage capacity of approximately
a) 650 to 700 MB
b) 1.44 MB
c) 4.7 GB
d) 1 GB

46. DVD Digital Versatile *Disc stores* at least
a) 1.44 MB
b) 700 MB

c) video
d) digital form

34. MICR Magnetic Ink Character Recognition is widely used in
a) supermarkets
b) banks to process cheques
c) hospitals
d) schools

35. OMR Optical Mark *Recognition senses* presence or absence of marks and is used in
a) aptitude tests and answer sheets
b) printing
c) banking
d) navigation

36. Monitor is also called VDU which stands for
a) Virtual Display Utility
b) Visual Display Unit
c) Video Data Unit
d) Virtual Data Uplink

37. *Dot matrix* printer prints characters in dotted pattern through printer ribbon and its speed is
a) 6 to 12 PPM
b) 300 to 600 LPM
c) 200 to 700 characters per second CPS
d) 1 page per minute

38. *Laser printer* uses laser beam to produce images and its speed is
a) 6 to 12 pages per minute PPM
b) 200 to 700 CPS
c) 300 to 600 LPM
d) 90 CPS

c) 4.7 GB
d) 100 GB

47. *Flash drive* or pen drive connects to computer via
a) serial port
b) parallel port
c) USB port and stores data in flash memory
d) audio port

48. *Cache memory* is a small high-speed memory between CPU and RAM that stores
a) all data permanently
b) frequently used data for faster access
c) only programs
d) only output

49. *Input devices* receive data and send to computer while output devices
a) also receive data
b) receive processed information from CPU and present to user
c) store data
d) process data

50. Webcam is an input device used for
a) printing
b) scanning documents
c) capturing video and images for input to computer
d) measuring temperature

Topic 3: Software Operating Systems and Programming

51. Software is a program or set of instructions that causes hardware to function in a desired way and the difference from hardware is that software cannot be
a) used
b) programmed
c) upgraded
d) touched or physically seen unlike hardware

52. *Five categories* of software are Operating System Translators Utility Programs Application *Programs and*
a) Input Programs
b) Output Programs
c) General Purpose Packages
d) Hardware Programs

53. Operating *System manages* resources of computer system and acts as interface between
a) only programs
b) only user and internet
c) hardware and user programs facilitating execution of programs
d) only memory and disk

54. DOS Disk Operating *System was* developed by Bill Gates in 1980 and is a
a) multi-user multi-task system
b) single user single task operating system for personal computers
c) server operating system
d) network operating system

55. *Windows operating* system supports

64. FORTRAN stands for
a) Formula *Translation used* for scientific calculations
b) Forward *Transfer and* Navigation
c) File *Organization and* Table Range Analysis
d) Function *Operations and* Terminal Range Networking

65. COBOL stands for
a) Computer Oriented Business Output Language
b) Common Business Oriented *Language used* for business applications
c) Coded Binary *Operations and* Logic
d) Collective Business Oriented Library

66. BASIC stands for
a) Binary Arithmetic Standard Input Code
b) Basic *Arithmetic and* Scientific Input Computer
c) Beginner's All-purpose Symbolic Instruction Code
d) Batch Automated Standard Interface Code

67. *Utility programs* are pre-written programs supplied by manufacturer for maintaining day to day activities examples are
a) payroll programs
b) COPY SORT MAILING and virus scanning software
c) billing programs
d) reservation programs

68. *Application programs* are written for specific jobs examples include
a) COPY and SORT utilities
b) FORTRAN compilers
c) payroll billing and railway reservation programs
d) DOS commands

a) single user single task
b) single user multitask system with graphical user interface GUI
c) only command line
d) only servers

56. UNIX is suited for
a) single user only
b) personal computers only
c) DOS only
d) multi-user multi-task systems

57. CUI Character User Interface is operated with
a) only mouse
b) only touchscreen
c) keyboard only example MS-DOS
d) only joystick

58. GUI Graphical User *Interface can* be operated with mouse and keyboard example is
a) MS-DOS
b) Windows 95 Windows XP
c) UNIX text mode
d) command prompt

59. *Compiler translates*
a) the entire source program into machine language before execution
b) one statement at a time
c) only output to input
d) graphics to text

60. *Interpreter translates* and executes
a) the whole program at once
b) only mathematical expressions
c) only graphics
d) one statement at a time stopping at errors

61. *Machine language* uses
a) binary codes 0 and 1 directly understood by computer
b) *English like* statements
c) SQL commands
d) high level syntax

62. *Assembly language* uses mnemonic codes instead of binary and example is
a) FORTRAN
b) ADD or A symbol for addition which is machine dependent
c) COBOL
d) BASIC

63. High Level Language HLL is machine independent and follows rules like *English language* examples include
a) FORTRAN COBOL BASIC and C
b) machine language
c) assembly language
d) binary codes

69. SPSS stands for Statistical *Package for* Social *Science used* for
a) data analysis and statistics
b) word processing
c) spreadsheet
d) database management

70. MS-Word is an example of
a) spreadsheet software
b) database software
c) word processing software
d) graphics software

71. MS-Excel is an example of
a) word processor
b) spreadsheet software for tabular calculations and charts
c) database
d) presentation software

72. MS-PowerPoint is used for creating
a) databases
b) presentations with slides and graphics
c) spreadsheets
d) word documents

73. MS-Access is an example of
a) database management software
b) word processing software
c) spreadsheet
d) presentation software

74. Internet is a global network connecting millions of computers enabling sharing of
a) only emails
b) only files
c) only hardware
d) information resources and services worldwide

75. *Computer applications* in agriculture include soil testing farm management data analysis crop simulation and
a) GIS remote sensing and precision farming
b) no agricultural applications
c) only word processing
d) only email

Topic 4: Internet Database and Agricultural Applications

76. WWW World Wide Web is a system of interlinked hypertext documents accessed via
a) internet using web browser
b) television
c) radio
d) telephone only

77. URL Uniform Resource Locator is the
a) computer speed
b) address of a resource on the internet

89. mKisan portal provides agricultural advisories through
a) SMS to farmers mobile phones
b) television only
c) radio only
d) newspaper

90. *Agrimarket app* and commodity market websites help farmers access
a) weather forecast only
b) soil information

c) password of website
d) file name

78. Email Electronic *Mail allows* sending and receiving messages through
a) postal service
b) telephone call
c) fax only
d) internet almost instantly

79. *Search engine* is a website that helps users find information by
a) storing all websites
b) creating websites
c) searching and indexing web pages based on keywords example Google
d) deleting websites

80. Browser is software used to
a) create websites
b) design databases
c) access and navigate websites on internet example Chrome Firefox
d) write programs

81. GIS Geographic Information *System integrates* geographical data with
a) only maps
b) only GPS
c) attribute data for spatial analysis mapping and decision support
d) only remote sensing

82. *Remote sensing* uses satellite or aerial sensors to collect information about
a) soil pH in lab
b) weather in office
c) earth surface without direct contact for crop monitoring and land use
d) market prices

83. GPS Global Positioning *System provides* precise
a) soil type information
b) rainfall data
c) temperature data
d) location coordinates using signals from satellites

84. *Precision agriculture* uses GPS GIS and remote sensing to
a) apply variable rate inputs based on field variability reducing waste
b) apply uniform inputs everywhere
c) avoid all technology
d) do manual farming

85. Database is an organized collection of
a) programs
b) hardware components
c) data stored and managed for easy retrieval and manipulation
d) network cables

86. DBMS Database Management *System software* manages databases examples include
a) MS-Word
b) dBASE Oracle MS-*Access and* MySQL
c) MS-*Excel only*
d) MS-PowerPoint

87. SQL Structured Query Language is used for
a) querying and managing relational databases
b) creating presentations
c) writing letters
d) creating spreadsheets

88. Kisan Call *Centre number* 1800-180-1551 provides farmers

c) current market prices of agricultural commodities
d) crop diseases only

91. e-Krishi *Kosh national* crop variety and germplasm database helps in accessing information on
a) released varieties germplasm and genetic resources
b) market prices
c) weather
d) soil maps

92. *Drone technology* in agriculture is used for
a) only photography
b) only transport
c) only irrigation
d) crop monitoring spraying fertilizers and pesticides and yield estimation

93. *Artificial intelligence* AI in agriculture is used for
a) crop disease detection yield prediction and advisory systems through machine learning
b) manual data entry only
c) only hardware control
d) only word processing

94. IoT Internet of Things in agriculture connects sensors and devices for
a) entertainment
b) real-time monitoring of soil moisture temperature and crop conditions
c) only weather
d) only storage

95. *Computer simulation* models in agriculture are used to
a) manually calculate yields
b) create websites
c) manage accounts only
d) simulate crop growth under different conditions for planning and decision making

96. *Cloud computing* allows agricultural data to be
a) stored only locally
b) printed only
c) stored in floppy only
d) stored and accessed from anywhere using internet-connected devices

97. FASAL Forecasting Agricultural output using Space *Agrometeorological and* Land based observations is a programme for
a) soil testing
b) market intelligence
c) irrigation scheduling
d) crop yield forecasting using remote sensing and weather data

98. *Digital soil* mapping uses laboratory soil data and remote sensing to produce
a) physical soil maps manually
b) only field boundaries
c) only topographic maps
d) continuous spatial prediction maps of soil properties

99. Virus in computer terminology is a
a) plant pathogen
b) hardware defect
c) malicious program that can damage data and spread to other computers
d) internet error

100. *Antivirus software* protects computer from
a) hardware damage
b) malicious programs viruses worms and trojans
c) power failure
d) internet disconnection

a) free seeds
b) free insurance
c) toll-free agricultural advisory service in local languages
d) free credit

ANSWER KEY — COMPUTER APPLICATIONS IN AGRICULTURE

1	2	3	4	5	6	7	8	9	10
D	A	B	C	A	A	C	A	C	C
11	12	13	14	15	16	17	18	19	20
A	C	A	C	A	C	B	D	A	A
21	22	23	24	25	26	27	28	29	30
D	D	A	D	A	D	A	C	C	A
31	32	33	34	35	36	37	38	39	40
B	B	D	B	A	B	C	A	A	A
41	42	43	44	45	46	47	48	49	50
A	B	A	D	A	C	C	B	B	C
51	52	53	54	55	56	57	58	59	60
D	C	C	B	B	D	C	B	A	D
61	62	63	64	65	66	67	68	69	70
A	B	A	A	B	C	B	C	A	C
71	72	73	74	75	76	77	78	79	80
B	B	A	D	A	A	B	D	C	C
81	82	83	84	85	86	87	88	89	90
C	C	D	A	C	B	A	C	A	C
91	92	93	94	95	96	97	98	99	100
A	D	A	B	D	D	D	D	C	B

SUBJECT 15: COMMUNICATION SKILLS IN ENGLISH

Code: DA 263 | Total Questions: 100

Topic 1: Parts of Speech – Noun Pronoun and Verb

1. Parts of speech in *English are* classified into how many categories
   a) 5
   b) 10
   c) 12
   d) 8

2. Noun is any word that names a person place thing quality or
   a) action only
   b) idea event or feeling
   c) movement only
   d) colour only

3. *Proper nouns* are names of specific persons places days months and
   a) common things
   b) colours
   c) actions
   d) festivals languages and events and begin with capital letter

4. Common nouns are names given to all persons or things of the
   a) specific individual
   b) different groups
   c) same class or kind
   d) opposite class

5. *Material nouns* are names of substances that cannot be divided into separate objects like
   a) boy and girl
   b) table and chair
   c) city and country
   d) rice sugar sand water and gold

6. *Collective nouns* are names of groups like committee army team audience and
   a) rice and water
   b) flock herd gang and crew
   c) joy and anger
   d) walk and run

7. *Abstract nouns* are names of qualities states or ideas that cannot be
   a) written or spoken
   b) counted
   c) measured in kg
   d) seen or touched like goodness bravery love and hatred

8. *Count nouns* can be counted and have both singular and plural forms example is
   a) water
   b) milk
   c) gold
   d) boy boys pen pens

9. *Uncount nouns* cannot be counted and have no plural form examples are
   a) book books
   b) pen pens
   c) table tables
   d) water milk advice information and furniture

10. Pronoun is a word used in place of a noun to avoid repetition examples include
   a) book pen chair
   b) he she it they we you I and me
   c) run walk jump

14. *Transitive verb* requires a direct object to complete its meaning example in
   a) He sleeps
   b) *She reads* a book where book is the object
   c) It rains
   d) The sun shines

15. *Intransitive verb* does not require a direct object example is
   a) The baby sleeps which needs no object
   b) *She eats* rice
   c) He reads a book
   d) I bought a pen

16. Adjective is a word that describes or modifies a noun examples are
   a) beautiful tall intelligent red and heavy
   b) quickly slowly
   c) run eat sleep
   d) he she they

17. *Adverb modifies* a verb adjective or another adverb examples are
   a) quickly slowly very beautifully and quite
   b) tall red heavy
   c) he she it
   d) book pen table

18. *Preposition shows* relationship between a noun or pronoun and other words in sentence examples are
   a) and but or
   b) on in at by from under over through and between
   c) he she we
   d) run walk eat

19. *Conjunction joins* words phrases or clauses examples are
   a) quickly slowly
   b) and but or so because although when and if
   c) on in at
   d) he she they

20. Interjection is a word or phrase expressing sudden emotion or exclamation examples are
   a) on in at by
   b) and but or
   c) he she they
   d) Oh Ah Alas *Hurrah and* Bravo

21. Articles in *English are* A An and
   a) By
   b) On
   c) In
   d) The

22. *Indefinite article* A is used before words beginning with consonant sound and An before words beginning with
   a) consonant sound
   b) any sound
   c) vowel sound a e i o u
   d) only E

23. *Definite article* The is used before
   a) any noun first mention
   b) all nouns always
   c) no noun
   d) specific or particular nouns already known to both speaker and listener

d) slowly quickly

11. *Personal pronouns* in subject position are I you he she it we and
   a) me him her
   b) us them
   c) my his her
   d) they

12. Verb is a word that expresses an action state or occurrence and is the
   a) least important word
   b) most essential part of every sentence
   c) optional part
   d) describing word

13. *Auxiliary verbs* or helping verbs include am is are was were can could may might have has had will would should and
   a) run walk eat
   b) happy sad good
   c) quickly slowly
   d) shall must do did does

24. *Proper nouns* names of specific persons places do not usually take article A or An but may take
   a) An always
   b) A always
   c) The in specific contexts like The Ganges The Himalayas
   d) no article ever

25. Which sentence uses correct article usage
   a) She is an honest woman which correctly uses An before H with vowel sound
   b) He is a honest man
   c) I saw the elephant at zoo
   d) Give me a water

Topic 2: Tenses Voices and Sentence Structures

26. *Tense shows* the time of action in a sentence and *English has* how many main tenses
   a) 3 – Present *Past and* Future
   b) 2
   c) 5
   d) 8

27. Simple *Present tense* is used for habitual actions general truths and
   a) actions completed in past
   b) past completed actions
   c) future actions only
   d) scheduled future events and permanent facts

28. Simple *Present tense* form for third person singular adds
   a) ed to verb
   b) ing to verb
   c) s or es to the base verb
   d) will before verb

29. Present Continuous tense is formed with is am or are plus
   a) past participle
   b) base verb
   c) verb plus ing indicating action happening right now
   d) had plus verb

30. Present *Perfect tense* uses has or have plus
   a) past participle indicating action completed with present relevance
   b) base verb
   c) ing form
   d) will plus verb

31. Simple *Past tense* uses
   a) will plus verb
   b) is plus ing
   c) past form of verb ed for regular verbs or irregular past form
   d) has plus past participle

32. Past Continuous tense uses was or were plus
   a) base verb
   b) past participle
   c) verb ing indicating action in progress at a past time
   d) will plus verb

33. Past *Perfect tense* uses had plus

39. In Indirect or *Reported speech* the tense usually shifts one step back meaning *Present becomes*
   a) Past Simple
   b) *Present perfect*
   c) Future
   d) Present Continuous

40. *Question tags* are short questions added at the end of statements to seek confirmation example She is kind
   a) is she not
   b) isn't she – using auxiliary and opposite positive or negative
   c) she is
   d) isn't he

41. *Conditional sentences* with If have different types: *Zero conditional* uses simple present in both clauses for
   a) hypothetical situations
   b) general truths and scientific facts
   c) past events
   d) impossible situations

42. First conditional uses if plus simple present and will plus base verb for
   a) impossible situations
   b) past habits
   c) real or possible future situations
   d) general truths

43. Second conditional uses if plus past simple and would plus base verb for
   a) real future situations
   b) past facts
   c) general truths
   d) hypothetical or unlikely present or future situations

44. Third conditional uses if plus past perfect and would have plus past participle for
   a) real future events
   b) present habits
   c) unreal or hypothetical past situations and their results
   d) future plans

45. *Synthesis combining* sentences can be done using relative clauses for example two sentences about a man who is tall become
   a) the man is tall the man is kind

a) base verb
b) ing form
c) will plus verb
d) past participle indicating action completed before another past action

34. Simple *Future tense* uses will or shall plus
a) base verb indicating future action
b) past form
c) ing form
d) past participle

35. Active voice is when the subject performs the action example
a) The book was read by him
b) Mango is eaten by me
c) Rice is grown by farmers
d) He reads the book – subject He does the action

36. Passive voice is when the subject receives the action and is formed with appropriate form of be plus
a) past participle of main verb
b) base verb
c) ing form
d) will plus verb

37. To convert active to passive the object of active sentence becomes
a) the predicate
b) the verb
c) the subject of passive sentence
d) the adjective

38. In Direct speech the exact words of the speaker are reported inside
a) brackets
b) quotation marks
c) parentheses
d) dashes

b) The man who is tall is also kind – using relative pronoun who
c) tall man kind
d) no combination possible

46. *Punctuation mark* full stop period is used at end of
a) questions
b) exclamations
c) lists only
d) declarative sentences and after abbreviations

47. Comma is used to separate items in a list introduce a clause or
a) end a sentence
b) replace a period
c) separate coordinate clauses joined by conjunctions
d) start a paragraph

48. *Question mark* is placed at end of
a) direct questions and interrogative sentences
b) statements
c) exclamations
d) commands

49. Apostrophe is used to show
a) list items
b) question
c) possession Ram's book and contractions like don't won't can't
d) exclamation

50. *Exclamation mark* is used after
a) ordinary statements
b) interjections and exclamatory sentences showing strong emotion
c) questions
d) lists

Topic 3: Vocabulary and Communication Skills

51. *Synonyms are* words that have similar or the same meaning as another word example synonyms of happy are
a) joyful cheerful glad and pleased
b) sad unhappy
c) angry tired
d) quick fast

52. *Antonyms are* words that have opposite meaning example antonym of hot is
a) warm
b) cold
c) cool
d) very hot

53. *Homophones are* words that sound the same but have different spelling and meaning example
a) big and large
b) there their and they are – all sound like ther
c) fast and quick
d) happy and glad

54. *Homonyms are* words that have same spelling and pronunciation but different meanings example
a) to two and too
b) there and their
c) fast and quick
d) bank can mean river bank or financial institution

64. *Letter writing* in *English has* two main types *Formal and*
a) *Official only*
b) *Legal only*
c) Informal or *Personal letters*
d) *Government only*

65. *Formal letter* begins with the sender's address date receiver's address salutation and ends with
a) casual greeting
b) formal complimentary close like *Yours faithfully* or *Sincerely followed* by signature
c) nickname
d) abbreviation

66. Curriculum Vitae CV or Resume is a document that summarizes
a) personal details education work experience skills and achievements for job application
b) only educational qualifications
c) only hobbies
d) only references

67. Types of reading include skimming which is reading for
a) detailed understanding
b) overall gist or main idea quickly
c) critical analysis
d) word by word study

68. Scanning is reading to find

55. Syllable is a unit of pronunciation with one vowel sound and knowing syllables helps in correct
- a) stress and pronunciation of words
- b) grammar
- c) sentence formation
- d) article usage

56. *Word stress* means emphasis placed on a particular syllable example in the word *Agriculture stress* is on
- a) second syllable Ag-RI-cul-ture
- b) last syllable
- c) first syllable only
- d) every syllable equally

57. Communication is the process of conveying meaning from sender to receiver through
- a) silence only
- b) verbal non-verbal and written channels
- c) thought only
- d) action only

58. *Verbal communication* uses
- a) gestures only
- b) facial expressions only
- c) spoken words and language
- d) written text only

59. Non-verbal communication uses
- a) spoken words
- b) body language gestures facial expressions eye contact and posture
- c) written words only
- d) telephone only

60. Listening is an active process of receiving understanding and
- a) talking
- b) writing
- c) reading
- d) interpreting spoken messages which is different from simply hearing

61. *Reading comprehension* involves understanding the meaning of written text beyond just
- a) meaning
- b) vocabulary only
- c) grammar only
- d) word recognition by identifying main ideas inferences and relationships

62. *Precis writing* is the art of summarizing a passage in
- a) longer form
- b) same length
- c) about one third its original length in clear concise language
- d) question form

63. *Report writing* presents facts information and findings in a structured format with
- a) clear objective language headings subheadings and conclusions
- b) fictional content
- c) only opinions
- d) story format

- a) overall meaning
- b) specific information or details quickly
- c) enjoyment only
- d) word meaning only

69. *Extensive reading* is reading
- a) word by word
- b) only grammar books
- c) widely for general understanding enjoyment or information
- d) only dictionaries

70. *Intensive reading* is reading
- a) carefully for detailed understanding and analysis
- b) quickly for gist
- c) only for pleasure
- d) randomly

71. *Dictionary usage* helps in finding word meaning spelling pronunciation part of speech and
- a) crop information
- b) usage examples etymology and synonyms
- c) mathematical formulas
- d) chemical formulas

72. *Guide words* at top of dictionary page help in
- a) understanding grammar
- b) finding synonyms
- c) checking spelling only
- d) quickly locating words by showing first and last entry on that page

73. *Sentence synthesis* by using participles example He was tired and he slept becomes
- a) *Being tired* he slept – using participial phrase
- b) He was tired he slept
- c) *Tired and* sleeping
- d) He tired sleeps

74. Using *Too and Enough example* He is too weak to lift the box means
- a) He is so weak that he cannot lift the box
- b) He can lift the box
- c) He will lift the box
- d) He lifts the box easily

75. Using So and Such example it was such a hot day that everyone suffered or It was so hot that everyone suffered both mean
- a) different things entirely
- b) the same thing expressing result of extreme condition
- c) opposite meanings
- d) unrelated things

Topic 4: Writing Skills and Agricultural Communication

76. Technical writing in agriculture aims to communicate agricultural information in
- a) literary poetic form
- b) complex scientific jargon only

89. Interview in agriculture may be conducted for
- a) only entertainment
- b) job selection field data collection or extension contact with farmers

c) narrative story form
d) clear precise and objective language understandable to target audience

77. *News story* in agricultural journalism should answer five Ws which are Who What Where When and
a) Why – the five W questions of journalism
b) Which
c) Whom
d) Whether

78. *Extension bulletin* is a periodic publication providing detailed information on a specific agricultural topic for
a) only scientists
b) farmers extension workers and students
c) only government officials
d) only researchers

79. Folder is a multi-page printed material providing more detailed information than a
a) bulletin
b) pamphlet
c) leaflet or single sheet material
d) book

80. Circular letter is a letter sent to many people simultaneously with same content for
a) mass communication on agricultural schemes and recommendations
b) personal correspondence
c) legal proceedings
d) individual advice

81. Newsletter is a periodic publication of an organization providing updates on activities research and
a) only entertainment
b) legal notices only
c) recommendations relevant news and information to members and stakeholders
d) advertisements only

82. *Paragraph writing* requires a topic sentence supporting sentences and a
a) random list
b) new topic introduction
c) concluding sentence that ties together the ideas
d) repeated topic sentence

83. *Essay writing* structure typically includes introduction body paragraphs and
a) another introduction
b) bibliography always
c) conclusion that summarizes main points and provides closing thought
d) appendix always

84. SWOT analysis in farm business planning stands for Strengths Weaknesses *Opportunities and*
a) Tasks
b) Options
c) Threats
d) Targets

85. *Effective communication* requires clarity brevity accuracy and
a) complexity
b) use of technical jargon always
c) relevance to the target audience
d) very long sentences

86. Barriers to effective communication include language differences noise and
a) clear message
b) prejudice poor listening and inappropriate medium
c) good feedback
c) legal disputes only
d) scientific laboratory work only

90. *Presentation skills* involve organizing content clearly using visual aids and
a) reading from notes without looking up
b) engaging the audience through eye contact and confident delivery
c) speaking very fast
d) using all technical terms

91. Minutes of a meeting are the official written record of
a) discussions decisions and action points taken during a meeting
b) casual conversations
c) weather observations
d) field observations

92. Notice is a written communication informing people about an upcoming event meeting or
a) personal matter only
b) secret information
c) important announcement requiring their attention or action
d) historical event

93. *Advertisement writing* should be
a) concise attention-grabbing highlighting key benefits and including contact details
b) lengthy and complex
c) completely factual only
d) written in passive voice only

94. *Slogans are* short memorable phrases used to
a) convey a key message or promote a product scheme or idea effectively
b) explain complex ideas at length
c) write essays
d) prepare reports

95. Editing a written document involves checking for grammar spelling
a) only style
b) only length
c) punctuation clarity coherence and factual accuracy
d) only vocabulary

96. *Proof reading* is the final stage of checking a document for
a) content revision
b) adding new content
c) typographical errors spelling mistakes and formatting errors before publication
d) changing the topic

97. In agricultural communication use of local language or mother tongue is important because
a) English is not important
b) it is cheaper
c) it ensures better understanding and acceptance by farming community
d) it requires less skill

98. *Visual communication* tools in extension include posters charts flip charts and
a) only text books
b) only radio
c) only television
d) photographs drawings models and film slides

99. Good agricultural writing should use active voice short sentences and
a) complex technical terminology always
b) passive voice always
c) simple everyday words familiar to farmers avoiding unnecessary jargon
d) very long paragraphs

d) proper channel

87. Feedback in communication is the
a) original message
b) response of receiver showing whether message was understood
c) noise
d) medium

88. *Group discussion* involves multiple participants exchanging ideas on a topic and requires
a) one person speaking only
b) written submission only
c) active listening turn taking and respectful disagreement
d) silent observation

100. *Mobile phone* based agricultural extension reaches farmers through
a) SMS voice messages WhatsApp groups and agricultural apps
b) only voice calls
c) only internet
d) only television

ANSWER KEY — COMMUNICATION SKILLS IN ENGLISH

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	**B**	**D**	**C**	**D**	**B**	**D**	**D**	**D**	**B**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	**B**	**D**	**B**	**A**	**A**	**A**	**B**	**B**	**D**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	**C**	**D**	**C**	**A**	**A**	**D**	**C**	**C**	**A**
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
C	**C**	**D**	**A**	**D**	**A**	**C**	**B**	**A**	**B**
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
B	**C**	**D**	**C**	**B**	**D**	**C**	**A**	**C**	**B**
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
A	**B**	**B**	**D**	**A**	**A**	**B**	**C**	**B**	**D**
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
D	**C**	**A**	**C**	**B**	**A**	**B**	**B**	**C**	**A**
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
B	**D**	**A**	**A**	**B**	**D**	**A**	**B**	**C**	**A**
81	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**
C	**C**	**C**	**A**	**C**	**B**	**B**	**C**	**B**	**B**
91	**92**	**93**	**94**	**95**	**96**	**97**	**98**	**99**	**100**
A	**C**	**A**	**A**	**C**	**C**	**C**	**D**	**C**	**A**

SUBJECT 16: FORESTRY MEDICINAL AND AROMATIC PLANTS

Code: DA 281 | Total Questions: 100

Topic 1: Introduction to Forestry and Forest Types

1. The word Forest is derived from *Latin word Foris which* means
   a) inside
   b) trees
   c) outside
   d) shade

2. Forest is defined as an area set aside for production of timber and other forest produce or maintained under woody vegetation for
   a) certain indirect benefits like climatic or protective benefits
   b) agriculture only
   c) industrial use only
   d) mining

3. Forestry is defined as the theory and practice of creation conservation and scientific management of forests and utilization of their
   a) agricultural resources
   b) water resources
   c) resources for service of the nation
   d) mineral resources

4. *Forest area* in India covers approximately
   a) 10 percent
   b) 50 percent
   c) 23 percent of total land area of 327.7 million hectares
   d) 35 percent

5. National Forest Policy 1988 recommends that India should have how much of its total land area under forest
   a) one quarter
   b) one half
   c) one fifth
   d) one third

6. *Forest area* in Telangana state covers approximately
   a) 10 percent
   b) 50 percent
   c) 35 percent
   d) 25.11 percent of 112077 sq km total geographical area

7. Which district in Telangana has the highest forest area
   a) *Khammam district* with 7945 sq km of forest area
   b) Nalgonda
   c) Warangal
   d) Hyderabad

8. India's per capita forest area is only 0.11 ha compared to world average of
   a) 1.6 hectares per capita
   b) 0.5 ha per capita
   c) 3 ha per capita
   d) 5 ha per capita

9. Silviculture is the branch of forestry dealing with establishment development care and reproduction of stands of
   a) agricultural crops
   b) medicinal plants only
   c) timber forests
   d) grass only

10. Dendrology is the study of
    a) tree planting only
    b) description classification and identification of various tree species
    c) forest management

14. *Forests reduce* wind velocity and studies show that foliage at crown reduces wind by
    a) 10 percent
    b) 43 percent on outer side with middle of foliage reducing by 14 percent
    c) 80 percent
    d) 20 percent only

15. *Forests help* in flood reduction by utilizing moisture for evapotranspiration increasing infiltration capacity and
    a) increasing runoff
    b) increasing surface runoff
    c) waterlogging fields
    d) reducing silting and controlling soil erosion

16. *One hectare* of pine trees can filter how many tonnes of dust from air
    a) 32 tonnes of dust
    b) 5 tonnes
    c) 100 tonnes
    d) 1 tonne

17. *Forests absorb* CO_2 and release O_2 and also filter hazardous gases like
    a) N_2
    b) CO_2 only
    c) SO_2 sulphur dioxide
    d) O_2

18. A park with 50 metres width can reduce traffic sound by
    a) 20 to 30 decibels
    b) 5 to 10 decibels
    c) 50 to 60 decibels
    d) 100 decibels

19. The first National Forest Policy of India was enacted on
    a) 1947 after independence
    b) 19th October 1894 during *British period*
    c) 1952
    d) 1988

20. The second National Forest Policy 1952 was based on six dominant needs and recommended India should maintain forest on
    a) one fourth of land
    b) half of land
    c) one fifth of land
    d) one third of total land area

21. The National Forest Policy 1988 emphasizes conservation of forest ecosystems biodiversity and
    a) revenue maximization
    b) only timber production
    c) only wildlife
    d) meeting basic needs of people living near forests while promoting production forestry

22. Indian Forest Act 1972 consolidates laws relating to forests transit of forest produce and duties on
    a) agricultural produce
    b) only wildlife
    c) timber and other forest produce
    d) only mining

23. Violation of Indian Forest *Act can* lead to penalty of Rs 500 fine or imprisonment of
    a) 1 month

d) forest economics

11. *Forest mensuration* deals with
   a) forest policy
   b) forest protection
   c) measurement of forest produce
   d) tree planting

12. Agroforestry is cultivation of crops and forest plants either sequentially or simultaneously on the
   a) separate plots
   b) different states
   c) different seasons only
   d) same piece of land for sustainable land management

13. *Forests influence* climate by reducing maximum temperature raising minimum temperature and
   a) increasing soil erosion
   b) reducing rainfall
   c) increasing drought
   d) controlling wind velocity and maintaining humidity

b) 5 years
c) 6 months or both
d) 10 years

24. Deforestation in India during 1951 to 1982 averaged about
   a) 1.5 million hectares per year
   b) 0.1 million ha per year
   c) 5 million ha per year
   d) 10 million ha per year

25. Main reasons for loss of forest area include undue biotic interference excessive cutting and
   a) tree planting
   b) diversion of area for agriculture river valley projects and industries
   c) conservation efforts
   d) afforestation

Topic 2: Silviculture Nursery and Afforestation

26. Stages of tree development from youngest to oldest are Seedling Sapling *Pole and*
   a) Shrub
   b) Bush
   c) Herb
   d) Tree

27. *Seedling stage* is defined as germination to height of
   a) 5 metres
   b) 10 metres
   c) 50 cm
   d) 3 feet or 1 metre

28. Crown of a tree is the
   a) root system
   b) lower trunk
   c) upper branchy part above bole comprising branches and foliage
   d) bark

29. Bole of a tree is the
   a) crown
   b) root
   c) lower portion of stem up to point where main branches are given off
   d) leaf

30. Mycorrhiza is a composite structure formed by fine extremities of rootlets invaded by specific non-pathogenic soil
   a) bacteria
   b) nematodes
   c) fungi helping trees absorb water and nutrients
   d) algae

31. *Mangrove roots* include pneumatophores which are peg-like projections helping in
   a) water absorption only
   b) exchange of gases in waterlogged soils preventing plant rotting
   c) photosynthesis
   d) pollination

32. Principles of forest nursery raising include producing healthy vigorous plants producing seedlings for distribution and
   a) selling timber
   b) only cutting trees
   c) only harvesting

39. *Coastal sea* sand suitable tree species include *Casuarina equisetifolia* and
   a) Teak
   b) Oak
   c) Sal
   d) *Avicennia officinalis* and Eucalyptus

40. *Roadside trees* should be quick growing evergreen with compact crown tolerant to wind and
   a) very tall above 30 metres
   b) invasive roots
   c) thorny branches
   d) not of climbing nature protected from stray animals examples *Mangifera and* Tamarindus

41. Bio-aesthetic plantation is growing plants for
   a) timber production
   b) food production
   c) medicine
   d) beautification of surroundings in different shapes

42. *Delonix regia* Gulmohar is an example of tree used for
   a) timber
   b) food production
   c) paper
   d) flowering bio-aesthetic plantation providing shade and colour

43. *Social forestry* involves planting trees on community lands road sides canal banks and
   a) farm land as farm forestry for local needs of fuel fodder and small timber
   b) only forest land
   c) only government land
   d) mining land

44. Joint Forest Management JFM involves
   a) only government
   b) only timber companies
   c) partnership between forest department and local communities for forest conservation and benefit sharing
   d) only NGOs

45. Van Mahotsav is a national festival celebrated annually in India to promote
   a) harvest festival

d) raising exotic species facilitating initial care in nursery conditions

33. Temporary forest nurseries are also called field nurseries as they are
a) permanent structures
b) in cities
c) in research stations
d) established adjacent to planting area and abandoned once stock is met

34. Permanent forest nurseries have advantage of
a) better infrastructure skilled staff and efficient supervision and quality control
b) low investment
c) no supervision
d) no record keeping

35. Afforestation on shifting sand dunes uses drought-tolerant species like *Acacia nilotica* and
a) *Ziziphus and* shelter belts to reduce wind velocity
b) *Tectona grandis*
c) *Eucalyptus only* in wet areas
d) paddy in sand

36. *Saline and* alkaline soils suitable tree species include *Acacia nilotica Prosopis juliflora Azadirachta indica* and
a) *Butea monosperma Pongamea pinnata* and *Ailanthus excelsa*
b) paddy
c) wheat
d) sugarcane

37. *Ravine lands* are mainly found in
a) Kerala
b) Uttar Pradesh *Gujarat and* Madhya Pradesh
c) Assam
d) *Coastal areas*

38. *Tectona grandis* teak grows well in
a) sandy soils only
b) lateritic soils of West *Bengal and* other tropical soils
c) very cold alpine soils
d) saline soils

b) tree planting and afforestation with millions of trees planted
c) forest harvesting
d) wildlife hunting

46. *Shifting cultivation* jhum is a traditional practice of clearing forest growing crops for few years and then
a) permanent cultivation
b) abandoning and moving to new area allowing forest regrowth
c) deep ploughing
d) continuous cropping

47. Timber is used for construction furniture paper pulp and
a) fuelwood charcoal railway sleepers and other industrial purposes
b) food
c) only paper
d) only furniture

48. Non-timber forest products NTFP include bamboo cane resins gums honey and
a) only timber
b) only fuelwood
c) only wildlife
d) medicinal plants beeswax tendu leaves and edible forest foods

49. *Carbon sequestration* by forests is important for
a) increasing CO2
b) increasing temperature
c) absorbing and storing atmospheric CO2 helping to mitigate climate change
d) reducing rainfall

50. *Biodiversity hotspots* are regions with high species richness and high levels of
a) endemism and habitat threat requiring urgent conservation
b) agricultural productivity
c) forest harvesting
d) mining activity

Topic 3: Medicinal Plants and Their Importance

51. *Medicinal plants* are plants that are rich in secondary metabolites and are potential source for
a) food only
b) drugs medicines and pharmaceuticals
c) timber only
d) ornamental use only

52. India ranks among top countries in traditional medicine systems like Ayurveda *Siddha and*
a) *Western medicine* only
b) only modern pharmacy
c) only surgery
d) *Unani which* use hundreds of medicinal plant species

53. Tulsi Holy *Basil botanical* name is
a) *Withania somnifera*
b) *Centella asiatica*
c) *Azadirachta indica*
d) *Ocimum tenuiflorum* sanctum used as antiseptic and for respiratory ailments

54. *Neem botanical* name *Azadirachta indica* has medicinal properties and is used for

64. *Zingiber officinale* Ginger is used for
a) only spice
b) treatment of nausea digestive problems cold and inflammation
c) only ornamental
d) only timber

65. *Andrographis paniculata* Kalmegh is used as
a) food only
b) bitter tonic for liver disorders fever and infections
c) only fuelwood
d) only ornamental

66. WHO World Health *Organization has* recognized traditional medicine and estimates that about
a) 10 percent
b) 80 percent of world population in developing countries use traditional medicine
c) 5 percent
d) 50 percent in developed countries

67. GMP Good Manufacturing *Practices for* herbal medicines ensure
a) maximum profit

a) antibacterial antifungal insecticidal properties and treatment of skin diseases
b) only timber
c) only ornamental purpose
d) only shade

55. *Ashwagandha botanical* name *Withania somnifera* is used as a
a) food crop
b) only ornamental plant
c) adaptogen and tonic for stress relief immunity and nervous system
d) only fuelwood

56. *Senna botanical* name *Cassia angustifolia* is used as a
a) stimulant
b) laxative for constipation treatment
c) analgesic
d) antibiotic

57. Periwinkle *Catharanthus roseus* contains alkaloids vincristine and vinblastine used for treatment of
a) diabetes
b) hypertension
c) cancer especially leukemia
d) malaria

58. *Aloe vera* is used for
a) timber
b) only ornamental
c) only fuel
d) treatment of burns wounds skin conditions and digestive health

59. *Terminalia arjuna* arjuna is used in *Ayurveda for* treatment of
a) skin diseases
b) fever only
c) diabetes only
d) heart conditions and cardiovascular health

60. *Phyllanthus emblica* Amalaki Aonla is very rich in Vitamin C and used in *Ayurveda for*
a) fever only
b) skin whitening only
c) general health tonic hair care and immunity
d) blood pressure only

61. *Tinospora cordifolia* Guduchi Giloy is used for
a) only ornamental
b) only fuelwood
c) only food
d) immunity boosting anti-inflammatory and treatment of fever and jaundice

62. *Bacopa monnieri* Brahmi is used to improve
a) digestion only
b) blood pressure only
c) liver health only
d) memory cognitive function and as a nerve tonic in Ayurveda

63. *Curcuma longa Turmeric contains* curcumin and is used for its
a) only food flavouring
b) only yellow colour
c) anti-inflammatory antiseptic antioxidant properties and treatment of various ailments
d) only religious use

b) quality safety and efficacy of herbal preparations
c) minimum cost
d) maximum extraction

68. Primary collection of medicinal plants should preferably be done from
a) polluted areas
b) roadsides
c) waste dumps
d) natural healthy habitats away from pollution for best quality and potency

69. Post-harvest processing of medicinal plants includes drying which should be done in
a) shade or controlled temperature to preserve active compounds
b) direct harsh sunlight
c) high humidity
d) wet conditions

70. Standardization of herbal medicines involves
a) no testing needed
b) only colour check
c) only weight measurement
d) defining quality parameters like active ingredient content physical and chemical characteristics

71. Cultivation of medicinal plants under Good Agricultural Practices GAP ensures
a) maximum yield at any cost
b) no quality control
c) only profit
d) consistent quality safety and sustainability of raw material supply

72. In-situ conservation of medicinal plants means
a) growing in gardens
b) laboratory preservation
c) cold storage
d) conserving plants in their natural habitat through protected areas and sanctuaries

73. Ex-situ conservation of medicinal plants means
a) in natural habitat only
b) conserving outside natural habitat in botanical gardens seed banks and tissue culture
c) in situ only
d) in forests only

74. *Medicinal plant* board at national level coordinates activities related to
a) timber only
b) agricultural crops only
c) development production processing and marketing of medicinal plants in India
d) only research

75. Intellectual Property Rights IPR and patents for medicinal plant knowledge are important to
a) prevent all research
b) increase biopiracy
c) reduce conservation
d) protect traditional knowledge from biopiracy and ensure fair benefits to indigenous communities

Topic 4: Aromatic Plants and Essential Oils

76. *Aromatic plants* are those that contain essential oils which are
a) fixed non-volatile oils like coconut oil
b) odoriferous steam volatile constituents giving characteristic fragrance
c) fatty acids
d) alkaloids only

77. Examples of aromatic plants grown in India include Lemongrass Citronella Palmarosa Vetiver *Geranium and*
a) *Neem and Tulsi only*
b) *Teak and* Sal
c) Davanam *Rose and* Peppermint
d) *Mango and* Guava

78. *Essential oils* from aromatic plants are extracted mainly by
a) cold pressing only
b) steam distillation which is the most common method
c) solvent extraction only
d) boiling in water

79. Lemongrass *Cymbopogon flexuosus* essential oil contains mainly
a) menthol
b) geraniol
c) citral used in perfumery soap cosmetics and flavouring
d) eugenol

80. Palmarosa *Cymbopogon martinii* essential oil contains high amount of
a) geraniol used in perfumery and soap industry
b) citral
c) menthol
d) camphor

81. Vetiver *Vetiveria zizanioides* roots yield vetiver oil used in
a) food only
b) high-quality perfumes and for soil conservation due to its deep root system
c) only medicine
d) only timber

82. Peppermint *Mentha piperita* essential oil contains mainly
a) menthol used in medicines confectionery and cosmetics
b) citral
c) geraniol
d) eugenol

83. Rose *Rosa spp* essential oil called Attar of Rose or Rose Otto is one of the
a) cheapest essential oils
b) most common oils
c) most valuable and expensive essential oils used in fine perfumery
d) least useful

84. *Eucalyptus essential* oil contains cineole eucalyptol and is used for
a) food flavouring mainly
b) antiseptic and decongestant properties in medicines and inhalants
c) only perfumery
d) only insecticides

85. Clove *Syzygium aromaticum* essential oil contains eugenol and is used as
a) sweetener
b) dental analgesic antiseptic and flavouring agent
c) only ornamental
d) only food color

86. Sandalwood *Santalum album* yields sandalwood oil used in
a) construction timber only
b) only medicine

89. *Solvent extraction* method uses organic solvents to extract
a) only essential oils
b) only alkaloids
c) delicate floral oils that cannot withstand steam distillation like jasmine and rose
d) only fixed oils

90. *Hydrodistillation method* submerges plant material directly in water and then
a) cold presses
b) distills the mixture separating essential oil from water condensate
c) filters
d) centrifuges only

91. Quality of essential oil is determined by its
a) colour only
b) chemical composition particularly percentage of key active constituents
c) weight only
d) country of origin only

92. GC-MS Gas Chromatography Mass Spectrometry is used for
a) soil testing
b) fertilizer analysis
c) water testing
d) identifying and quantifying individual chemical constituents of essential oils

93. *Aromatherapy uses* essential oils for
a) only cooking
b) only cleaning
c) only industrial use
d) therapeutic purposes to promote physical and psychological well-being

94. *Global essential* oil market is dominated by oils like peppermint orange eucalyptus lavender and
a) citronella lemon and tea tree oil
b) only neem
c) only sandalwood
d) only vetiver

95. India is a major producer and exporter of essential oils from
a) lemongrass palmarosa mint menthol geranium and sandalwood
b) only neem
c) only coconut oil
d) only food oils

96. *Spices are* natural plant products used to improve flavor aroma taste and colour of food examples are Pepper Cardamom *Clove and*
a) Rice and wheat
b) only Pepper
c) only Cardamom
d) Nutmeg Turmeric *Coriander and* Cumin

97. *Condiments are* plants used to add only taste to food examples include
a) *Pepper and Cardamom which* add aroma
b) *Clove and* Nutmeg
c) *Vanilla and* Saffron
d) *Coriander and* Cumin

98. Spice Board of India promotes cultivation processing and export of spices and is headquartered at
a) New Delhi
b) Mumbai
c) Chennai
d) Kochi Kerala

c) only paper
d) perfumery cosmetics and religious ceremonies being one of valuable woods

87. *Steam distillation* process extracts essential oils by passing steam through plant material which volatilizes the oil and
a) leaves oil in plant
b) carries it over to condenser where it separates from water
c) destroys the oil
d) dissolves in water permanently

88. *Cold pressing* method is used for extraction of essential oils from
a) leaves and stems
b) citrus fruit peels like orange lemon and lime by mechanical pressing
c) roots
d) seeds only

99. Black pepper botanical name *Piper nigrum* is called the King of *Spices and* is mainly grown in
a) Rajasthan
b) Punjab
c) West Bengal
d) Kerala *Karnataka and* Tamil Nadu

100. Cardamom *Elettaria cardamomum* is called the Queen of *Spices and* is mainly grown in
a) *Rajasthan and* Punjab
b) *Kerala and Karnataka hills*
c) Andhra Pradesh
d) Bihar

ANSWER KEY — FORESTRY MEDICINAL AND AROMATIC PLANTS

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	A	C	C	D	D	A	A	C	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	D	D	B	D	A	C	A	B	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	C	C	A	B	D	D	C	C	C
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	D	D	A	A	A	B	B	D	D
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
D	D	A	C	B	B	A	D	C	A
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
B	D	D	A	C	B	C	D	D	C
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
D	D	C	B	B	B	B	D	A	D
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
D	D	B	C	D	B	C	B	C	A
81	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**
B	A	C	B	B	D	B	B	C	B
91	**92**	**93**	**94**	**95**	**96**	**97**	**98**	**99**	**100**
B	D	D	A	A	D	D	D	D	B

SUBJECT 17: HORTICULTURE

Code: DA 282 | Total Questions: 100

Topic 1: Introduction and Fruit Crop Cultivation

1. Horticulture is the science art and business of cultivating fruits vegetables flowers and ornamental plants with
   a) extensive low input farming
   b) intensive skilled cultivation methods
   c) only food crops
   d) only field crops

2. Pomology is the study or cultivation of
   a) vegetables
   b) flowers
   c) fruit crops like mango sapota guava grape and banana
   d) plantation crops

3. Olericulture is the science of cultivation of
   a) fruit crops
   b) flowers
   c) vegetable crops like brinjal okra tomato pumpkin
   d) spices

4. Floriculture is the science of
   a) fruit production
   b) flower production
   c) vegetable production
   d) timber production

5. *Plantation crops* are extensively cultivated perennial trees including commercial crops like Coconut Arecanut Cashewnut Tea *Coffee and*
   a) Rubber
   b) *Mango and* Banana
   c) only Coconut
   d) only Coffee

6. *Spices are* natural plant products used to improve flavor aroma taste and colour examples are Pepper Cardamom *Clove and*
   a) Brinjal Tomato
   b) Nutmeg Turmeric
   c) Mango Banana
   d) Paddy Wheat

7. *Medicinal plants* are those rich in secondary metabolites which are potential sources for
   a) food production
   b) timber
   c) drugs and pharmaceuticals
   d) ornamental use only

8. *Aromatic plants* contain essential oils which are odoriferous steam volatile constituents examples include Lemongrass *Citronella and*
   a) Paddy and Wheat
   b) *Mango and* Guava
   c) Palmarosa *Vetiver and* Geranium
   d) Rice and Maize

9. *Arboriculture deals* with raising of perennial trees for shade avenue or ornamental purposes examples include
   a) Polyalthia Spathodea *Cassia and* Gulmohar
   b) Paddy and Wheat
   c) *Mango and Banana food*
   d) Rice and Maize

10. India ranks second in the world in fruits and vegetables production after
   a) USA
   b) China

14. Pomum is the *Latin word* from which Pomology is derived and it means
   a) soil
   b) leaf
   c) fruit
   d) root

15. *Fruits are* considered protective foods because they contain vitamins proteins and
   a) starch mainly
   b) minerals that protect the human body
   c) fats mainly
   d) cellulose only

16. *Mango and Papaya are* rich sources of
   a) Vitamin A important for eyesight and growth
   b) Vitamin C
   c) Vitamin B
   d) Vitamin D

17. Aonla Amla *Guava and Citrus fruits* are very rich sources of
   a) Vitamin A
   b) Vitamin D
   c) Vitamin C ascorbic acid improving immunity
   d) Vitamin B12

18. *Fruit trees* cover ground as plantation and help maintain
   a) only aesthetic beauty
   b) only market value
   c) ecological balance in addition to providing nutrition and economic benefit
   d) only shade

19. *Reasons for* low fruit production in India include high perishability lack of transport facilities absence of organized marketing and
   a) lack of awareness technology and industrial development for processing
   b) too much technology
   c) excess storage
   d) too many varieties

20. India's share in global fruit export market is approximately
   a) 20 percent
   b) 1 percent despite having suitable climate and soils for year-round fruit cultivation
   c) 50 percent
   d) 10 percent

21. *Fruit crops* are more profitable to farmers than field crops because they have
   a) lower input cost always
   b) less risk
   c) no diseases
   d) more productivity per hectare higher yield value and multiple uses

22. *Banana contains* 16 to 17 percent dextrose sugars making it a good source of
   a) protein only
   b) fat only
   c) minerals only
   d) easily digestible energy

23. *Papaya contains* pectin which helps in
   a) easy digestion of food
   b) eyesight only

c) Brazil
d) Japan

11. India produced approximately 99.07 million MT of fruits with area under cultivation of
a) 6.66 million hectares as per National Horticulture Board 2019-20
b) 1 million ha
c) 50 million ha
d) 10 million ha

12. India accounts for about what percent of world fruit production
a) 5 percent
b) 25 percent
c) 30 percent
d) 10.9 percent

13. Post-harvest losses of horticultural produce in India are estimated at
a) 30 to 40 percent due to improper post-harvest management
b) 5 percent
c) 10 percent
d) 1 percent

c) building muscles
d) blood clotting

24. *Cashew nut* export from India contributes significantly to
a) no economic value
b) only domestic market
c) national economy and foreign exchange earnings
d) no export

25. Phytochemicals in fruits act as antioxidants giving protection from
a) soil erosion
b) crop pests
c) weather damage
d) different diseases like cancer and cardiovascular diseases

Topic 2: Orchard Layout Planting Systems and Management

26. *Square system* of orchard layout has plant to plant and row to row distances equal and each four plants form a
a) triangle
b) hexagon
c) rectangle
d) square facilitating intercultural operations in both directions

27. *Rectangular system* of orchard layout has more space between rows with plants closer
a) in row to row direction
b) in all directions
c) within the row accommodating more plants per row
d) randomly

28. *Hexagonal system* or septule accommodates how much more plants than square system
a) 5 percent more
b) 15 percent more plants by placing seventh tree in center of hexagon of six trees
c) 10 percent more
d) 20 percent more

29. Quincunx or diagonal or filler system modifies square system by adding a filler tree in
a) corners
b) rows
c) boundaries
d) center of each square doubling the plant population

30. Common filler trees used in quincunx system include
a) *Mango and* Sapota
b) *Coconut and* Arecanut
c) only annual crops
d) *Guava phalsa* plum papaya peaches as they are precocious and shorter duration

31. *Contour system* of orchard layout is followed on hills where plants are planted along
a) contour lines across slope to minimize soil erosion and conserve moisture
b) straight rows down slope
c) random positions
d) any direction

39. *Pits are* filled with mixture of top soil FYM leaf mould and
a) clay only
b) river sand
c) bone meal filled a few inches above ground level for settlement
d) gravel

40. Planting of fruit trees is ideally done during
a) summer peak heat
b) winter peak cold
c) onset of monsoon in moderate rainfall areas or close of monsoon in heavy rainfall
d) any time equally

41. The bud union graft joint of transplanted fruit plants should
a) remain slightly above ground level and not be covered with soil
b) be buried in soil
c) be very deep
d) be at surface only

42. Extra 10 percent plants are maintained in nursery to use for
a) extra sale
b) gap filling when some plants die after transplanting
c) making shade
d) boundary

43. Training of fruit trees refers to
a) only pruning
b) giving desired shape and structure to the tree for proper development and light distribution
c) only fertilizing
d) only irrigation

44. Pruning of fruit trees removes dead diseased or unproductive wood to
a) reduce plant size only
b) reduce irrigation
c) reduce fertilizer use
d) improve light penetration air circulation and overall health and productivity

45. Intercropping in orchards during early years before fruit trees establish is done with

32. *Paired row* system brings two plant rows close together followed by
a) another pair immediately
b) random spacing
c) single rows
d) a wide gap before next set of two rows

33. Selection of orchard site should consider climate soil depth availability of irrigation and
a) cost alone
b) only appearance
c) nearness to market transport facilities power supply and availability of labour
d) only soil type

34. *Water table* in orchard site should be below
a) 0.5 metre
b) 2 metres depth for proper root growth and drainage
c) 5 metres
d) 10 metres

35. Windbreaks in orchard are planted on
a) south side only
b) north-west side being sensitive to cold and hot winds with gap of 30 to 40 feet from first row
c) east side
d) south-east side

36. Live fencing plants suitable for orchard boundaries include Agave *Prosopis juliflora Pithecolobium dulce* and
a) Mango
b) Thevetia
c) Coconut
d) Banana

37. *Pits for* fruit planting are generally dug how many months in advance of planting
a) 2 to 3 months in advance typically March to May
b) 1 week before
c) 1 month before
d) 1 year before

38. *Standard pit* size for fruit planting is generally
a) 25 cm x 25 cm
b) 50 cm x 50 cm
c) 1 cubic metre or 1m x 1m x 1m
d) 2m x 2m x 2m

a) only trees
b) only grass
c) short duration annual vegetable or cereal crops for extra income
d) nothing

46. High density planting HDP of fruit trees accommodates more plants per unit area using
a) no change in management
b) only pruning
c) only training
d) dwarfing rootstocks drip irrigation and intensive management for early and higher yields

47. *Drip irrigation* in orchards provides water and nutrients directly to
a) whole field flooded
b) root zone of each tree with high efficiency and reducing disease
c) only soil surface
d) only leaves

48. *Fertigation through* drip is the practice of applying fertilizers dissolved in
a) rain water
b) irrigation water through drip system ensuring precise nutrient delivery
c) any water
d) river water only

49. Mulching in orchards conserves soil moisture controls weeds moderates soil temperature and
a) adds organic matter when organic mulch decomposes
b) increases erosion
c) increases weed growth
d) compacts soil

50. Inter-culture or clean cultivation in orchards removes weeds by
a) herbicide only
b) only hand weeding
c) tillage between trees and rows to keep the orchard weed free and aerated
d) only mulching

Topic 3: Propagation Methods and Important Fruit Crops

51. *Vegetative propagation* of fruit crops produces plants genetically identical to mother plant called
a) clones ensuring uniformity in yield quality and earliness
b) hybrid plants
c) seedlings
d) cross-pollinated plants

52. Budding is a vegetative propagation method where a bud of desired variety is inserted onto a
a) seed
b) leaf cutting
c) air layer
d) rootstock of compatible plant and allowed to grow

53. Grafting is vegetative propagation where scion shoot of desired variety is joined to
a) rootstock and the union forms a new plant
b) seed directly
c) soil only
d) leaf

64. Papaya *Carica papaya* belongs to family
a) Rosaceae
b) Rutaceae
c) Musaceae
d) Caricaceae

65. *Papaya has* sex types male female and hermaphrodite and commercial cultivation prefers
a) hermaphrodite plants which bear fruit and are self pollinating
b) male plants for fruit
c) female plants with male pollinators
d) only male plants

66. Coconut *Cocos nucifera* belongs to family
a) Arecaceae Palmae
b) Rosaceae
c) Rutaceae
d) Anacardiaceae

67. Coconut is propagated by

54. *Air layering* or gootee involves inducing roots on an intact branch by
a) cutting branch completely
b) planting branch in soil
c) immersing in water
d) removing ring of bark applying rooting hormone and wrapping with moist medium

55. Mango *Mangifera indica* belongs to family
a) Anacardiaceae
b) Rosaceae
c) Rutaceae
d) Moraceae

56. Mango is called the King of Fruits in India and India is the
a) third largest producer
b) second largest producer
c) fourth largest producer
d) largest producer of mango in the world

57. Mango is propagated vegetatively mainly by
a) stone grafting or veneer grafting for true to type varieties
b) seed
c) cutting
d) division

58. Banana *Musa spp* belongs to family
a) Musaceae
b) Rosaceae
c) Rutaceae
d) Anacardiaceae

59. Banana is propagated by
a) seeds
b) grafting
c) suckers rhizome bits and tissue culture plants
d) budding

60. Sapota *Manilkara zapota* is a fruit crop propagated by
a) seeds only
b) air layering only
c) cuttings only
d) budding and grafting on wild sapota seedling rootstock

61. Guava *Psidium guajava* is very rich in
a) Vitamin A only
b) Vitamin D
c) Vitamin C being one of richest plant sources
d) Vitamin B12

62. Grape *Vitis vinifera* is mainly propagated by
a) stem cuttings called hardwood cuttings
b) seeds
c) budding
d) air layering

63. *Citrus fruits* are propagated mainly by
a) seeds always
b) T-budding on rootstocks like *Rangpur lime* and rough lemon
c) grafting and air layering
d) division

a) stem cuttings
b) seeds nuts selecting from mother palm with high yield
c) budding
d) air layering

68. Pineapple *Ananas comosus* is propagated by
a) seeds
b) grafting
c) crowns slips ratoons and suckers
d) budding

69. Pomegranate *Punica granatum* is propagated mainly by
a) seeds
b) budding
c) hardwood stem cuttings
d) air layering

70. Aonla Indian gooseberry *Phyllanthus emblica* is very rich in Vitamin C and is used extensively in
a) only food
b) only ornamental
c) only timber
d) *Ayurvedic preparations* and food products like murabba pickle and dried pieces

71. Jackfruit *Artocarpus heterophyllus* is the national fruit of
a) India
b) China
c) *Bangladesh and* Sri Lanka
d) Indonesia

72. Ber Indian jujube *Ziziphus mauritiana* is a hardy drought tolerant fruit suitable for
a) high rainfall humid areas
b) arid and semi-arid dryland regions
c) waterlogged areas
d) only hill areas

73. Watermelon *Citrullus lanatus* belongs to family
a) Cucurbitaceae
b) Rosaceae
c) Rutaceae
d) Solanaceae

74. Muskmelon *Cucumis melo* and *Watermelon are* both
a) perennial crops
b) warm season annual vine crops belonging to Cucurbitaceae
c) cool season crops
d) biennial crops

75. *Fruit maturity* indices include days from flowering change in skin colour development of aroma and
a) only size
b) firmness specific gravity starch iodine test and ease of separation from plant
c) only weight
d) only colour

Topic 4: Vegetable Cultivation and Post-harvest Management

76. *Vegetables are* herbaceous plants of which some part like root stem leaf flower or fruit is used as
a) only fodder
b) food in fresh or cooked form
c) only medicine
d) only ornamental

89. Grading of fruits and vegetables is done based on size weight colour and
a) only price
b) only variety
c) only origin

77. *Cool season* vegetables that prefer cool temperatures include Cabbage Cauliflower Peas *Spinach and*
a) Carrot Radish *Beetroot and* Potato
b) Okra Brinjal
c) Tomato Cucumber
d) *Bitter gourd*

78. *Warm season* vegetables that require high temperatures include Tomato Brinjal Okra *Cucumber and*
a) *Cauliflower and* Cabbage
b) *Bitter gourd* ridge gourd pumpkin and chilli
c) *Pea and* Spinach
d) *Carrot and* Radish

79. Hardening of vegetable seedlings before transplanting is done by gradually
a) adding more fertilizer
b) increasing shade
c) increasing irrigation
d) reducing irrigation and exposing to outdoor conditions to acclimatize seedlings

80. Tomato *Solanum lycopersicum* belongs to family
a) Cucurbitaceae
b) Solanaceae
c) Fabaceae
d) Asteraceae

81. *Damping off* in tomato nursery is caused by *Pythium and Rhizoctonia and* is managed by
a) increasing irrigation
b) only pesticide spray
c) seed treatment soil drenching with *Metalaxyl and* maintaining proper drainage
d) only fertilizer

82. Anthracnose of tomato is caused by
a) *Alternaria solani*
b) *Fusarium oxysporum*
c) Xanthomonas
d) *Colletotrichum phomoides* affecting ripening fruits

83. Onion *Allium cepa* belongs to family
a) Solanaceae
b) Cucurbitaceae
c) Liliaceae Alliaceae
d) Asteraceae

84. Potato *Solanum tuberosum* is propagated by
a) true seeds
b) seed tubers cut pieces with eyes
c) cuttings
d) grafting

85. Late blight of potato caused by *Phytophthora infestans* caused the Great Irish Famine in
a) 1900
b) 1920
c) 1980
d) 1845 to 1849

86. Cucumber *Cucumis sativus* belongs to family
a) Solanaceae
b) Fabaceae
c) Asteraceae
d) Cucurbitaceae

87. Bittergourd *Momordica charantia* is cultivated for its
a) sweet fruits
b) seeds only
c) leaves only
d) bitter fruits used as vegetable and medicine for diabetes management

88. Chilli *Capsicum annuum* pungency is due to the compound

d) freedom from blemishes and defects for better market price

90. Pre-cooling of fruits and vegetables immediately after harvest removes
a) only dirt
b) only moisture
c) only dust
d) field heat rapidly to slow respiration and extend shelf life

91. *Controlled atmosphere* CA storage maintains specific levels of CO2 O2 and N2 to
a) ripen fruits faster
b) extend shelf life of fruits and vegetables by reducing respiration rate
c) increase respiration
d) improve colour artificially

92. *Cold storage* at low temperature reduces
a) respiration transpiration and microbial activity extending storage life
b) vitamin content only
c) market value
d) consumer demand

93. Waxing of fruits like apple and citrus is done to
a) reduce moisture loss and improve appearance
b) add flavour
c) add colour artificially
d) only for export

94. MAP Modified Atmosphere *Packaging uses* selective permeability films to maintain
a) high oxygen levels
b) no atmosphere control
c) modified gas atmosphere around produce extending shelf life
d) high CO2 only

95. *Value addition* in horticulture includes processing into jams jellies pickles juices dried products and
a) no processing
b) only fresh sale
c) canned products frozen vegetables and dehydrated products adding economic value
d) only local sale

96. FSSAI Food *Safety and* Standards Authority of India regulates
a) only agricultural inputs
b) only exports
c) food safety quality and labelling of all food products including horticultural produce
d) only imports

97. *Landscaping uses* horticultural plants to beautify
a) gardens parks roadsides public spaces and private properties
b) only farms
c) only forests
d) only agricultural land

98. *Floriculture products* like cut flowers loose flowers and potted plants have growing
a) no market
b) only local demand
c) domestic and export market with carnation rose gerbera and chrysanthemum being major cut flowers
d) only rural demand

99. *Tissue culture* or micro-propagation produces
a) hybrid seeds
b) disease-free true to type planting material in large numbers in short time
c) only seeds

a) menthol
b) capsaicin present in placenta and inner walls of fruit
c) eugenol
d) citral

d) only cuttings

100. National Horticulture Mission NHM and Horticulture *Mission for* North *East and* Himalayan States HMNEH are government programmes to
a) promote holistic development of horticulture sector through area expansion productivity and post-harvest management
b) reduce horticulture
c) reduce fruit production
d) promote only exports

ANSWER KEY — HORTICULTURE

1	2	3	4	5	6	7	8	9	10
B	C	C	B	A	B	C	C	A	B
11	12	13	14	15	16	17	18	19	20
A	D	A	C	B	A	C	C	A	B
21	22	23	24	25	26	27	28	29	30
D	D	A	C	D	D	C	B	D	D
31	32	33	34	35	36	37	38	39	40
A	D	C	B	B	B	A	C	C	C
41	42	43	44	45	46	47	48	49	50
A	B	B	D	C	D	B	B	A	C
51	52	53	54	55	56	57	58	59	60
A	D	A	D	A	D	A	A	C	D
61	62	63	64	65	66	67	68	69	70
C	A	B	D	A	A	B	C	C	D
71	72	73	74	75	76	77	78	79	80
C	B	A	B	B	B	A	B	D	B
81	82	83	84	85	86	87	88	89	90
C	D	C	B	D	D	D	B	D	D
91	92	93	94	95	96	97	98	99	100
B	A	A	C	C	C	A	C	B	A

GENERAL AGRICULTURE

Code: GA 001 | Total Questions: 100 | Mixed Topics from All 17 Subjects

All questions sourced exclusively from uploaded PJTAU PDF notes | Answer key at end

Group 1: Research Institutes & Agricultural Meteorology (DA 101)

1. The headquarters of the Food and Agricultural Organization (FAO) is in:
   a) Geneva
   b) Rome
   c) Paris
   d) New York

2. ICAR was established in:
   a) 1929
   b) 1947
   c) 1954
   d) 1965

3. ICAR headquarters is located at:
   a) Hyderabad
   b) Pusa, New Delhi
   c) Pune
   d) Mumbai

4. PJTSAU stands for:
   a) Professor Jawaharlal Telangana State Agricultural University
   b) Professor Jayashankar Telangana State Agricultural University
   c) Punjab Jayashankar Telangana State Agricultural University
   d) Pusa Jayashankar Telangana State Agricultural University

5. PJTSAU was established in the year:
   a) 1972
   b) 2014
   c) 2016
   d) 1988

6. Father of Green Revolution in India is:
   a) Norman Borlaug
   b) Dr. M.S. Swaminathan
   c) Dr. B.P. Pal
   d) Dr. Verghese Kurien

7. Father of Green *Revolution globally* is:
   a) Dr. M.S. Swaminathan
   b) Norman Borlaug
   c) Dr. Verghese Kurien
   d) Dr. B.P. Pal

8. Father of White Revolution (Operation Flood) in India is:
   a) Dr. M.S. Swaminathan
   b) Dr. Verghese Kurien
   c) Dr. B.P. Pal
   d) Sir M. Visvesvaraya

9. Father of Yellow Revolution (Oilseeds) in India is:
   a) Dr. M.S. Swaminathan
   b) Dr. M.V. Rao
   c) Dr. Sam Pitroda
   d) Dr. K. Pratap Reddy

10. India's National Agricultural Research System (NARS) is led by:
   a) FAO
   b) ICAR
   c) NABARD
   d) Ministry of Commerce

14. India promotes millets under which brand name?
   a) Pulses
   b) Shree Anna (Millets)
   c) Sustainable Agriculture
   d) Natural Farming

15. NABARD was established in:
   a) 1969
   b) 1982
   c) 1995
   d) 2000

16. NABARD headquarters is located at:
   a) Delhi
   b) Pune
   c) Mumbai
   d) Hyderabad

17. Pradhan Mantri Fasal Bima Yojana (PMFBY) provides:
   a) Loans
   b) Crop insurance
   c) Subsidies
   d) Seed distribution

18. PM-KISAN scheme provides direct income support of:
   a) 4000 per year
   b) 6000 per year
   c) 10000 per year
   d) 12000 per year

19. e-NAM stands for:
   a) e-National Agricultural Market
   b) e-National Agricultural Mission
   c) e-National Agriculture Money
   d) e-Network Agricultural Market

20. Soil Health *Card scheme* was launched in:
   a) 2014
   b) 2015
   c) 2016
   d) 2018

21. Pradhan Mantri Krishi Sinchayee Yojana (PMKSY) is for:
   a) Insurance
   b) *Irrigation and* water management
   c) Fertilizers
   d) Seeds

22. The Largest Producer of Milk in the world is:
   a) USA
   b) India
   c) China
   d) New Zealand

23. India ranks 1st globally in production of:
   a) Wheat
   b) Pulses, Milk and Jute
   c) Maize
   d) Rice

24. India is the largest producer of which spice in the world?
   a) Black pepper
   b) Cardamom
   c) Most spices including chilli, ginger, turmeric, cumin
   d) Saffron

11. The Krishi Vigyan Kendra (KVK) is at which administrative level?
a) State
b) Block
c) District
d) National

12. Which year was declared as International Year of Millets by UN?
a) 2020
b) 2023
c) 2025
d) 2030

13. Which year was declared as International Year of Co-operatives by UN?
a) 2012
b) 2020
c) 2023
d) 2025

25. World Food Day is celebrated on:
a) October 16
b) July 1
c) June 5
d) March 22

Group 2: Soil Science & Soil Properties (DA 121)

26. World Water Day is observed on:
a) March 22
b) April 22
c) June 5
d) October 16

27. World Environment Day is celebrated on:
a) June 5
b) April 22
c) October 16
d) March 22

28. World Soil Day is observed on:
a) December 5
b) October 5
c) November 5
d) June 5

29. National Farmers Day (Kisan Diwas) is celebrated on:
a) October 16
b) December 23
c) July 1
d) August 15

30. World Food Prize 2024 was awarded to:
a) Dr. Geoffrey *Hawtin and* Dr. Cary Fowler
b) Dr. M.S. Swaminathan
c) Dr. Norman Borlaug
d) Dr. Mariangela Hungria

31. World Food Prize 2025 was awarded to:
a) Dr. Mariangela Hungria
b) Dr. Geoffrey Hawtin
c) Dr. Cary Fowler
d) Dr. Heidi Kuhn

32. The first World Food *Prize laureate* was:
a) Norman Borlaug
b) Dr. M.S. Swaminathan
c) Verghese Kurien
d) B.P. Pal

33. India's first Biochar Centre of *Excellence was* inaugurated at:
a) Punjab Agricultural University
b) Kanha Shanti Vanam, Hyderabad
c) IARI New Delhi
d) NDRI Karnal

39. Minimum Support Price (MSP) is recommended by:
a) RBI
b) NITI Aayog
c) CACP (*Commission for* Agricultural *Costs and* Prices)
d) FCI

40. Currently MSP is announced for how many crops?
a) 10
b) 23
c) 50
d) *All crops*

41. FRP for sugarcane stands for:
a) Fixed Rate Price
b) *Fair and* Remunerative Price
c) Final Rate Procurement
d) Farmer's Reservation Price

42. Buffer Stock of food grains in India is maintained by:
a) NAFED
b) CCI
c) FCI (Food Corporation of India)
d) NABARD

43. Public Distribution System (PDS) is operated by:
a) State *Government only*
b) Central *Government only*
c) Both *Centre and* State
d) RBI

44. India ranks ___ in world wheat production:
a) 1st
b) 2nd
c) 3rd
d) 4th

45. India ranks ___ in world rice production:
a) 1st
b) 2nd
c) 3rd
d) 4th

46. The largest mango-producing state in India is:
a) Maharashtra
b) Uttar Pradesh
c) Andhra Pradesh
d) Telangana

34. First National Co-operative University of India was established at:
a) Anand, Gujarat
b) New Delhi
c) Mumbai
d) Hyderabad

35. NaBFID (National *Bank for* Financing *Infrastructure and* Development) is for:
a) Crop loans
b) Long-term infrastructure financing
c) Microfinance
d) Subsidies

36. VKSA stands for:
a) Vikas Krishi Surakshit Abhiyan
b) Viksit Krishi Sankalp Abhiyan
c) Viksit Kisan Samagra Abhiyan
d) Vikasit Kisan Sahaayata Abhiyan

37. MGNREGA guarantees how many days of employment per year?
a) 50
b) 100
c) 150
d) 365

38. Which of these is NOT an agricultural input subsidy?
a) *Fertilizer subsidy*
b) Seed subsidy
c) *Power subsidy* for irrigation
d) GST

47. The Father of Hybrid Rice is:
a) Dr. M.S. Swaminathan
b) Yuan Long Ping
c) Norman Borlaug
d) Dr. B.P. Pal

48. India's largest producer state of cotton is:
a) Punjab
b) Maharashtra
c) Gujarat
d) Telangana

49. *Largest producer* state of sugarcane in India is:
a) Maharashtra
b) Uttar Pradesh
c) Karnataka
d) Tamil Nadu

50. The largest tea-producing state in India is:
a) Assam
b) West Bengal
c) Kerala
d) Tamil Nadu

Group 3: Horticulture & Forestry (DA 282 & DA 281)

51. The largest coffee-producing state in India is:
a) Karnataka
b) Kerala
c) Tamil Nadu
d) Andhra Pradesh

52. *Largest fish*-producing state in India is:
a) Andhra Pradesh
b) Kerala
c) West Bengal
d) Tamil Nadu

53. Indian Council of Agricultural Research (ICAR) is under which Ministry?
a) Education
b) Science
c) DARE Ministry of Agriculture
d) Commerce

54. Indian Agricultural Research Institute (IARI) is located at:
a) Mumbai
b) New Delhi (Pusa)
c) Pune
d) Hyderabad

55. NIPHM (National Institute of Plant Health Management) is located at:
a) Pusa
b) Hyderabad
c) Bangalore
d) New Delhi

56. Central Rice Research Institute (CRRI/NRRI) is at:
a) Cuttack, Odisha
b) New Delhi
c) Hyderabad

64. India's first agricultural college was established at:
a) Pune (1879)
b) Pusa
c) Coimbatore
d) New Delhi

65. India's first State Agricultural *University was* established at:
a) Coimbatore
b) Pantnagar (1960)
c) Hyderabad
d) Bangalore

66. The 73rd Constitutional Amendment Act 1992 is related to:
a) *Urban local* bodies
b) Panchayati Raj (*Rural local* self-government)
c) *Cooperative societies*
d) Land reforms

67. The 74th Constitutional Amendment is related to:
a) Panchayati Raj
b) *Urban local* bodies (Municipalities)
c) Co-operatives
d) Land

68. India's National *Mission for* Sustainable Agriculture (NMSA) focuses on:
a) Heavy fertilizer use
b) Climate-resilient and sustainable farming
c) *Plantation crops*
d) *Marine fishing*

69. Genome-edited Rice varieties developed by ICAR (announced 2025):
a) DRR Dhan 100 and Pusa DST Rice 1
b) IR-8 and IR-36
c) Basmati 370 and Sona Masuri

d) Coimbatore

57. Indian Institute of Pulses Research (IIPR) is at:
a) Hyderabad
b) Kanpur, UP
c) Pusa
d) Coimbatore

58. Sugarcane Breeding Institute (SBI) is located at:
a) Cuttack
b) Coimbatore
c) Pune
d) Nagpur

59. ICRISAT is located at:
a) New Delhi
b) Hyderabad (Patancheru)
c) Pune
d) Coimbatore

60. IRRI (International Rice Research Institute) is in:
a) India
b) Philippines (Los Banos)
c) China
d) Thailand

61. CIMMYT (International Maize and Wheat Improvement Center) is in:
a) USA
b) Mexico
c) India
d) Italy

62. The Svalbard Global Seed Vault is in:
a) USA
b) Norway
c) Italy
d) India

63. India's National Gene *Bank for* plant genetic resources is at:
a) IARI New Delhi
b) NBPGR New Delhi
c) NBAII Bangalore
d) CTCRI Trivandrum

d) Pusa 1121 and Pusa 1509

70. Bt Cotton (genetically modified) was approved in India in:
a) 1995
b) 2002
c) 2010
d) 2015

71. GEAC stands for:
a) Global Environment Assessment Committee
b) Genetic Engineering Appraisal Committee
c) Government Environmental Authority of Crops
d) Genetic *Evaluation and* Certification Committee

72. India's per capita arable land is approximately:
a) 1.5 ha
b) 0.5 ha
c) 0.12 ha
d) 2.0 ha

73. The largest source of agricultural credit in India is:
a) Cooperatives
b) Commercial banks
c) *Money lenders*
d) Regional Rural Banks (RRBs)

74. Kisan Credit Card (KCC) was launched in:
a) 1995
b) 1998-99
c) 2005
d) 2014

75. *Largest organic* farming state in India (first fully organic state) is:
a) Sikkim
b) Maharashtra
c) Madhya Pradesh
d) Karnataka

Group 4: Farm Management, Surveying, Extension & Farm Power (DA 241, DA 252, DA 291, DA 151, DA 262, DA 263)

76. The current N:P:K consumption ratio in India is approximately:
a) 4:2:1 (ideal)
b) 11:4:1 (actual, skewed)
c) 1:1:1
d) 10:5:1

77. From which year was 100% neem coating of subsidised urea made mandatory in India?
a) 2012
b) 2015
c) 2018
d) 2020

78. India's first genome-edited rice was named:
a) Kamala (DRR Dhan 100)
b) Sona Masuri
c) Pusa Basmati
d) IR-64

79. India's National Food Security *Act was* enacted in:
a) 2005
b) 2009
c) 2013

87. Rabi season is:
a) June to October
b) October/November to March/April (winter)
c) July to September
d) *All year* round

88. Zaid season refers to:
a) *Monsoon crops*
b) Summer crops grown between Rabi and Kharif (March to June)
c) Winter crops
d) Perennial crops

89. India's largest exported agricultural commodity by value is:
a) Tea
b) Spices
c) *Marine products* and *Basmati rice*
d) Sugar

90. India's highest agricultural import (by value) is:
a) Wheat
b) Sugar
c) *Edible oils*
d) Fertilizers

d) 2016

80. Pradhan Mantri Krishi Sinchayee *Yojana motto* is:
a) *Health for* all
b) Har Khet Ko *Pani and* Per Drop More Crop
c) Jai Jawan Jai Kisan
d) Land for landless

81. PJTSAU recently organized a state-wide campaign Rythu Mungitlo *Shastravethalu during*:
a) May 2025
b) July 2025
c) October 2025
d) 2024

82. The current Director General of ICAR is:
a) Dr. Himanshu Pathak
b) Dr. Trilochan Mohapatra
c) Dr. M.L. Jat
d) Dr. R.S. Paroda

83. ATMA (Agricultural Technology Management Agency) operates at which level?
a) National level
b) State level
c) *District level*
d) *Block level*

84. Aspirational Districts Programme (ADP) was launched in:
a) 2014
b) 2018
c) 2020
d) 2022

85. PM-AASHA is for:
a) Soil health
b) *Income protection* of farmers at MSP
c) Irrigation
d) Insurance

86. In Indian *Agriculture kharif* season is approximately:
a) October to March
b) June to October (monsoon-dependent)
c) April to June
d) *Throughout the* year

91. *Largest consumer* of urea in the world is:
a) USA
b) China
c) India
d) Brazil

92. Agriculture's share in India's GDP is approximately:
a) 5 percent
b) 17 to 18 percent
c) 30 percent
d) 45 percent

93. World Pulses Day is celebrated on:
a) February 10
b) June 5
c) March 22
d) October 16

94. The first Five Year Plan in India primarily focused on:
a) Industry
b) *Agriculture and* irrigation
c) *Service sector*
d) Defence

95. AGRICET (Agriculture Common Entrance Test) is conducted by:
a) ICAR
b) PJTSAU (for Telangana state)
c) SCERT
d) Central Government

96. Father of Modern Agriculture in India is regarded as:
a) Norman Borlaug
b) Dr. M.S. Swaminathan
c) Dr. Verghese Kurien
d) Sir Albert Howard

ANSWER KEY — GENERAL AGRICULTURE

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	A	B	B	B	B	B	B	B	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	B	D	B	B	C	B	B	A	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	B	B	C	A	A	A	A	B	A
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
A	B	B	A	B	B	B	D	C	B
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
B	C	C	B	B	B	B	B	B	A
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
A	A	C	B	B	A	B	B	B	B
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
B	B	B	A	B	B	B	B	A	B
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
B	C	B	B	A	B	B	A	C	B
81	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**
A	A	C	B	B	B	B	B	C	C
91	**92**	**93**	**94**	**95**	**96**				
C	B	A	B	B	B				

PREVIOUS YEAR AGRICET QUESTION PAPERS

PJTSU AGRICET — Diploma in Agriculture | 2023 · 2024 · 2025 | 300 Questions with Answers

Note: The 2023 AGRICET original question paper (QP) is a scanned image PDF and cannot be reproduced as text. The 2023 paper below is reproduced from the official Answer Key (SET II) which contains full questions and answers. The original scanned QP is provided separately as AGRICET2023_DA_QP.pdf.

AGRICET 2023 — AGRICULTURE SET II

100 Questions | Answers given below each question | Correct option shown in BOLD

1. *Precipitation occurs* from one of the following clouds
- **a) Nimbus**
- b) Cirrus
- c) Stratus
- d) Cumulus

2. The following weed induces hypersensitivity to light
- a) *Sorghum halepense*
- b) Rhododendron sp
- c) Ambrosia sp
- **d) *Lantana camara***

3. Short cut key to insert hyper link is
- **a) Ctrl + K**
- b) Ctrl + V
- c) Ctrl + R
- d) Ctrl + N

4. Anti- erosion value of the following crop is highest
- a) Cotton
- **b) Pillipesara**
- c) Sugarcane
- d) Maize

5. The planting method that doesn't require intercultural operations is
- a) *Border strip* method
- b) *Furrow method*
- **c) Corrugations**
- d) *Check basin* method

6. One of the following classes of soils are suitable for forage crop cultivation (c)
- **a) I to II**
- b) II to III
- c) III to VII
- d) Above VII

7. Which among the following varieties is not correctly matched A. Redgram : PRG 176 B. Bengalgram : Kranthi ICCC37 C. Greengram : DHM -107 D. Blackgram : WGG-42
- a) A and B only
- **b) C and D only**
- c) A and C only
- d) B and D only

8. In the context of seed rates of different crops, choose the correct order of (b) seed rate per acre in descending order
- **a) Rice- Greengram Soybean Bengalgram**
- b) Rice Soybean Bengalgram Greengram

51. The part connected to cylinder head is
- a) Piston
- b) *Connecting rod*
- **c) *Spark plug***
- d) *Oil rings*

52. *Grain storage* structure which is made up of wood and raised on pillars is (a) known as
- **a) Kothari**
- b) Morai
- c) Bukhari
- d) Bin

53. *Causal organism* of powdery mildew in chilli
- a) *Erisiphe polygoni*
- **b) *Leveillula taurica***
- c) *Pythium aphanidermatum*
- d) *Uncinula necator*

54. *Groundnut bud* necrosis is transmitted by
- **a) Thrips**
- b) Whitefly
- c) Green leaf hopper
- d) Mite

55. *Wider spacing* of 90 x 60 cm is adopted for management of which disease (a) in castor
- **a) *Grey rot***
- b) Root rot
- c) Wilt
- d) Blight

56. The floral parts are transformed into green leafy structure due to this disease in sesamum
- a) Powdery mildew
- **b) Phyllody**
- c) *Alternaria leaf* spot
- d) Root and stem rot

57. In citrus the following disease is managed by double ring method of irrigation
- a) Tristeza
- **b) Gummosis**
- c) Root rot
- d) Canker

58. The word destiny has --- syllables
- a) One
- b) Two
- **c) Three**
- d) Four

c) Greengram Rice Bengalgram Soybean
d) Bengalgram Soybean Greengram - Rice

9. *Carbohydrates are* high in --- type of corn
a) *Pop corn*
b) *Flint corn*
c) *Waxy corn*
d) *Pod corn*

10. *Bloat disease* in animals is caused due to consumption of which fodder grass
a) *Para grass*
b) Berseem
c) *Guinea grass*
d) Lucerne

11. Which of the following crop can be used as pulse, fodder and green manure?
a) Paddy
b) Redgram
c) Maize
d) Cowpea

12. L & Y shaped cracks on pomegranate is the symptom of LY
a) Anthracnose
b) *Bacterial leaf* spot
c) *Cercospora leaf* spot
d) *Die back*

13. *Azure blue* colour seed tag is attached to one of the following class of seed (d)
a) *Foundation seed*
b) *Breeder seed*
c) *Nucleus seed*
d) *Certified seed*

14. The sample drawn with the help of triers / hand from a bag or in bulk is called
a) Primary sample
b) *Composite sample*
c) *Submitted sample*
d) *Working sample*

15. *Example for* protogyny crop
a) Paddy
b) Greengram
c) Bajra
d) Sesamum

16. *Economic threshold* level for rice hispa per hill is
a) 1
b) 2
c) 3
d) 4

17. Robinson's pipette method works on the principle of
a) Continuous decrease in density
b) Sedimentation
c) Formation of aggregates
d) *Tyndal effect*

18. Eutrophication of water bodies occurs due to accumulation of higher amounts of

59. An example of fungicide which belongs to copper sulphate group is
a) *Bordeaux mixture*
b) *Lime sulphur*
c) Captan
d) Carbendazim

60. "Hollow Heart" in groundnut is caused due to deficiency of one of the following element?
a) Nitrogen
b) Calcium
c) Boron
d) Potassium

61. The cultivated cotton with longest and finest fibre is
a) *Fine cotton*
b) *Arboreum cotton*
c) *Sea island* cotton
d) *Herbaceum cotton*

62. High presence of which element in soil induces Magnesium (Mg) deficiency (d) in crops
a) Boron
b) Nitrogen
c) Sulphur
d) Calcium

63. One of the following components is used in second generation of computers (b)
a) *Vacuum tubes*
b) Transistors
c) *Integrated circuits*
d) Very large scale *Integrated circuits*

64. Which of the following is VDU of the computer
a) Printer
b) Keyboard
c) Monitor
d) RAM

65. The fund launched by Government of India to increase the number of agricultural startups in rural areas is
a) Startup India Seed Fund Scheme
b) Agriculture Accelerator Fund
c) Atal Innovation Mission
d) Startup India

66. Government of India promotes millets with one of the following brand name
a) *Shree anna*
b) Nutricereals
c) *Millets give* away
d) Poorman's food

67. From the following choose a mismatched option
a) Sesamum Pedaliaceae
b) Cotton Malvaceae
c) Tobacco Compositae
d) Castor Euphorbiaceae

68. Law of proportionality explains that, an increase in some inputs relative to (a) other fixed inputs will be in a given state of technology cause output to increase, but after a

a) *Nitrate nitrogen*
b) *Nitrogen and* Phosphorous
c) *Radio isotopes*
d) Pesticides

19. *Predominant soil* structure present in arid and semi arid regions of Rajasthan is
a) Plates
b) Prismatic
c) Crumb
d) Vesicular

20. Under anaerobic decomposition process the end products are
a) CO2
b) NH4+
c) CH4
d) NO3-

21. *For plants*, molybdenum is available in the form of
a) MoO2+
b) MoO22-
c) MoO42-
d) MoO32-

22. Soil moisture constant suitable for optimum plant growth is
a) Permanent wilting point
b) *Hygroscopic coefficient*
c) Saturation
d) *Field capacity*

23. *Disease resistance* is improved by this element
a) Nitrogen
b) Phosphorous
c) Sulphur
d) Potassium

24. *Match the* following and choose the correct answer
a) Bacteria
b) Algae
c) Actinomycetes
d) A-III; B-I; C-II c A-I; B-II; C-III

25. Which of the following is a compost making fungi
a) Thermanospora
b) Chaetomium
c) Nocardia
d) Streptomyces

26. In mixed fertilizers the material used to reduce crystal nitting is
a) Coal
b) Silica
c) Gypsum
d) Lime

27. *Bone meal* is highly useful for ___ soils
a) *Saline soils*
b) *Acid soils*
c) *Alkali soils*
d) Saline-alkali soils

28. Which of the following is not an *Urease inhibitor*

point the extra output resulting from the same addition of extra inputs will become ----
a) Less
b) Increase
c) Stable
d) No change

69. *Match the* following
a) Deccan Agriculture Relief Act 1 1912
b) Cooperative Societies Act
c) Madras Cooperative Land
d) Land Improvement Loan Act 4 1934

70. Lead Bank *Scheme was* recommended by which of the following committee? ?
a) Prof. D.R. Gadgil Study Group by National Credit Council (NCC) 1969
b) Sri F.K.F Nariman Committee by Reserve Bank of India 1969
c) Sri B.P. Patel Committee on Taccavi 1961-62
d) Sri R.G. Saraiya Cooperative Planning Committee

71. Break Even *Analysis relies* on the assumption of --- (a) Constant factor prices (b) Constant technology (c) Constant selling prices (d) Constant supply & demand
a) a and b only
b) b and c only
c) a,b and c only
d) a and c only

72. *Choose correct* formula for calculating Average Variable Cost (AVC)
a) TVC/Q
b) TC/Q
c) MC/Q
d) ATC/Q

73. *Fore bearing* is measured in the direction of
a) Progress of survey
b) Opposite to progress of survey
c) North
d) South

74. *For application* of trapezoidal and simpson's rule in measuring area of irregular boundaries, on a base line the interval between successive ordinates must be
a) Uniform
b) Varied by half
c) Varied by twice
d) Varied by 10 times

75. The longest chain line in chain surveying is called as
a) *Base line*
b) *Check line*
c) *Tie line*
d) Closed line

76. The valve used to prevent reverse flow in pipes is
a) *Flush valve*
b) *Gate valve*
c) *Check valve*
d) *Pressure valve*

77. The impeller which doesn't have side wall and shroud is called as

a) *Hydroxamic acid*
b) Thiourea
c) *Methyl urea*
d) Dicyandiamide

29. Pyrethrum is extracted from
a) *Azadiracta indica*
b) *Nicotiana tabacum*
c) Chrysanthemum cinerarifolium
d) *Tephrosia macropoda*

30. Small *Image that* represents a file, folder or programme is
a) Chip
b) Floppy
c) Icon
d) RAM

31. *For control* of which pest marigold can be used as trap crop in cotton
a) *Tobacco caterpillar*
b) *American boll* worm
c) Whiteflies
d) Mite

32. Which of the following chemical blocks acetylcholinesterase and affects nervous system
a) Organophosphates & carbamates
b) Organophosphates & *Arsenical compounds*
c) Organophosphates & Precocenes
d) Carbamates & organochlorines

33. Which of the following is not related to rodenticides
a) *Zinc phosphide* in the form of black powder
b) *Aluminium phosphide* releases phosphine gas
c) Bromadiolone is used in poison bait
d) *Methyl bromide* belongs to anticoagulant type

34. Which of the following chemical affects fecundity of mite & whiteflies
a) Tebufenozide
b) Spiromesifen
c) Pymetrozine
d) Indoxacarb

35. *Match the* following parts of an insect body
a) Pulvilli
b) Palpifer
c) Scape
d) Palpiger

36. *Stone flies* belong to one of the following orders
a) Hemiptera
b) Odonota
c) Mallophaga
d) Plecoptera

37. What is the diapausing stage of red hairy caterpillar
a) Egg
b) Larval
c) Pupal
d) Adult

a) Closed
b) Open
c) *Semi open*
d) *Semi closed*

78. *Geodetic survey* is also called as
a) *Plane survey*
b) Trigonometric survey
c) *Compass survey*
d) *Chain survey*

79. Which leader believes that the work will be completed quickly if the workers (d) were left alone
a) Lay leader
b) *Autocratic leader*
c) *Democratic leader*
d) Laissez *Faire leader*

80. Muriate of *Potash cannot* be used in one of the following soils
a) Saline
b) Red sandy
c) Black cotton
d) Loamy

81. The developmental programme in which Government of India has given (a) high priority to water conservation and management is
a) Prime Minister Krishi Sinchayee Yojana
b) Paramparagat Krishi Vikas Yojana
c) Prime Minister Fasal Bheema Yojana
d) National Mission on Oil Seeds & Oil Palm

82. The example of an interjection is
a) of
b) an
c) oh
d) but

83. *Forest that* reduces bronchitis disease
a) Teak
b) Sal
c) Coniferous
d) Aerobusta

84. Generally, wind breaks are planted in --- direction
a) North West
b) North South
c) East West
d) South North

85. Pashanbedi is commonly known as---
a) Coleus
b) Senna
c) *Lemon grass*
d) Citronella

86. Which of the following oil is used in preparation of mosquito repellent creams
a) Vettiver
b) Lemon grass
c) Kalmegh
d) *Sacred basil*

38. Root grub adult after emergence from soil are preferably found on this tree (b)
- **a) Coconut**
- b) Neem
- c) Guava
- d) Pongamia

39. Which adult is dark brown moth with black hind wings having white band (a) medially and three large white spots on the center margins
- a) *Castor semilooper*
- **b) *Castor shoot* & capsule borer**
- c) *Castor stem* borer
- d) White grub

40. *Identify the* sugarcane pest with milky white coloured nymphs and a pair of (b) characteristic processes/ filament covered by wax
- **a) Early shoot borer**
- b) Leaf hopper
- c) *Internodal borer*
- d) Scales

41. In chillies the pest causing downward curling belongs to the order
- a) Lepidoptera
- **b) Acarina**
- c) Diptera
- d) Thysanoptera

42. In cashew, the infestation of the following insect pest leads to holes on the (b) lower leaves with withered appearance on the leaves
- **a) *Shoot borer***
- b) Leaf webber
- c) Stem borer
- d) White grub

43. In onion, the infestation of one of the following pest leads to white blotches on leaves which later turn to brownish in colour
- **a) Thrips**
- b) Leaf eating caterpillar
- c) Redmite
- d) Leaf folder

44. In coconut, infestation of the following pest leads to "V" shaped cutting on (a) fronds
- **a) *Rhinoceros beetle***
- b) Black headed caterpillar
- c) Red palm weevil
- d) *Eriophyid mite*

45. In a power thresher, the effect of reciprocating motion of top sieve on lighter materials is
- a) *Falls through* sieve
- b) Doesn't get separated
- c) *Blows out*
- **d) Accumulate at the top**

46. Which of the following is an intercultural operation tool / equipment
- a) *Bund former*
- b) Ridger
- **c) *Wheel hoe***

87. Which of the following tree bark yields oxalic acid
- a) Bamboo
- b) Subabul
- c) Karakkaya
- **d) Eucalyptus**

88. Which of the following plant is called as poor man's rose
- a) Davana
- **b) Geranium**
- c) Citronella
- d) *Lemon grass*

89. In case of sunk costs, the TFC curve is
- a) Inverse `S' shaped
- **b) *Straight line* parallel to X-axis**
- c) *Rectangular hyperbola*
- d) *Straight line* parallel to Y-axis

90. The chemical used to over come the seed dormancy is
- a) *Citric acid*
- b) *Acetic acid*
- **c) H2SO4**
- d) HNO3

91. The method of grafting, in which scion shoot is still attached to the mother (a) plant till the grafting union takes place is known as ___
- **a) Inarching**
- b) *Side grafting*
- c) *Veneer grafting*
- d) *Epicotyl grafting*

92. In Cucurbits, the female flower ratio can be increased by spraying of
- a) IAA
- b) GA
- **c) Ethrel**
- d) ABA

93. In Cauliflower, `Whiptail' a physiological disorder is due to deficiency of
- a) Nitrogen
- b) Boron
- c) Calcium
- **d) Molybdenum**

94. The fourth step in extension educational process is
- a) Situation & Problems
- **b) Evaluation**
- c) Reconsideration
- d) *Teaching plan* of work

95. The location of audio visual aids which gives direct experience to audience (b) in cone of experience given by Dr. Edgar Dale is
- **a) Top**
- b) Bottom
- c) Middle
- d) Side

96. Using wide range of information methods to cross check the collected data is
- a) PRA
- b) *Statistical replicability*

d) Puddler

47. A disc harrow assembly of concave discs mounted on a common shaft with (c) spools in between is called

a) Bearing
b) Arbor
c) Gang
d) Spacer

48. One of the following is a component of a *Mould board* plough

a) Scraper
b) Disc
c) *Furrow wheel*
d) Jointer

49. The main purpose of power transmission system in a tractor is

a) *Speed variable* to rear wheels
b) *Reduce fuel* consumption
c) Speed up lubricating
d) *Monitor engine* functioning

50. The type of sprayer which develops a pressure of about 4 to 10 kg/cm2 is (a) kg/cm2

a) Bucket sprayer
b) *Knapsack sprayer*
c) *Foot operated* sprayer
d) *Power sprayer*

c) Mix of techniques
d) Triangulation

97. To meet the location specific needs of the farmers at district level, the (c) following extension reform is taken up

a) Diagnosis
b) Integration
c) Strategic *Research and* Extension Plan
d) *Group formation*

98. Antonym of dismal is

a) Intelligent
b) Agitate
c) Egress
d) Shrivel

99. Which of the following is not a function of the gram panchayat ?

a) *Water supply*
b) *Birth and Death registration*
c) *Women and* child care
d) Rural housing

100. Indian Council of Agricultural Research (ICAR) has signed MoU with one (b) of the following digital platforms to guide farmers on scientific cultivation

a) Kisan sarathi
b) *Amazon kisan*
c) *Digital green*
d) Flipkart

AGRICET 2024 — DIPLOMA IN AGRICULTURE SET I

100 Questions | Answers given below each question | Correct option shown in BOLD

1. Commercial bread wheat is an example of which ploidy?
a) Octaploid
b) Triploid
c) Hexaploid
d) Tetraploid

2. *Match the* following and choose the correct answer (Paddy, Maize, Bajra, *Groundnut with* cultivation operations)
a) I-b-4, II-a-3, III-d-2, IV-c-1
b) I-c-4, II-a-2, III-d-3, IV-b-1
c) I-b-1, II-d-3, III-a-2, IV-c-4
d) I-c-1, II-a-3, III-d-2, IV-b-4

3. Which of the following species bear male & female flowers on separate plants
a) Rice
b) Maize
c) Pea
d) Date palm

4. *Match the* following 1 Stolon
a) Colocasia
b) Strawberry
c) Banana
d) Mint

5. Physical purity in Groundnut
a) 96%
b) 95%
c) 98%
d) 99%

6. Which of the following are objectionable weeds in mustard seed production (c)
a) Chicoruim intybus
b) Melilotus sp.
c) *Argemone mexicana*
d) *Lactuca sativa*

7. Which feature of MS-*Word helps* to create a list in a document
a) *Word art*
b) *Hyper link*
c) Scaling
d) *Bullets and* numbering

8. Which are the nutrients absorbed by plants in the form of anion?
a) Boron, Chlorine, Copper & Manganese
b) Copper, Iron, Sulphur & Zinc
c) Potassium, Manganese, Zinc & Phosphorus
d) Chlorine, Molybdenum, Phosphorus & Sulphur

9. The size of micro pores is
a) >0.06 mm
b) <0.006 mm <
c) <0.06 mm
d) > 0.6 mm

51. *Match the* following and choose the correct answer Disease Vector 1 *Bhendi yellow* mosaic
a) Mite
b) Thrips
c) Aphid
d) Whitefly

52. *Recommended resistant* root stock to tristeza disease in citrus is
a) Rangapur lime
b) *Tenali selection*
c) Jamberi
d) Balaji

53. *Tissue along* the vein is green and tissue between the veins is chlorotic is (b) known as
a) Vein clearing
b) *Vein banding*
c) Distortion
d) Enation

54. Which among the following is not correctly matched Disease Resistant variety a *Turmeric leaf* spot PCT -13 PCT -13 b *Rose dieback Bebe bune* c *Castor grey* mold Jwala d *Groundnut bud* necrosis Vemana
a) a&b
b) a, b & c
c) a, b & d
d) c&d

55. Which of the following comes under self liquidating loan?
a) *Term loan*
b) *Home loan*
c) Open loan
d) Crop loan

56. As per the balance sheet livestock loan comes under--------------
a) Long term liabilities
b) Intermediate liabilities
c) Short term liabilities
d) Long term Assets

57. NABARD head office is located at______
a) Delhi
b) Chennai
c) Pune
d) Mumbai

58. Cash on hand comes under which of the following assets
a) Long term
b) Intermediate
c) Current
d) Working

59. *Arrange the* following in chronological order according to their year of (a) establishment
a) *Cooperative credit* societies act b *Lead bank* scheme
b) NABARD
c) Kisan Credit Card

10. ***Match the*** **following List - I I List - II II List - III III A. Sugarcane *Top sickness I. zinc* B. Maize Whiptail II. Boron C. Tobacco *Pahala blight* III. Molybdenum D. Cauliflower White bud IV. Manganese**

a) A-2-I, B-4-II, C-3-III, D-1-IV
b) A-2-IV, B-3-II, C-4-I, D-2-III
c) A-3-IV, B-4-I, C-1-II, D-2-III
d) A-4-I, B-1-III, C-3-II, D-2-IV

11. Which of the following are not correctly matched?

a) Pillipesara — Triple purpose green manure crop
b) Diancha — Green manure crop that corrects sodic soils
c) Wild Indigo — suitable for waterlogged conditions
d) *Sesbania rostrata* — stem nodulating green manure crop

12. What is the bulk density of soils in general?

a) 1.33 g/cc
b) 1.98 g/cc
c) 2.56 g/cc
d) 1.08 g/cc

13. What is the soil problem that is encountered due to excessive use of irrigation water over a prolonged period of time Assumption i : *Water logging* and excessive crop yield i Assumption ii : Development of salinity and alkalinity in soils ii

a) Both assumptions are true
b) Only (i) is true but (ii) is false
c) Both assumptions are false
d) Only (ii) is true but (i) is false

14. Who is father of Soil Science

a) Jenny
b) Jacob
c) Docuchaev
d) Buckman

15. Which of the following soils is deficient in organic matter?

a) Black soil
b) *Alluvial soil*
c) Sandy soil
d) Red soils

16. Electrical conductivity meter (EC meter) can be standardized with the (b) following (EC meter)

a) Mercuric Chloride
b) Potassium Chloride
c) *Deionised water*
d) 7.00 Buffer

17. How much of DAP a farmer has to apply if he has to supplement 60 kg (a) N per hectare through it

a) 333.33 kg DAP
b) 285.50 kg DAP
c) 300.00 kg DAP
d) 450.50 kg DAP

18. ***Calculate porosity*** **of soil when bulk density and particle density are 1.54 (b) g/cc and 2.62 g/cc, respectively**

a) 20.38%
b) 41.30%
c) 72.08%
d) 58.02%

d) c, b, a, d

60. In the context of seed rates in different method of rice cultivation practices, choose the correct order of seed rate per acre in descending order a. SRI method b. Seed drill (rainfed) c. *Dry paddy* d. Broadcasting a.b. c. d.

a) a>b>c>d
b) d>c>b>a
c) b>d>c>a
d) d>a>c>b

61. ***Bajra variety*** **suitable to cultivate only during kharif season is**

a) ICTP 8203
b) ICMV 221
c) APS 1
d) ICMH-451

62. ***Choose the*** **correct statements about *Horsegram from* the following a. Kharif pulse crop b. *Test crop* for drought c. Crop of virgin soils d. Poorman's pulse crop**

a) a, b, c, d
b) a, b, c
c) b, c, d
d) a, b, d

63. Which is not correct as per paddy harvesting index

a) Green grains < 4-9%
b) Milk grains < 2.5%
c) *Grain moisture* < 20%
d) 80% panicles are straw coloured

64. ***Striga emerges*** **__ days after sorghum seed germination**

a) 25-30
b) 35-40
c) 45-50
d) 50-55

65. ***Match the*** **following**

a) Trishulatha
b) WCC-75
c) Hima
d) Palnadu

66. The cultivated cotton with longest and finest fibre is

a) *Fine cotton*
b) *Arboreum cotton*
c) *Sea island* cotton
d) *Herbaceum cotton*

67. A solid chemical fertilizer with high nitrogen content is

a) *Ammonium chloride*
b) *Calcium nitrate*
c) Urea
d) *Calcium ammonium* nitrate

68. HDBRG (Guntur) *Tobacco contain* __ % of Nicotine HDBRG

a) 1.3
b) 3.5
c) 4.8
d) 3.9

69. Family of sesame crop

19. Which group of insecticides disrupts ATP formation? ATP)

a) Organophosphates
b) Oxadiazines
c) *Phenyl pyrazoles*
d) Thio-urea derivaties

20. The neem seed extract or neem oil contains

a) Azadirachtin, Pyrethrum
b) Salannin, Ryania
c) Epinimbin, Pyrethrum
d) Salannin, Nimbin

21. The hind legs in grasshopper are modified as ____ type ___

a) Fossorial
b) Raptorial
c) Natatorial
d) Saltatorial

22. Which of the following insects comes under order Hymenoptera i Wasps, Ants, Midges, Sawflies ii. Wasps, Ants, Bees, *Sawflies iii.* True flies, Mosquitoes, Wasps, Ants iv. True flies, Ants, Bees, Sawflies

a) ii,iii
b) ii only
c) i,ii
d) iv only

23. Examples of internal feeder stored grain pest

a) Red flour beetle, Pulse beetle
b) Pulse beetle, *Grain weevil*
c) Rice moth, Pulse beetle
d) *Saw toothed* beetle, *Cigarette beetle*

24. The egg is whitish cigar shaped or flattened boat shaped with wing like lateral projection in

a) *Sorghum earhead* bug
b) *Sorghum shoot* fly
c) *Sorghum midge*
d) *Sorghum aphid*

25. *Chlorotic patches* at tips of leaflets in a typical "V" shape symptoms present on groundnut "V"

a) Aphid
b) Leaf hopper
c) Thrips
d) White grub

26. Which of the following pest of citrus larva resembles bird's drop

a) *Citrus leaf* miner
b) *Citrus bark* eating caterpillar
c) *Citrus butterfly*
d) *Citrus fruit* sucking moth

27. *Violent destructive* storm of small horizontal dimensions on land is

a) Cyclone
b) Anticyclone
c) Tornado
d) Waterspout

28. Anti-erosion value of the following crop is highest

a) Asteraceae
b) Leguminosae
c) Pedaliaceae
d) Malvaceae

70. *Consider the* following operations carried out in sugarcane crop a. Propping b. Nipping c. Ratoon d. Detrashing

a) a, b, c and d
b) a, c and d
c) a, b and c
d) a and b

71. *Nipping means* A. Removal of the auxiliary buds B. Removal of the root tips C. Removal of the lateral roots

a) Only A is correct
b) Only A & C are correct
c) Only B and C are correct
d) Only C is correct

72. Alkaloids in Indian-ginseng are

a) Withanine A & D, Somniferine
b) & D
c) Withanine A & B only
d) & B

73. Sona is high yielding variety of __ crop

a) Coleus
b) Aloevera
c) Senna
d) Geranium

74. *Per capita* forest area in world is ___ ha.

a) 1.2
b) 1.4
c) 1.5
d) 1.6

75. One kg of teak seed contains around __ number of seeds

a) 500
b) 600-1000
c) 1200-3000
d) 3200-4000

76. TSS in Jelly

a) 33%
b) 40%
c) 65%
d) 68%

77. In sacred basil (Ocimum) seed required to transplant in one acre area is (c) (Ocimum)

a) 50-100g
b) 100-150g
c) 200-300g
d) 500-600g

78. Which of the following pair of fruit pigment found is correct? I Onion Anthocyanin II Chilli Capsanthin III Carrot Carthamin

a) Only I and III
b) Only III
c) Only I and II
d) I, II and III

a) Cotton
b) Pillipesara
c) Sugarcane
d) Maize

29. Maize crop is sown with a spacing of 80 cm x 40 cm. *Calculate the* plant (b) population per hectare
a) 41250 plants
b) 31250 plants
c) 41500 plants
d) 35250 plants

30. *Choose the* correct answer from the following statements A. *Multi storeyed* cropping is growing of plants of different heights on same piece of land at same time B. *Inter cropping* is growing two or more crops on the same piece of land simultaneously
a) Neither A nor B
b) A only
c) B only
d) Both A and B

31. Assertion : The temperature in the upper parts of stratosphere is almost (c) as high as near the earth's surface Reason : U.V radiation from the sun is absorbed by ozone in the stratosphere region
a) Assertion is true & reason is false
b) Both assertion & reason are false
c) Assertion is correct & reason
d) Assertion is correct & reason

32. Statement 1 : The crop to be grown influences the preparatory tillage Statement 2 : *Tilth influences* the crop emergence
a) Statement 1 is correct, 2 is
b) Both statements are correct
c) Both statements are incorrect
d) Statement 1 is incorrect, 2 is correct

33. One of the following is a drought tolerant crop?
a) Bajra
b) Rice
c) Wheat
d) Maize

34. *Match the* following List i List ii
a) IIPR
b) CRRI
c) IIMR
d) SBI

35. What is the shortcut key to redo the last action
a) Ctrl +Q
b) Ctrl + X
c) Ctrl + Y
d) Ctrl + Z

36. *Match the* following and choose the correct answer List i List ii 1 Secondary Tillage
a) *Thermo siphon* method
b) 9-35 Tonnes
c) Weeding
d) 1-E, 2-C, 3-D,4-B, 5-A

79. Amrutha is a variety of
a) Karakkaya
b) Tamarind
c) Pongamia
d) Jatropha

80. The following phytohormone is antagonistic to auxin
a) Cytokinins
b) Gibberellins
c) *Abscissic acid*
d) Ethylene

81. *Match the* following and identify the correct option 1 NABARD
a) Sri Venkatappaiah
b) V. L. Mehtha
c) Mac Lagan Committee
d) 1-a,2-b,3-c,4-d

82. A market in which commodities traded in these markets are durable in (c) nature and can be stored for many years
a) *Cash market*
b) *Forward market*
c) *Secular market*
d) *Capital market*

83. The process of bringing desirable change in the human behavior is
a) Education
b) *Programme planning*
c) Extension
d) Evaluation

84. *Arrange the* stages of Extension *Educational process* in correct order (b) A. Evaluation B. Reconsideration C. Objective & Solution D. *Teaching plan* of work E. Situation & problems
a) E,C,D,B,A
b) E,C,D,A,B
c) A,B,C,D,E
d) A,C,B,D,E

85. *Match the* following and identify the correct match Column- I - I Column-II - II 1 *Marthandam project* i Albert Mayer 2 *Etawah project* ii F.L.Brown 3 *Srinikethan project* iii Dr. Spencer Hatch 4 Gurugram Experiment iv Rabindranath Tagore
a) 1-iii, 2-i, 3-ii, 4-iv
b) 1-iii, 2-ii, 3-i, 4-iv
c) 1-iii, 2-i, 3-iv, 4-ii
d) 1-ii, 2-i, 3-iii, 4-iv

86. *Match the* following Method of pruning
a) Pollarding
b) *Heading back*
c) Nipping
d) II, I, III

87. *Pusa parvathi* is a variety of ___ crop
a) Tomato
b) Okra
c) Brinjal
d) *French bean*

37. Which type of threshing cylinder is most commonly used for manually (a) operated paddy thresher

a) *Wire loop*
b) *Angle bar*
c) *Peg tooth*
d) *Hammer mill*

38. *Match the* following and choose the correct answer List i List ii 1 *Hand operated* sprayer

a) 20-55 kg/cm2
b) 1-7 kg/cm2
c) *Petrol engine*
d) *Diesel engine*

39. *Fly wheel* is made of?

a) *Alloy steel*
b) *Forged steel*
c) *Cast iron*
d) Aluminium

40. The horizontal distance between the front and rear wheels of a tractor is (b) known as

a) Track
b) *Wheel base*
c) *Ground clearance*
d) *Turning space*

41. The vertical face of the share which slides along the furrow is called

a) *Share point*
b) *Cutting edge*
c) Wing of the share
d) Gunnel

42. The storage capacity of *Morai type* structure is ___ tonnes

a) 3.5-18
b) 10-40
c) 9-35
d) 10-15

43. The word 'Always' is

a) Adverb
b) Verb
c) Adjective
d) Preposition

44. Antonym of word `diligent'

a) Lazy
b) Cleverness
c) Careful
d) Different

45. Which of the following are NOT correctly matched?

a) PKM-1 — Alphonso x Neelum
b) Ratna — Neelum x Alphonso
c) Amrapalli — Dashehari x Neelum
d) Mallika — Neelum x Dashehari

46. 10 *Gunter chain* =

a) 1 mile
b) 1 furlong
c) 1 acre
d) 0.5 acre

88. *Match the* following and identify the correct option List I (Crop) I List II (Scientific name) II List-III (Variety) III 1 *Bottle gourd*

a) *Momordica charantia* i Arka Sheetal
b) *Trichosanthes anguina* ii Arka Bahar
c) *Lagenaria siceraria*
d) *Luffa acutangula*

89. *Phomopsis blight* in brinjal can be controlled by treating with

a) Captan
b) *Trisodium orthophosphate*
c) Aureofungin
d) Imidachloprid

90. *World water* day is celebrated on

a) March 22nd
b) May 22nd
c) June 22nd
d) February 22nd

91. *Write the* full form of SREP

a) Strategic *Research and* Education Planning
b) Strategic *Research and* Extension Programme
c) Strategic *Research and* Extension Planning
d) Strategic *Rural and* Employment Programme

92. *Projected visual* aid____

a) Model
b) Epidiascope
c) *Panel graph*
d) *Flash cards*

93. Standing committee - 5 is related to _____

a) Education
b) works
c) Development
d) Taxes

94. What are the main pillars of `Viksit Bharat 2047' ?

a) Farmers, women, Children, Youth
b) Women, Entrepreneurs, Youth, Poor
c) Entrepreneurs, Women, Children, Youth
d) Poor, Farmers, Women, Youth

95. Who is the first Agricultural Minister of the Country?

a) Babu Jagjivan Ram
b) Sardar Vallabhai Patel
c) Rafi Ahmed Kidwai
d) Dr. Rajendra Prasad

96. Which district in Telangana is largest producer of chillies

a) Warangal
b) Khammam
c) Mahaboobabad
d) Gadwal

97. What is the current ratio of N, P, K fertilizer use in the country? N, P, K

a) N:P:K :: 11:4:1
b) N:P:K :: 5:1:1
c) N:P:K :: 4:2:1
d) N:P:K :: 7:5:1

47. H and I mirrors in an optical square are set at ___ degree angle

a) 30°
b) 60°
c) 75°
d) 45°

48. *Current rate* of CO2 in atmosphere CO2

a) 100 ppm
b) 420 ppm
c) 145 ppm
d) 545 ppm

49. Type of greenhouse constructed on hilly terrain

a) *Ridge and* furrow
b) Quonset
c) Even span
d) Uneven span

50. Statement I: A disease that usually occurs widely but periodically in a destructive form is endemic disease
Statement II: A disease which is constantly present in a moderate to severe form and is confined to particular country or district is epidemic disease

a) Only statement I is true
b) Only statement II is true
c) Both statements are false
d) Both statements are true

98. Among the following which constitutes the highest percentage share of (c) India's agricultural imports?

a) Wheat
b) Sugar
c) *Edible oils*
d) Pulses

99. From which year Government of India made it mandatory to coat all subsidized agricultural grade urea with neem oils?

a) 2015
b) 2016
c) 2018
d) 2012

100. The headquarters of Food and Agricultural Organization (FAO) is located (b) in? (FAO)

a) Geneva
b) Rome
c) New York
d) Paris

AGRICET 2025 — DIPLOMA IN AGRICULTURE

100 Questions | Answers given below each question | Correct option shown in BOLD

1. Which of the following gases contribute to nearly 99% by volume of air in lower atmosphere?

a) Nitrogen + Carbondioxide
b) Oxygen + Carbondioxide
c) Argon + Carbondioxide
d) Nitrogen + Oxygen

2. Which concept is more practically relevant to explain Indian monsoons?

a) Thermal concept
b) *Aerological concept*
c) *Humidity concept*
d) *Flohn concept*

3. In which land capability class, the land use is restricted to recreation

a) Class -II
b) Class IV
c) Class - VI
d) Class VIII

4. Diameter of USWB class A open pan evaporimeter is

a) 20 cm
b) 120 cm
c) 100 cm
d) 200 cm

5. Which one of the following is not an intercultivation practice

a) Thinning
b) Weeding
c) Levelling
d) *Gap filling*

6. *Calculate the* plant population per acre, if crop spacing is 80 cm between the rows and 50 cm between the plants

a) 1,000 plants/acre
b) 10,000 plants/ acre
c) 1,00,000 plants/acre
d) 4,000 plants /acre

7. Which of the following is a pop corn?

a) *Zea mays* var amylacea
b) *Zea mays* var indurate
c) *Zea mays* var everta
d) *Zea mays* var ceratina

8. The origin of *Bengalgram crop* is

a) China
b) Turkey
c) Africa
d) Brazil

9. Which of the following fertilizer is applied for better decomposition of green manure crops

a) Murate of Potash
b) Sulphate of Potash
c) Single Super Phosphate
d) Urea

51. Match Column A with Column B Market Structure Characteristic

a) *Perfect competition*
b) *Monopolistic competition* B *Few sellers*-homogenous product
c) Pure Oligopoly
d) *Differentiated oligopoly* D Many sellers -homogenous product

52. The market information pertaining to market arrivals, prices, demand for the commodities etc., in the past period refers to

a) *Market news*
b) *Market intelligence*
c) Communication
d) *Market extension*

53. In a four stroke engine, the speed of camshaft is

a) Double the speed of crank shaft
b) *Half the* speed of crank shaft
c) Triple the speed of crank shaft
d) Equal to speed of crank shaft

54. *Walking type* tractor is fitted with

a) 8-12 hp engine
b) 25-80 hp engine
c) 1.5-5 hp engine
d) 150-200 hp engine

55. *Disc angle* of harrow plough is

a) 15-25o
b) 5-10 o
c) 50-65o
d) 40-45o

56. The type of threshing cylinder mostly used for paddy crop is

a) *Peg type*
b) *Wire loop*
c) *Angle bar*
d) *Hammer mill* type

57. In thin layer drying, grain depth should not be more than

a) 20 mm
b) 20 cm
c) 10 mm
d) 20 feet

58. *Rotary tiller* is also called as

a) Puddler
b) *Chisel plough*
c) *Auger plow*
d) Rotavator

59. In a rectangular weir if the crest length is same as that of the channel width it is known as

a) *Cipoletti weir*
b) *Triangular weir*
c) *Suppressed weir*
d) Nappe

10. Striga is a parasitic weed seen in

a) Rice
b) Sorghum
c) Redgram
d) Soybean

11. Which of the following is a *Townsville stylo*?

a) *Stylosanthes hamata*
b) *Stylosanthes scabra*
c) Stylosanthes humilis
d) *Stylosanthes fruiticole*

12. The desirable level of nicotine in FCV tobacco is

a) 1.7 to 2.0 %
b) 4.7 to 6.0 %
c) 2.7 to 4.0 %
d) 6.7 to 8.0 %

13. Which of the following crop is highly sensitive to waterlogging

a) Rice
b) Sugarcane
c) Sesame
d) Jute

14. Which of the following sugarcane varieties are poor ratoons

a) Early varieties
b) *Mid late* varieties
c) Late varieties
d) Both *Mid and* Late varieties

15. Which type of tobacco is cultivated in *Odisha state*.

a) *Bidi tobacco*
b) *Hookah tobacco*
c) Pikka tobacco
d) *Chewing tobacco*

16. Which method of tobacco curing is a slow process and takes 6-8 weeks

a) Air curing
b) *Pit curing*
c) *Sun curing*
d) *Fire curing*

17. Triticale is an example of which type of hybridization

a) Intergeneric hybridization
b) *Interspecific hybridization*
c) *Intervarietal hybridization*
d) *Intrageneric hybridization*

18. The sequence of events in release of new crop varieties in India 1. Identification of superior strains 2. Evaluation of performance 3. Development of new strains 4. *Release and* notification

a) 1,2,3,4
b) 3,1,2,4
c) 3,2,1,4
d) 4,3,2,1

19. *Arrange the* events in seed certification sequentially 1. Inspection of seed field 2. *Post harvest* inspection 3. Receipt & scrutiny of application 4. Verification of seed source 5. Grant of certificate 6. Seed sampling & testing

60. In subsurface drip irrigation system the discharge rate is generally in the range of

a) 0.6 to 3 l/h
b) 6 to 13 l/h
c) 16 to 23 l/h
d) 0.006 to 0.3 l/h

61. If a orifice flows partially then it behaves like a

a) Sluice
b) Spillway
c) Weir
d) Aqueduct

62. In weirs the depth of flow over the crest is known as

a) Nappe
b) *End contraction*
c) Head
d) Flume

63. To obtain reasonable degree of uniformity in the discharge of each sprinkler the mains should run in the direction of the

a) *Parallel slope*
b) Steepest slope
c) *Grid lines*
d) *Counter lines*

64. The area covered by a rotating head sprinkler can be estimated from the formula

a) R= 1.35
b) R=1.95
c) R=1.05
d) R=1.98

65. ROM stands for

a) Read Only Memory
b) Refer Only Memory
c) Refer Only Magnetic
d) Record Only Memory

66. Which of these is not a computer language

a) Fortran
b) Cobol
c) Basic
d) Jargon

67. *Homophones occur*

a) Single
b) In pairs
c) In triplets
d) In quadruples

68. A part of word that can pronounced in one breath is called

a) Phrase
b) Syllable
c) Clause
d) Vowel

69. What are the conditions for hot water treatment for managing grassy shoot of sugarcane

a) 52oC for 30 min
b) 50oC for 30 min
c) 52oC for 10 min

a) 1,2,3,4,5,6
b) 3,4,6,1,2,5
c) 3,2,4,5,6,1
d) 3,4,1,2,6,5

20. The development of embryo directly from the nucellus or integuments is called as ---
a) Apospory
b) Adventive embryony
c) Apogamy
d) Parthenogenesis

21. The standard physical purity of *Groundnut certified* seed is ----
a) 98%
b) 99%
c) 96%
d) 100%

22. In which of the following crops bloom, internode length, sex expression and branching pattern are used for identification of rogues during field inspection
a) Sunflower
b) Cotton
c) Sesame
d) Castor

23. The seed coat peroxidase test is used for genetic purity testing in which of the following crops
a) Rice
b) Wheat
c) Soybean
d) Oats

24. *Match the* following soil particles based on size (in mm)
a) Clay
b) Fine Sand
c) Silt
d) Gravel Pebbles

25. *Match the* following units correctly
a) *Bulk density*
b) Electrical conductivity
c) ESP
d) Gypsum requirement

26. The effect of light scattering in colloidal dispersion while showing no light in a true solution is called as
a) *Brownian effect*
b) *Flocculation effect*
c) *Deflocculation effect*
d) Tyndall effect

27. The pH concept was given by ----
a) Robinson
b) Dokuchaev
c) Buckman& Brady
d) Sorensen

28. *Match the* following elements with their deficiency symptoms in crops
a) Magnesium
b) Zinc
c) Copper
d) Molybdenum

d) 50oC for 10 min

70. The fungal genus causing damping off in vegetables
a) Phytophthora
b) Pythium
c) Verticillium
d) Fusarium

71. The symptoms of corky growth and localized death of cortical tissues of fruit is due to
a) Scab
b) Cankers
c) Blast
d) Warts

72. The scientific name of fungi causing downy mildew in sunflower
a) *Plasmopara viticola*
b) *Pseudoperenospora cubensis*
c) *Selerospora gramincola*
d) Plasmopara halstedii

73. The bacterial leaf blight of rice spreads from one field to another field through
a) Irrigation water
b) Soil
c) Insects
d) Air

74. The association of nematode infestation is prominent in causing the following disease in banana
a) *Moko wilt*
b) Panama wilt
c) *Heart rot*
d) *Sigatoka leaf* spot

75. Which one of the following rootstocks are resistant against gummosis of citrus
a) *Rangapur lime*
b) Balaji
c) Jambheri
d) *Tenali selection*

76. *Match the* following — Crop: Coleus, Kalmegh, Ashwagandha, *Senna with* Secondary metabolite: Withamine A&B, Sennosides, Diterpenoids, Forskolin
a) 1-i, 2-ii, 3-iii, 4-iv
b) 1-iv, 2-iii, 3-i, 4-ii
c) 1-ii, 2-iii, 3-i, 4-iv
d) 1-iii, 2-i, 3-ii, 4-iv

77. Application of the following nutrient increases the number of tillers in palmarosa
a) Mg
b) Ca
c) B (Boron)
d) Zn

78. The division of forestry dealing with measurement of forest produce
a) *Forest utilization*
b) *Wood technology*
c) Forest mensuration
d) *Forest management*

29. ***Guano manure*** **is a product of**

a) *Sea animal*
b) *Sea bird*
c) Marine Fish
d) *Sheep horns*

30. ***Find out*** **the free living nitrogen fixing bacteria from the following bio- fertilizers?**

a) Rhizobium
b) Azospirillum
c) Azorhizobium
d) Azatobacter

31. ***Match the*** **chemical formula with fertilizers and N content (%). Fertilizer Formula N content (%)**

a) Urea
b) Calcium Ammonium Nitrate B (NH4)2SO4
c) Calcium Nitrate
d) Ammonium Sulphate

32. ***Match the*** **following P fertilizers with chemical formula.**

a) Gypsum
b) *Rock phosphate*
c) *Basic slag*
d) *Dolomitic lime*

33. ***Biuret concentration*** **in urea should be below ----**

a) 10%
b) 1.5%
c) 5.0%
d) 50%

34. ***Rasping and*** **sucking type of mouth parts are asymmetrical due to the absence of**

a) Right Maxillae
b) Right Mandible
c) Left Maxillae
d) Left Mandible

35. Which of the following is anticoagulant rodenticide

a) *Zinc phosphide*
b) *Aluminium phosphide*
c) Bromodialone
d) *Methyl bromide*

36. ***Disease caused*** **by fungi in insects is commonly known as**

a) Septicemia
b) *Treetop disease*
c) Pebrine
d) Mycosis

37. ***Stridulatory organs*** **are the characteristic feature of the following order?**

a) Orthoptera
b) Hemiptera
c) Thysanoptera
d) Lepidoptera

38. The abdominal second segment is modified into pedicel or petiole in the following order

a) Isoptera
b) Hymenoptera
c) Thysanoptera

79. ***Match the*** **following Tree Common name**

a) Teak
b) Bamboo
c) Neem
d) 1-ii, 2 iii,3-i

80. Modified system of shifting cultivation is

a) *Random mix*
b) *Multistoried cropping*
c) *Taungya system*
d) *Strip system*

81. ***Katha and*** **cutch are produced from the wood of**

a) Khair
b) Neem
c) Bamboo
d) Teak

82. Examples of tuberous roots

a) Potato
b) *Sweet potato*
c) Ginger
d) Gladiolus

83. Which of the following is the attached scion method of grafting

a) *Veneer grafting*
b) Soft wood grafting
c) *Epicotyl grafting*
d) Inarching

84. In banana removal of heart or male bud is known as

a) Mattocking
b) Tipping
c) Propping
d) Trashing

85. ***Gynodioecious varieties*** **of papaya produces**

a) Only staminate flowers
b) Only hermaphrodite flowers
c) Only pistillate flowers
d) Both pistillate & hermaphrodite flowers

86. ***Match the*** **following — Crop: Tomato, Chilli, Onion,** ***Carrot with*** **Pigment: Capsanthin, Quercetin, Lycopene, Carotene**

a) 1-iii, 2-i, 3-ii, 4-iv
b) 1-i, 2-ii, 3-iii, 4-iv
c) 1-ii, 2-i, 3-iii, 4-iv
d) 1-iv, 2-iii, 3-i, 4-ii

87. ***Match the*** **following — Crop: Cabbage, Cauliflower,** ***Bottle gourd*****,** ***Colocasia with Economic product*****: Head, Curd, Pepo, Corms**

a) 1-i, 2-ii, 3-iii, 4-iv
b) 1-iii, 2-ii, 3-iv, 4-i
c) 1-ii, 2-i, 3-iii, 4-iv
d) 1-iv, 2-i, 3-iii, 4-ii

88. According to the principles of ***Extension what*** **is the basic unit of civilization**

a) Home
b) School
c) Nation

d) Diptera

39. Drie-*Die results* in death of storage pests due to
a) Loss of moisture
b) Increase in temperature
c) Decrease in RH
d) Loss of wax

40. *Fore wings* are divided into 2 parts and *Hind wings* are divided into 3 parts in the following insect pest ?
a) *Helicoverpa armigera*
b) *Spodoptera litura*
c) Exelastis atmosa
d) *Maruca vitrata*

41. *Chlorotic patches* at tips of leaflets in typical V shaped manner in groundnut is due to
a) White grub
b) Leaf miner
c) Leaf hopper
d) Thrips

42. *Lepidopteran predator* on sugarcane wooly aphid?
a) *Chrysoperla carnea*
b) Dipha aphidivora
c) *Epiricrania melanoleuca*
d) *Ceratovacuna lanigera*

43. *Pumpkin beetle* grubs feeds on
a) Stem
b) Leaves
c) Roots
d) Fruits

44. Maryland, USA does not allow mango imports from India because of the following pest?
a) Nut weevil
b) *Fruit fly*
c) *Fruit borer*
d) *Shoot borer*

45. *Ovaries distorted* into gall like structure in chillies due to
a) Thrips
b) Mites
c) Blossom midge
d) Aphids

46. *Mid larval* parasite on black headed caterpillar of Coconut
a) *Opisina arenosella*
b) Bracon brevicornis
c) *Cotesia taragami*
d) *Goniozus nephantidis*

47. What is the Elasticity of production (EP) in stage III of production function
a) EP<1
b) EP >1
c) EP <0
d) EP =0

48. The loans that are due for the repayment within a period of two to five years are called
a) *Current liabilities*

d) Community

89. The leaders who actually initiate action within the group regardless of whether (or) not they hold an elected office
a) *Ornamental leader*
b) *Prominent talent* leader
c) Operational leader
d) *Professional leader*

90. An outline of procedures pertaining to some phases of extension work is known as --- ?
a) Project
b) Calendar of work
c) Plan of work
d) Programme

91. Which PRA technique is used for analyzing recurring seasonal challenges ?
a) *Time line*
b) *Transact walk*
c) *Matrix ranking*
d) Trend analysis

92. A process of formation of opinion about work (or) an object depending on its values and use to make effective decisions rather than hasty decision is known as which form of Evaluation
a) *Informal evaluation*
b) *Project evaluation*
c) Formal evaluation
d) *Extension evaluation*

93. India's first Biochar Centre of *Excellence was* inaugurated at
a) Punjab Agricultural University, Ludhiana
b) Kanha Shanti Vanam, Hyderabad
c) Gujarat Natural Farming Science University
d) National Dairy Research Institute, Karnal

94. World Food *Prize for* the year 2025 was awarded to
a) Dr. Geoffrey Hawtin
b) Ms. Heidi Kuhn
c) Dr. Cary Fowler
d) Dr. Mariangela Hungria

95. United Nations (UN) designated 2025 as the year of
a) International Year of Co-operatives
b) International Year of Pulses
c) International Year of Millets
d) International Year of Women Farmers

96. First National Co-operative *University was* established at
a) Anand, Gujarat
b) New Delhi
c) Mumbai
d) Hyderabad

97. The current Director General of ICRISAT is
a) Dr. Himanshu Pathak
b) Dr. M. L. Jat
c) Dr. Jacqueline Hughes
d) Dr. T. Mohapatra

b) *Intermediate liabilities*
c) Long term Assets
d) Long term liabilities

49. *Kincked demand* curve is an important feature of
a) *Oligopoly market*
b) *Monopolistic competition*
c) *Duopoly market*
d) *Monopoly market*

50. Assertion (A): AFC curve is declining with increased output. Reason (R): The total fixed costs are constant.
a) A is true but R is false
b) A is false but R is true
c) Both A & R are true and R explains A
d) Both A & R are true but R does not explain A

98. *Full form* of VKSA, a 15-day nationwide agricultural outreach campaign
a) Vikas Krishi Sanarakshan Abhiyan
b) Viksit Krishi Sankalp Abhiyan
c) Viksit Kisan Samagra Abhiyan
d) Vikasit Krishi Surakshit Abhiyan

99. PJTAU recently organized a state wide campaign "Rythu Mungitlo Shastravethalu" during ------
a) 5th May -13th June, 2025
b) 12th May -24th June, 2025
c) 1st to 13th May, 2025
d) 1st to 31st May, 2025.

100. What are the names of the two genome-edited rice varieties developed by ICAR?
a) DRR Dhan 100 and Pusa DST Rice 1
b) Basmati 370 and Sona Masuri
c) DRR Dhan 64 and DRR Dhan 53
d) Pusa 1592 and Pusa Basmati 1728